

Comprehensive Literature Atlas and Environmental Histories of the MAUVE Virgo Galaxy Sample: Mechanisms, Timelines, and gFMR/FMR Implications

Version: 2026-08-12 **Scope:** 39 MAUVE target identifiers, representing 40 physical galaxies **Language:** English **Document type:** internal PhD-level research monograph and evidence atlas

AI disclosure. This atlas was produced with AI-assisted literature search, metadata reconciliation, evidence extraction, synthesis, figure generation and report drafting. Claims were checked against the cited source records and accessible full texts, but the document still requires domain-expert human review before publication or use as authoritative orbital history.

Abstract

The MAUVE stage labels compress a heterogeneous set of Virgo histories into pre-peak, near-peak and post-peak bins. This atlas restores the lost mechanism and time information for every target. It combines the live MAUVE catalogue and sample document with a 19,465-link SIMBAD bibliography snapshot, a curated registry of 78 environmentally informative works, and 259 paper-galaxy evidence records. The sample contains 39 target systems and 40 physical components because NGC4567_8 is treated as one observing target but two interacting galaxies. The adopted physical-galaxy counts are 12 pre-peak, 10 near-peak and 18 post-peak; target-system counts are 11, 10 and 18. The dominant result is that stage is not mechanism: secure ram-pressure cases coexist with tides, mergers, direct disk-disk collision, ancient accretion, starvation/old depletion, nuclear or stellar feedback, and ambiguous projected-phase-space assignments. NGC4192 and NGC4698 receive explicit reclassification audits. Each chapter separates observations from models and atlas inference, states timing limitations, and proposes MUSE/CO/H I tests relevant to resolved metallicity--SFR and gas-FMR work.

How to use this atlas

Use the adopted stage as a sample-design coordinate, not as a literal orbit. For any scientific interpretation, read the chapter's mechanism, timing and contradiction fields together. The evidence grade describes source coverage (tailored model, direct resolved observation, survey measurement or contextual inference); it is not the same as confidence in the reconstructed history. Every gFMR/FMR paragraph labelled **new atlas inference** is a synthesis proposed here, not a claim made by all cited authors.

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1. Executive synthesis

1.1 Sample accounting

Counting unit	Pre-peak	Near-peak	Post-peak	Total
MAUVE target systems	11	10	18	39
Physical galaxies	12	10	18	40

“Near-peak” is used throughout. “Close-to-peak” is a synonym appearing in earlier discussion. NGC4567_8 has one system chapter and two component evidence records; its two galaxies must not be merged when computing gas mass, metallicity, SFE or resolved relations.

1.2 Principal scientific conclusions

- **Secure recent/active ram pressure:** NGC4330, NGC4388, NGC4402, NGC4501, NGC4522 and NGC4569 have the strongest combinations of resolved gas morphology, stellar-population clocks, radio/X-ray constraints and/or tailored models.
- **Compound gravity plus ICM:** NGC4254, NGC4424 and NGC4654 require tidal/merger histories in addition to ram pressure. NGC4694 is primarily tidal; counting its bridge as a wind tail is misleading.
- **Direct galaxy collision:** NGC4567 and NGC4568 are an early disk--disk collision with a resolved molecular front but no secure absolute encounter clock and no strong present ICM-stripping diagnosis.
- **Ancient accretion is not a stripping clock:** NGC4698's polar nuclear component is older than 5 Gyr; it does not validate the adopted post-peak code. NGC4216's spectacular dwarf disruption proves gravitational processing but leaves the ICM epoch uncertain.
- **Feedback can masquerade as environment:** NGC4064, NGC4383, NGC4424 and NGC4569 contain ionized outflows or AGN-illuminated structures that must be separated from stripped gas and H II-region emission.
- **Chemical inference is path dependent:** gas removal, compression, phase conversion, inflow, mixing and feedback change gas mass, SFR and metallicity on different clocks. A cross-sectional fixed-Mstar FMR/gFMR is not the trajectory of one galaxy.

1.3 Mechanism-aware component classification

Component	Adopted stage	Yoon class	Köppen status	Primary mechanism	Secondary mechanism	History confidence
IC3392	post-peak	III	past	past ram pressure stripping	none clearly established	high
NGC4064	post-peak	III	past	past ram pressure plus gravitational disturbance	star-formation-driven outflow	high for past processing and current outflow; moderate for mechanism ordering
NGC4189	pre-peak	0	not included	weak hydrodynamic perturbation or group tide	internal warp	low-moderate
NGC4192	pre-peak	IV	active	weak or incipient environmental processing	bar/warp and projection	moderate for weak processing; low for orbital stage
NGC4216	post-peak	IV	past	long-term gas depletion plus satellite accretion	possible preprocessing/slow ICM process	high for minor accretion; low for gas-loss epoch
NGC4222	pre-peak	0	active	none established; possible weak ram pressure	warp	low
NGC4254	pre-peak	I	non-deficient	high-speed tidal encounter/harassment	weak ongoing ram pressure	high for a past fast encounter; moderate for current RPS
NGC4293	post-peak	III	past	past strong gas stripping	bar/interaction-driven central gas	moderate-high
NGC4294	pre-peak	I	active	early ram pressure stripping	tidal interaction with NGC 4299	moderate
NGC4298	near-peak	II	active	ram pressure plus interaction with NGC4302	pair tide	high for mixed processing; moderate for chronology
NGC4302	near-peak	II	active	active ram pressure stripping	interaction with NGC4298 and disk-halo flows	moderate-high
NGC4321	pre-peak	0	active	weak tidal perturbation	possible weak ram pressure	low-moderate
NGC4330	near-peak	II	active	active pre-peak ram pressure stripping	minor internal asymmetry	high
NGC4351	pre-peak	I	past	active/early ram pressure stripping	possible gravitational perturbation	moderate-high for active RPS; low for prior tide
NGC4380	post-peak	IV	past	old/slow gas depletion	starvation or weak past stripping	moderate for processed status; low for mechanism
NGC4383	pre-peak	0	non-deficient	recent gas acquisition and starburst feedback	tidal perturbation	high for feedback outflow; moderate for recent gas supply; low for orbital history
NGC4388	near-peak	II	active	strong recent ram pressure stripping	AGN illumination/outflow	very high for stripping; high-moderate for exact ICM/orbit
NGC4394	post-peak	IV	not included	slow/old gas depletion	possible weak M85 tide	low-moderate
NGC4396	pre-peak	I	active	early ram pressure stripping	possible high-speed tide	moderate
NGC4405	post-peak	III	past	past ram pressure stripping	none clearly established	high
NGC4402	near-peak	II	active	active ram pressure stripping	dense-cloud decoupling	high
NGC4419	post-peak	III	past	past strong ram pressure stripping	central bar/nuclear processing	high
NGC4424	near-peak	II	active	intermediate-ratio merger remnant plus ongoing ram pressure	central starburst outflow	high for compound history; moderate for detailed ordering
NGC4450	post-peak	IV	past	old/slow environmental depletion	bar/AGN/internal secular evolution	moderate-low
NGC4457	post-peak	III	not included	past/relaxed gas stripping	warp or minor accretion	moderate-low
NGC4501	near-peak	II	active	pre-peak ram pressure stripping	internal spiral/bar response	high
NGC4522	near-peak	II	active	near/post-peak ram pressure stripping	dense-cloud decoupling and ICM-ISM mixing	very high
NGC4535	pre-peak	I	active	weak ongoing ram pressure	internal spiral structure	moderate-low

Component	Adopted stage	Yoon class	Köppen status	Primary mechanism	Secondary mechanism	History confidence
NGC4548	post-peak	IV	past	slow or old stripping/starvation	bar dynamics	moderate-low
NGC4567	pre-peak	0	not included	direct galaxy-galaxy collision	no strong ICM stripping established	high for pair interaction; no secure pair or cluster clock
NGC4568	pre-peak	0	not included	direct galaxy-galaxy collision	tidal compression/shear; no strong ICM stripping established	high for pair interaction; no secure pair or cluster clock
NGC4569	post-peak	III	past	past strong ram pressure stripping	nuclear starburst/outflow	very high
NGC4579	post-peak	IV	past	old/slow gas depletion	bar and AGN	moderate-low
NGC4580	post-peak	III	past	past ram pressure stripping/backsplash candidate	none clearly established	high for past stripping; moderate for backsplash
NGC4606	post-peak	III	past	past severe gas stripping	gravitational disturbance	moderate
NGC4607	post-peak	III	past	past or ongoing ram pressure stripping	projected association with NGC4606	moderate
NGC4654	near-peak	II	non-deficient	tidal interaction plus weak ram pressure	stellar feedback	high
NGC4689	post-peak	IV	past	old/slow gas depletion	internal disk evolution	moderate-low
NGC4694	near-peak	II	past	tidal accretion/merger debris	possible secondary ICM reshaping	very high for tidal history; low for ICM history
NGC4698	post-peak	I	past	ancient external accretion plus uncertain weak cluster stripping	possible M49-substructure context	high for ancient accretion; low for adopted post-peak ICM status

1.4 Reclassification audits

NGC4192. Yoon class IV would map mechanically to post-peak, but MAUVE adopts pre-peak because the live sample document reports no H I deficiency or clear stripping. VIVA shows no severe classic truncation, while Köppen classifies it as a low-deficiency active stripper. The defensible reading is not “known first infall”; it is “no secure strong post-peak signature.” The code is retained with moderate-low orbital confidence.

NGC4698. Yoon class I and VIVA indicate weak/incipient processing and projection ambiguity, whereas Köppen's analytical model labels it a past stripper. The extended, mostly regular H I disk is unlike a secure post-stripping remnant. Its polar central structure proves an unrelated >5 Gyr accretion history. MAUVE's post-peak code is retained for provenance, but its ICM chronology is explicitly low confidence.

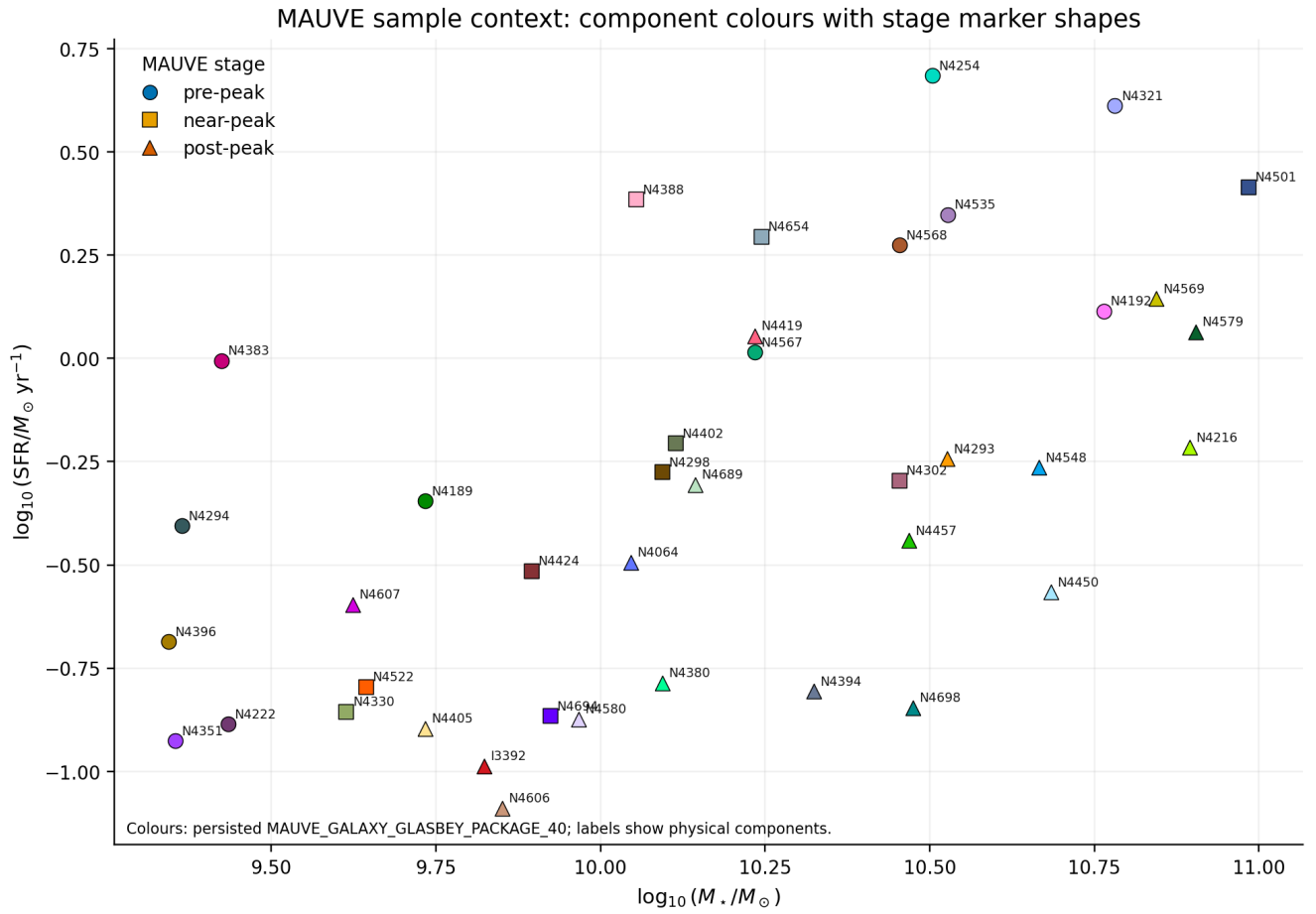


Figure 1. Stellar-mass–SFR context. Component identity uses the persisted 40-colour MAUVE palette; marker shape encodes the adopted stage.

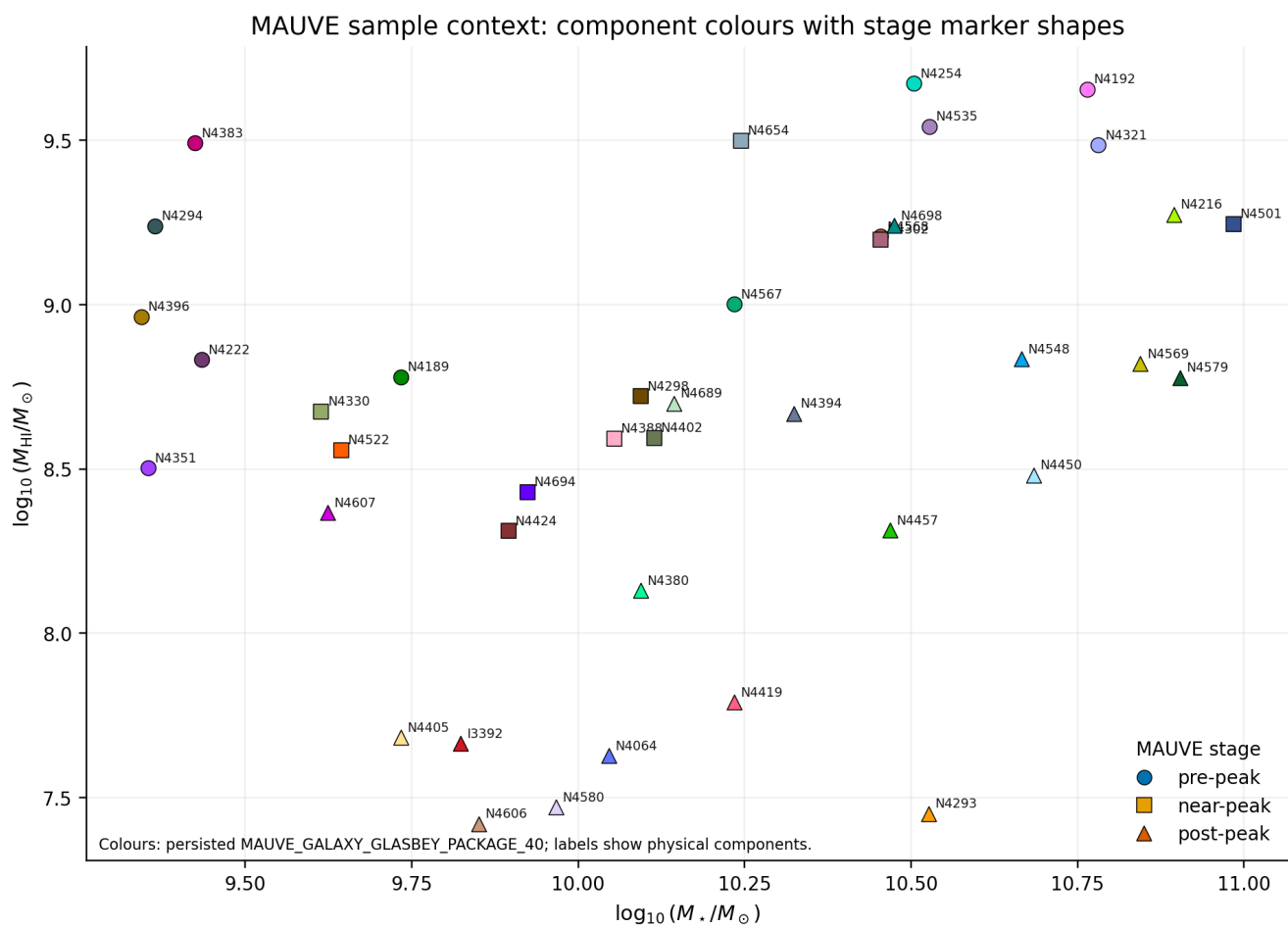


Figure 2. Stellar-mass–H I-mass context for the same physical components.

3. Research and evidence protocol

3.1 Search architecture and saturation

SIMBAD resolved all 40 physical components and returned 1,328 identifier rows. The complete execution-date snapshot contains 19,465 object--bibliography links and 5,213 unique bibcodes. It is preserved as [simbad_all_mentions_20260812.csv](#). The inventory includes catalogue-level appearances; its four title/abstract categories are triage, not evidence grades.

The curated environmental corpus began with VIVA, Yoon, Vollmer and Köppen, then expanded through VERTICO, VESTIGE, GUVICS, Halpha, CO, radio continuum/polarization, X-ray, stellar-population and dynamical studies. Two citation-chain passes were applied across the material corpus, followed by a component-alias/keyword pass. Journal/preprint duplicates were merged as one work. The stopping criterion was no new *material* source in the last chain/alias pass; this is an operational database snapshot through 2026-08-12, not universal completeness.

The full reproducible record is [query_log.md](#) and the DOI audit is [doi_resolution_audit.json](#).

3.2 Evidence hierarchy

1. **Tailored multiwavelength modelling:** object-specific simulations or SFHs constrained by several tracers. Strong for mechanisms/timing within model assumptions; not automatically a statistical error measurement.
2. **Direct resolved observations:** H I/CO/Halpha/UV/radio/X-ray/stellar morphology or kinematics that directly expose truncation, tails, bridges, compression, shocks or decoupled components.
3. **Survey-level measurement:** resolved or integrated measurements without a unique object-specific mechanism reconstruction.
4. **Phase-space/contextual inference:** projected radius/velocity, subcluster context or analytical pressure/deficiency classifications. Valuable statistically; intrinsically degenerate for individual orbits.

Every evidence record also labels claim type (observed, model-derived, inferred or speculative), locator, access, uncertainty and relevance. Source strength and final-history confidence are separate fields.

3.3 Adversarial quality checkpoints

- **Scope checkpoint:** “all papers” is not literally provable. The mitigation is a complete SIMBAD-linked snapshot plus a smaller, fully synthesized environmental corpus; inaccessible or incidental items never support detailed claims.
- **Causal checkpoint:** asymmetric gas is not automatically RPS; bridges, stellar disturbance, AGN cones, starburst winds and projection were tested as alternatives.
- **Timing checkpoint:** quenching lookback, encounter age, tail survival and time from peak pressure are kept distinct. Phase space is never treated as a unique orbit.
- **Selection checkpoint:** MAUVE's mechanism mix is not presented as a Virgo population fraction.
- **Ethics/integrity checkpoint:** no human subjects or dual-use concerns arise; the material risk is citation or causal overstatement, addressed through visible locators, access labels, DOI metadata checks and AI disclosure.

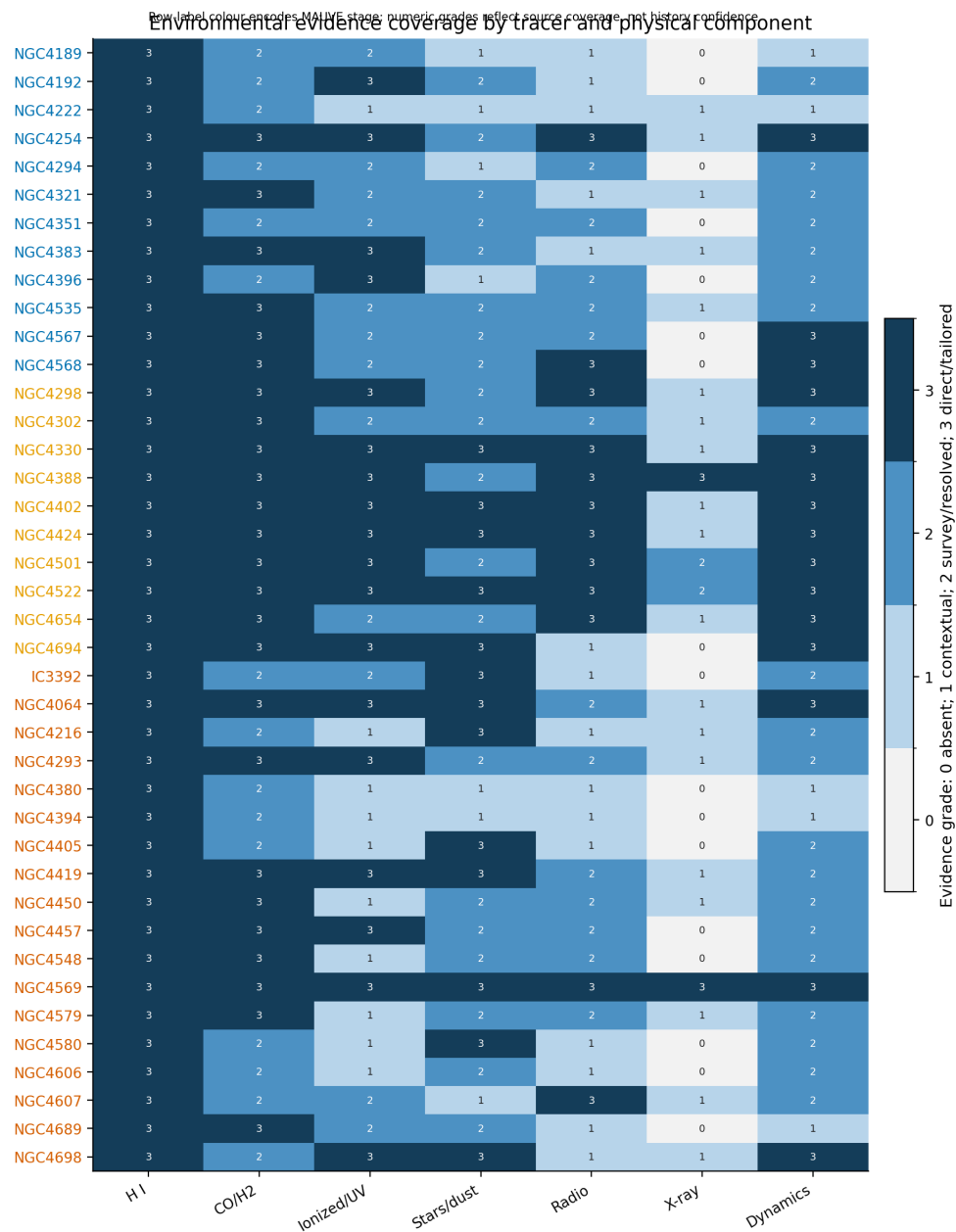


Figure 5. Evidence coverage by tracer. Grade and final-history confidence are deliberately separate.

4. Environmental and gFMR/FMR framework

4.1 What the three stages retain—and discard

VIVA provides resolved H I morphology; Yoon combines that morphology with projected phase space; Vollmer orders five tailored RPS models; Köppen compares present and time-integrated analytical pressures. These are complementary rather than interchangeable. A Yoon class is a morphology/context category, a Köppen label is model inference, and a Vollmer time is an object-specific model state.

Projected phase space uses only sky position and line-of-sight velocity. Here, projected radius is recomputed about M87 with $D=16.5$ Mpc and $R_{200}=1.55$ Mpc, and velocity is normalized by $v_{cl}=1088$ km/s and $\sigma=593$ km/s, matching the adopted Yoon/VIVA context. Unknown line-of-sight distance and transverse velocity make multiple orbits compatible with one point.

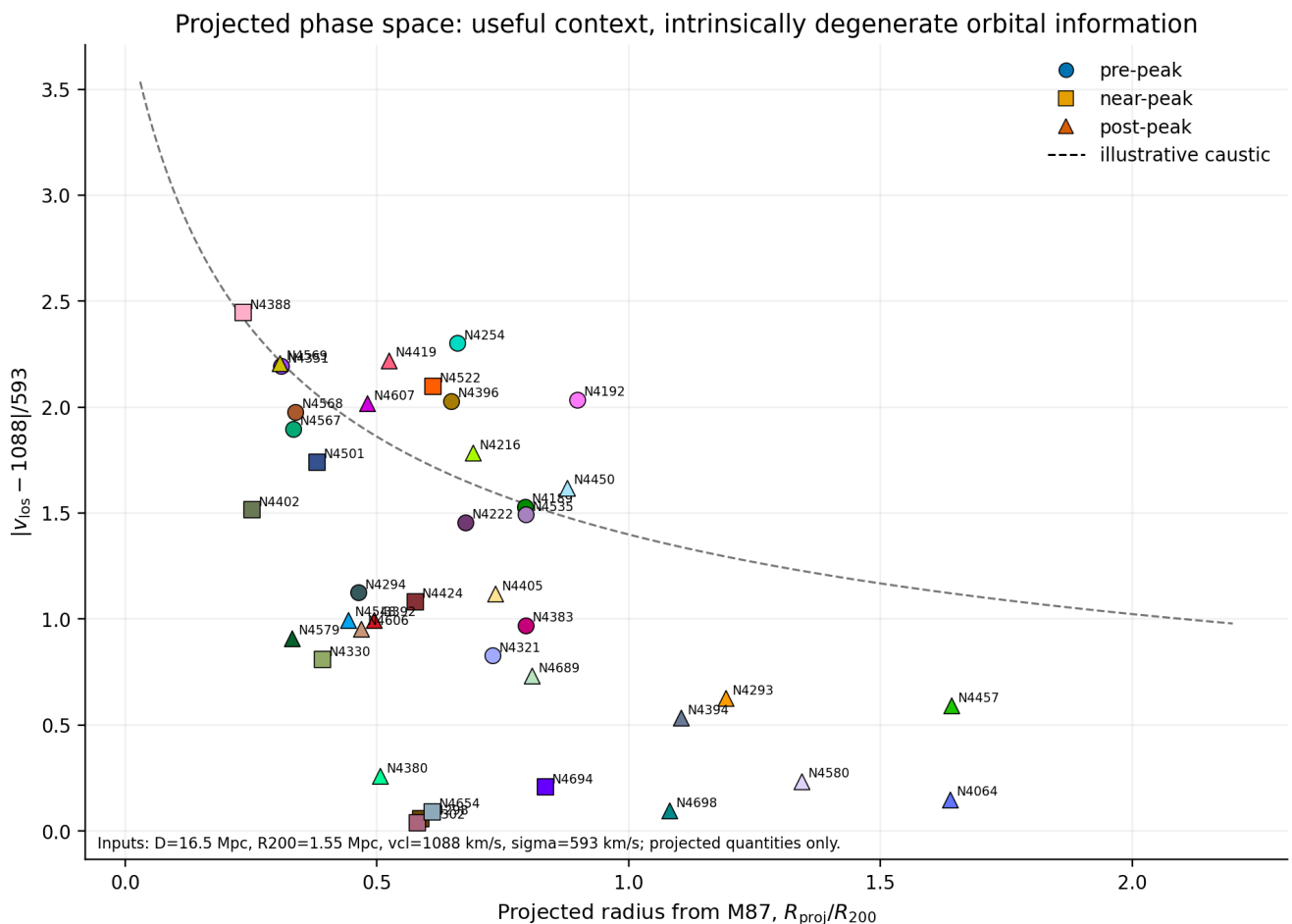


Figure 3. Projected phase space. This is statistical context, not a unique reconstruction of any orbit.

4.2 Mechanism vocabulary

Ram pressure preferentially removes diffuse outer gas, can compress the leading edge, and may permit fallback/re-accretion. **Tides, harassment and mergers** disturb stars as well as gas, produce bridges/streams, and can drive inflow or make gas easier to strip. **Preprocessing** requires evidence that a group or companion acted before/while entering the cluster; projected proximity alone is insufficient. **Starvation** is slow loss of replenishment and is difficult to distinguish from an old weak event in a relaxed disk. **Feedback** can launch ionized/radio/X-ray structures that resemble environmental tails but has a different origin.

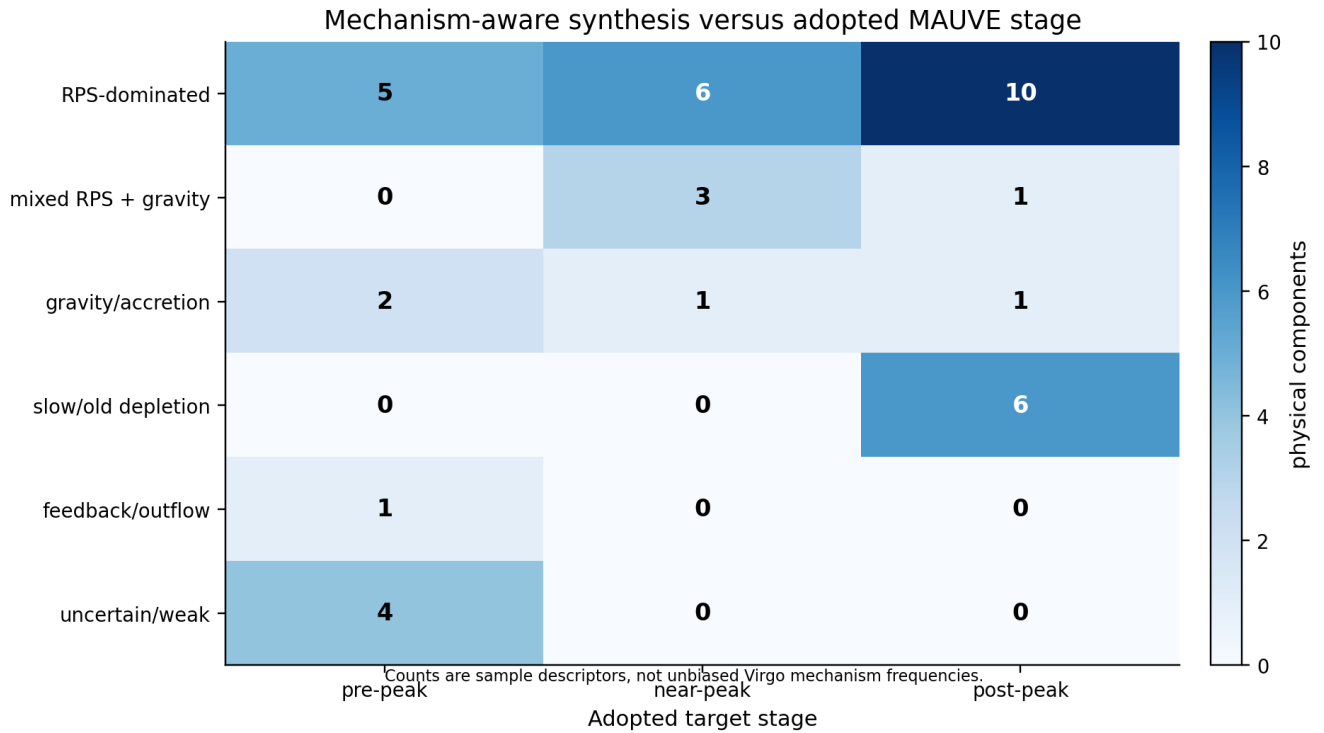


Figure 6. Mechanism-aware synthesis versus the adopted three-stage code.

Defensible published timelines: two different observables kept separate

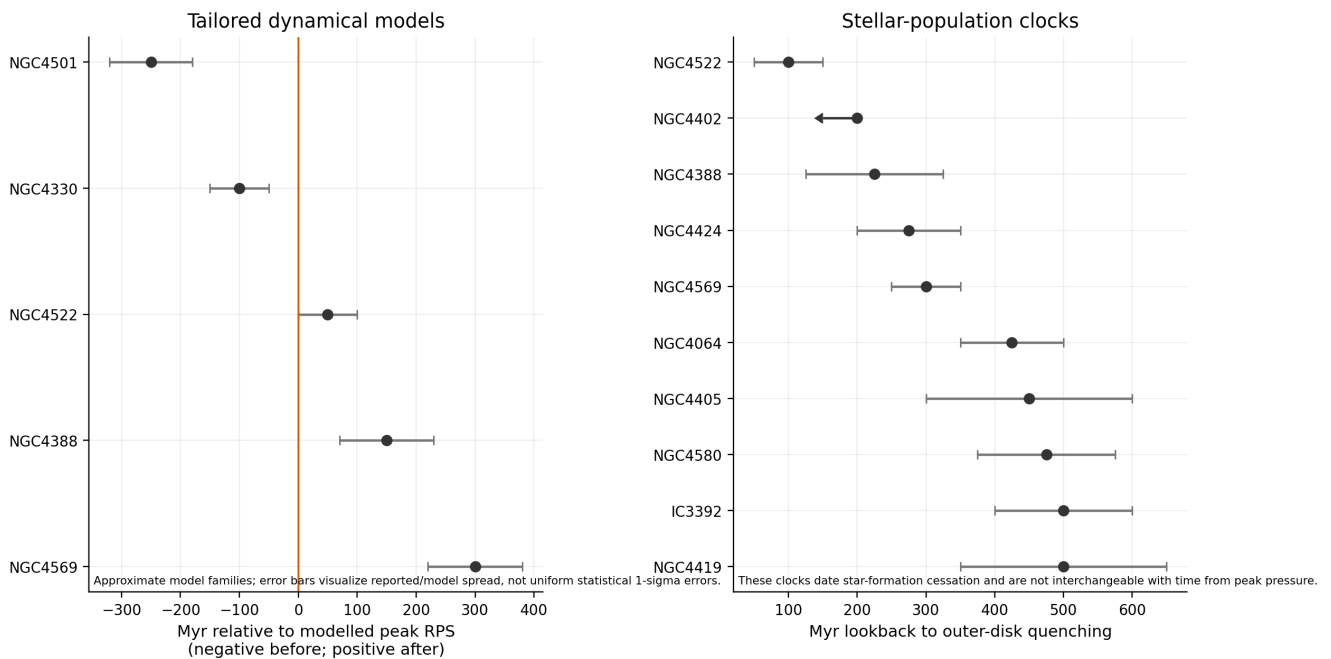


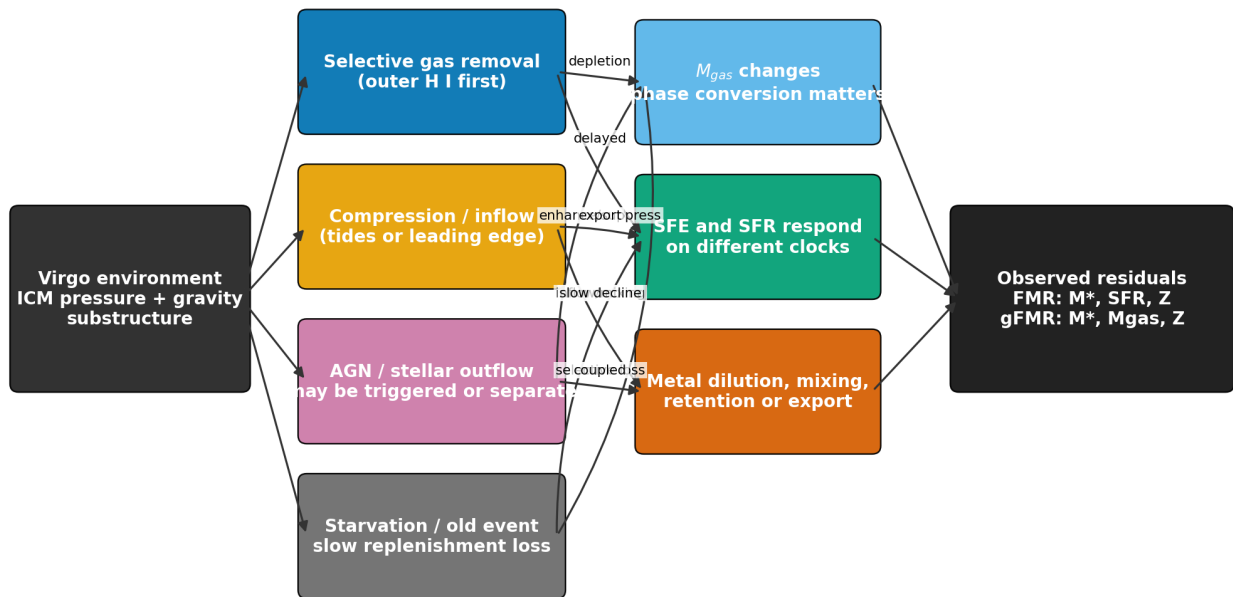
Figure 7. Published dynamical and stellar-population clocks, shown in separate panels because they date different physical events.

4.3 FMR and gas-FMR interpretation

The classical FMR is a population-level relation among stellar mass, SFR and gas-phase metallicity; gas-based extensions replace or augment SFR with atomic/molecular gas. The current Wang et al. framework ([arXiv:2608.04784](https://arxiv.org/abs/2608.04784), v1 preprint, not peer reviewed as of 2026-08-12) argues that gas mass is more fundamental and that SFR can enter through SFE. Huang et al. ([DOI](#)) show within MAUVE that resolved metallicity--SFR residuals are calibration- and galaxy-dependent. Therefore a fixed-Mstar population surface cannot be read as the time track of an individual environmentally processed disk.

Removal changes M_{gas} before or after SFR depending on phase and density; compression can increase SFR without immediate gas loss; tidal inflow can dilute central metallicity while raising molecular fraction; strangulation can increase metallicity as dilution stops; outflows can export gas and metals selectively. Integrated and resolved measurements can consequently move in different directions.

Mechanism-aware interpretation of FMR/gFMR offsets



A fixed- M^* population relation is not an evolutionary track: the same integrated residual can arise from different paths and timescales.

Figure 8. Environmental pathways through FMR/gFMR observable space; a population relation is not an evolutionary track.

5. Individual target environmental histories

Each target chapter is an evidence-constrained synthesis. Properties come from the live MAUVE table. Paper-level records below include source locators and access metadata. For galaxies without a dedicated modern environmental study, the chapter says so explicitly rather than promoting survey context into a tailored history.

5.1 IC3392

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
IC3392	VCC 1126; UGC 7602; IC 3392	9.82	-0.99	7.66	8.23	-0.76

Component	v (km/s)	MAUVE stage	Yoon	Köppen
IC3392	1678	post-peak	III	past

IC3392: observational record and environmental sequence

Adopted mechanism. past ram pressure stripping. Secondary/contextual process: none clearly established.

H I evidence (direct morphology first). The H I disk is severely, nearly symmetrically truncated while the stellar disk remains regular.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Outer-disk stars record cessation of star formation; no dedicated gravitational alternative is established.

Best-supported event sequence. A strong ICM event removed outer gas, after which the remaining disk relaxed.

Published timing and its meaning. Outer-disk quenching is about 500+/-100 Myr.

Competing interpretations and classification tension. Yoon III and Köppen past agree; the stellar clock is local and does not uniquely reconstruct the orbit.

History confidence. high.

gFMR/FMR implication — new atlas inference. The system anchors a clean post-stripping path in which low gas fraction and radial censorship precede continued central evolution.

Specific MAUVE-MUSE/VERTICO test. Measure the abundance step/gradient across the truncation radius and compare central molecular SFE to other post-peak disks.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
IC3392	3	2	2	3	1	0	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The H I disk is severely, nearly symmetrically truncated while the stellar disk remains regular. *Claim type*: observed. *Timing*: Outer-disk quenching is about 500+/-100 Myr.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Crowl et al. (2008), The stellar populations of stripped spiral galaxies in the Virgo cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: optical spectroscopy plus UV; truncated outer disks of ten Virgo spirals. **Locator**: Secs. 3-5; Tables 3-5. **Paper-level environmental claim**: Resolved outer-disk stellar populations yield galaxy-specific cessation ages from less than 200 to about 500 Myr, demonstrating rapid outside-in quenching. *Claim type*: model-derived from observations. *Timing*: component-specific values in Tables 3-5. *Uncertainty/alternative*: SFH priors, dust and aperture placement contribute uncertainty.

- **Huang et al. (2026), MAUVE-MUSE: when metallicity follows or fights star formation - a mass-dependent inversion in Virgo galaxies.** [DOI](#); [ADS](#). *Status/access*: peer reviewed; full text or accepted manuscript. *Tracer/method*: MUSE metallicity and Sigma_SFR; resolved HII-region spaxels in 14 MAUVE galaxies. **Locator**: Secs. 3-6; Figs. 3-6. **Paper-level environmental claim**: The residual metallicity-SFR relation changes sign with calibration and within galaxies; NGC4383 is a robust 0.1-0.2 dex low-metallicity outlier and NGC4192 changes internal sign. *Claim type*: observed analysis. *Timing*: current resolved state. *Uncertainty/alternative*: Strong-line calibration, DIG/HII selection and hierarchical sampling dominate interpretation.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: past; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: Outer-disk quenching is about 500+/-100 Myr.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class III; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.2 NGC4064

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4064	UGC 7054; NGC 4064	10.05	-0.49	7.63	8.27	-0.41

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4064	1000	post-peak	III	past

NGC4064: observational record and environmental sequence

Adopted mechanism. past ram pressure plus gravitational disturbance. Secondary/contextual process: star-formation-driven outflow.

H I evidence (direct morphology first). H I is confined to the inner few kiloparsecs and strongly deficient, while the outer stellar body is much less truncated.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Non-circular gas and central star formation indicate a gravitational perturbation. MAUVE-MUSE now directly resolves a bipolar ionized outflow with mass loading about 1.66 from the remaining central gas.

Best-supported event sequence. A past encounter/stripping episode removed the outer gas and concentrated what survived; approximately 0.4 Gyr later, residual central star formation is driving a wind that may accelerate final quenching.

Published timing and its meaning. Outer-disk stellar quenching is about 425+/-75 Myr. The outflow depletion argument is current and uncertain in electron density/geometry; much outflowing gas may remain bound.

Competing interpretations and classification tension. RPS alone does not explain all non-circular/stellar features. Feedback is a late-stage mechanism, not evidence that the original stripping was feedback-driven.

History confidence. high for past processing and current outflow; moderate for mechanism ordering.

gFMR/FMR implication — new atlas inference. This is a two-step path: environmental gas loss changes gas fraction, then feedback changes SFR and observable ionized gas. It can depart from equilibrium gFMR expectations in more than one direction.

Specific MAUVE-MUSE/VERTICO test. Measure outflow metallicity and escape fraction, central CO depletion, and stellar-age asymmetry to separate the prior gravitational and hydrodynamic events.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4064	3	3	3	3	2	1	3

Annotated environmental bibliography

- **Attwater et al. (2026), MAUVE-MUSE: A Star-formation-driven Outflow Caught in the Act of Quenching the Stripped Virgo Galaxy NGC 4064.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: MUSE ionized gas; resolved central disk and bipolar structure; outflow model. **Locator**: Secs. 3-5; Table 1. **Paper-level environmental claim**: NGC4064 hosts a central bipolar outflow with mass-loading factor about 1.66, superposed on a previously stripped and quenched disk. *Claim type*: observed plus model-derived. *Timing*: current outflow; earlier environmental quenching is not redated. *Uncertainty/alternative*: Much of the ionized material may remain bound; outflow is not itself the old stripping event.
- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: H I is confined to the inner few kiloparsecs and strongly deficient, while the outer stellar body is much less truncated. *Claim type*: observed. *Timing*: Outer-disk stellar quenching is about 425+/-75 Myr. The outflow depletion argument is current and uncertain in electron density/geometry; much outflowing gas may remain bound.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Cortés et al. (2006), The nature of the peculiar Virgo cluster galaxies NGC 4064 and NGC 4424.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: stellar, H α and CO kinematics; resolved NGC4064 and NGC4424 disks. **Locator**: Secs. 3-6. **Paper-level environmental claim**: Both peculiar disks show gravitational disturbance plus gas truncation; NGC4424 is best explained by an intermediate-ratio merger, while NGC4064 is consistent with tidal/bar inflow after stripping. *Claim type*: observed plus inferred. *Timing*: recent merger/encounter; no formal orbital time. *Uncertainty/alternative*: Ordering gravity and ICM stripping remains non-unique.
- **Crowl et al. (2008), The stellar populations of stripped spiral galaxies in the Virgo cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: optical spectroscopy plus UV; truncated outer disks of ten Virgo spirals. **Locator**: Secs. 3-5; Tables 3-5. **Paper-level environmental claim**: Resolved outer-disk stellar populations yield galaxy-specific cessation ages from less than 200 to about 500 Myr, demonstrating rapid outside-in quenching. *Claim type*: model-derived from observations. *Timing*: component-specific values in Tables 3-5. *Uncertainty/alternative*: SFH priors, dust and aperture placement contribute uncertainty.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: past; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: Outer-disk stellar quenching is about 425+/-75 Myr. The outflow depletion argument is current and uncertain in electron density/geometry; much outflowing gas may remain bound.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class III; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.3 NGC4189

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4189	VCC 89; UGC 7235; IC 3050; NGC 4189	9.73	-0.35	8.78	8.55	-0.06

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4189	1995	pre-peak	0	not included

NGC4189: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. weak hydrodynamic perturbation or group tide. Secondary/contextual process: internal warp.

H I evidence (direct morphology first). The H I disk extends slightly beyond the optical disk and is broadly symmetric but warped; the southeastern side has locally enhanced H I.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Halpha and radio emission are also enhanced to the southeast. VIVA notes its likely M-cloud/background location and nearby dwarf companions, so neither ICM compression nor a tidal origin is uniquely selected.

Best-supported event sequence. A largely gas-normal disk was mildly disturbed; no evidence establishes a completed stripping passage.

Published timing and its meaning. No published pericentric or quenching clock was found; only a present-day weak-disturbance state is defensible.

Competing interpretations and classification tension. Yoon class 0 and the absence of a Köppen entry agree that a strong stripping history is unproven, but do not distinguish tide, warp, or weak ram pressure.

History confidence. low-moderate.

gFMR/FMR implication — new atlas inference. Any metallicity-SFR offset should be interpreted as a possible local gas-supply or compression effect, not as a calibrated infall-time sequence.

Specific MAUVE-MUSE/VERTICO test. Use MUSE gas-star kinematic residuals and outer-disk abundance asymmetry; compare the southeastern sector to the opposite side and to resolved CO.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4189	3	2	2	1	1	0	1

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: CO(2-1); VERTICO ACA survey. Locator: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim:** Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. Claim type: survey measurement. Timing: none. Uncertainty/alternative: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: H I 21-cm; VLA imaging/atlas. Locator: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim:** The H I disk extends slightly beyond the optical disk and is broadly symmetric but warped; the southeastern side has locally enhanced H I. Claim type: observed. Timing: No published pericentric or quenching clock was found; only a present-day weak-disturbance state is defensible.. Uncertainty/alternative: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. Locator: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim:** Yoon class 0; provides a statistical stripping-stage context, not a unique orbit. Claim type: inferred. Timing: stage-scale only. Uncertainty/alternative: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.4 NGC4192

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4192	M 98; VCC 92; UGC 7231; NGC 4192	10.76	0.11	9.66	9.12	-0.28

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4192	-118	pre-peak	IV	active

NGC4192: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. weak or incipient environmental processing. Secondary/contextual process: bar/warp and projection.

H I evidence (direct morphology first). M98 has an extended, asymmetric southeastern H I distribution and a warp but is not strongly H I deficient in the adopted catalogue context.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. The central gas field is bar-distorted. MAUVE-MUSE finds resolved ionized-gas/metallicity structure, but these data do not by themselves prove a stripping epoch.

Best-supported event sequence. An intrinsically warped barred disk may be experiencing low-level pressure; a severe completed event is not required by its current gas reservoir.

Published timing and its meaning. No tailored clock. Köppen labels a low-deficiency active case, whereas Yoon class IV is morphology-based and MAUVE adopts pre-peak.

Competing interpretations and classification tension. This is a formal reclassification audit: Yoon IV would normally map to a slow/post category, while Köppen places it among low-deficiency active strippers. The retained pre-peak label records the lack of strong present truncation, not certainty about first infall.

History confidence. moderate for weak processing; low for orbital stage.

gFMR/FMR implication — new atlas inference. Because H I remains abundant, gas mass may still track inflow-like dilution more closely than SFR; bar-driven SFE changes and environmental gas removal are confounded.

Specific MAUVE-MUSE/VERTICO test. Measure leading/trailing metallicity and DIG asymmetries; combine MUSE with H I/CO velocity residuals before assigning a direction of motion.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4192	3	2	3	2	1	0	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: CO(2-1); VERTICO ACA survey. Locator: Sec. 2; Tables 1-2; Sec. 6. Paper-level environmental claim: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. Claim type: survey measurement. Timing: none. Uncertainty/alternative: Survey selection is environmentally targeted and not Virgo-population complete.
- **Brown et al. (2026), MAUVE-MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: MUSE emission-line ratios and velocity dispersion; resolved spaxels in 12 early-release MAUVE galaxies. Locator: Secs. 3-6; Table 2. Paper-level environmental claim: Non-HII excitation and high-dispersion gas are more prevalent than in field controls; diffuse ionized gas dominates much of the excess and shocks are secondary. Claim type: observed survey comparison. Timing: current ionized-gas state. Uncertainty/alternative: Early-release subset; excitation fractions depend on selections and spatial resolution.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: H I 21-cm; VLA imaging/atlas. Locator: Appendix pp. 14-26; Tables 1-4. Paper-level environmental claim: M98 has an extended, asymmetric southeastern H I distribution and a warp but is not strongly H I deficient in the adopted catalogue context. Claim type: observed. Timing: No tailored clock. Köppen labels a low-deficiency active case, whereas Yoon class IV is morphology-based and MAUVE adopts pre-peak.. Uncertainty/alternative: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Huang et al. (2026), MAUVE-MUSE: when metallicity follows or fights star formation - a mass-dependent inversion in Virgo galaxies.** DOI: ADS. Status/access: peer reviewed; full text or accepted manuscript. Tracer/method: MUSE metallicity and Sigma_SFR; resolved HII-region spaxels in 14 MAUVE galaxies. Locator: Secs. 3-6; Figs. 3-6. Paper-level environmental claim: The residual metallicity-SFR relation changes sign with calibration and within galaxies; NGC4383 is a robust 0.1-0.2 dex low-metallicity outlier and NGC4192 changes internal sign. Claim type: observed analysis. Timing: current resolved state. Uncertainty/alternative: Strong-line calibration, DIG/HII selection and hierarchical sampling dominate interpretation.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. Locator: Secs. 7.2-7.3; Tables E1-E5. Paper-level environmental claim: Analytical classification: active; compare with direct morphology before adoption. Claim type: model-derived. Timing: No tailored clock. Köppen labels a low-deficiency active case, whereas Yoon class IV is morphology-based and MAUVE adopts pre-peak.. Uncertainty/alternative: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. Locator: Sec. 2.1; Table 1; Secs. 4-5. Paper-level environmental claim: Yoon class IV; provides a statistical stripping-stage context, not a unique orbit. Claim type: inferred. Timing: stage-scale only. Uncertainty/alternative:

Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.5 NGC4216

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4216	VCC 167; UGC 7284; NGC 4216	10.89	-0.22	9.27	8.82	-0.70

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4216	30	post-peak	IV	past

NGC4216: observational record and environmental sequence

Adopted mechanism. long-term gas depletion plus satellite accretion. Secondary/contextual process: possible preprocessing/slow ICM process.

H I evidence (direct morphology first). The H I disk is fairly regular but low surface density and deficient, without a clear current stripping edge.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Deep optical imaging to about 29 mag arcsec⁻² reveals multiple streams, loops and disrupting dwarfs, directly proving ongoing gravitational bombardment; no extended group X-ray halo is detected.

Best-supported event sequence. Slow/older gas depletion and repeated minor accretion coexist; the streams need not have caused the H I deficiency, and an ICM or viscous-stripping history remains plausible.

Published timing and its meaning. No unique pressure clock. Stream survival is probably shorter than about 1 Gyr. A core passage at least roughly 1 Gyr ago is a travel-time scenario, not a fitted orbit.

Competing interpretations and classification tension. Yoon IV and Köppen past imply evolved gas processing, but Paudel establishes satellite interactions and leaves the stripping epoch ambiguous. Nearby NGC4222 has regular H I, so a simple bound-group preprocessing narrative is not demonstrated.

History confidence. high for minor accretion; low for gas-loss epoch.

gFMR/FMR implication — new atlas inference. Tidal accretion can add stars/gas while slow stripping removes diffuse H I, making global gas and metallicity histories non-monotonic; the streams alone do not demonstrate metal-poor inflow.

Specific MAUVE-MUSE/VERTICO test. Use MUSE stellar populations/kinematics at stream interfaces and compare outer-disk gas metallicity to test whether accreted material reached the disk.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4216	3	2	1	3	1	1	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The H I disk is fairly regular but low surface density and deficient, without a clear current stripping edge. *Claim type*: observed. *Timing*: No unique pressure clock. Stream survival is probably shorter than about 1 Gyr. A core passage at least roughly 1 Gyr ago is a travel-time scenario, not a fitted orbit.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: past; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: No unique pressure clock. Stream survival is probably shorter than about 1 Gyr. A core passage at least roughly 1 Gyr ago is a travel-time scenario, not a fitted orbit.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.

- **Paudel et al. (2013), The next generation Virgo cluster survey. IV. NGC 4216: a bombarded spiral in the Virgo cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: NGVS deep optical imaging plus H I context; NGC4216 halo and satellites. **Locator**: Secs. 2-4. **Paper-level environmental claim**: Multiple streams, loops and disrupting dwarfs prove active minor accretion around NGC4216; the ICM stripping epoch remains ambiguous. *Claim type*: observed plus inferred. *Timing*: debris survival likely <1 Gyr; RPS >=1 Gyr is only a scenario. *Uncertainty/alternative*: No extended group X-ray halo; pre- versus post-infall disruption is unresolved.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class IV; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.6 NGC4222

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4222	VCC 187; UGC 7291; IC 3087; NGC 4222	9.43	-0.89	8.83	7.93	-0.40

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4222	225	pre-peak	0	active

NGC4222: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. none established; possible weak ram pressure. Secondary/contextual process: warp.

H I evidence (direct morphology first). The extended, non-deficient H I disk is warped but lacks the bridge or one-sided truncation expected for a strong interaction with nearby NGC 4216.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. The stellar disk is comparatively regular. Survey radio/X-ray coverage supplies measurements but no dedicated causal history.

Best-supported event sequence. The galaxy is best treated as a minimally processed/weakly perturbed comparison object, with a possible early pressure signature left open.

Published timing and its meaning. No dedicated timing constraint.

Competing interpretations and classification tension. Yoon class 0 conflicts mildly with Köppen's low-deficiency active assignment; both are low-leverage classifications for this object.

History confidence. low.

gFMR/FMR implication — new atlas inference. A near-normal gas reservoir makes this a useful control for separating ordinary resolved MZR/SFR structure from removal-driven offsets.

Specific MAUVE-MUSE/VERTICO test. Search for faint one-sided ionized gas and compare outer-disk CO/H I truncation radii; do not use phase space alone as an orbit.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4222	3	2	1	1	1	1	1

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26;

Tables 1-4. **Paper-level environmental claim:** The extended, non-deficient H I disk is warped but lacks the bridge or one-sided truncation expected for a strong interaction with nearby NGC 4216. *Claim type:* observed. *Timing:* No dedicated timing constraint.. *Uncertainty/alternative:* H I morphology is direct; mechanism and viewing geometry remain interpretive.

- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator:** Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim:** Analytical classification: active; compare with direct morphology before adoption. *Claim type:* model-derived. *Timing:* No dedicated timing constraint.. *Uncertainty/alternative:* Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator:** Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim:** Yoon class 0; provides a statistical stripping-stage context, not a unique orbit. *Claim type:* inferred. *Timing:* stage-scale only. *Uncertainty/alternative:* Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.7 NGC4254

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4254	M 99; VCC 307; UGC 7345; NGC 4254	10.50	0.68	9.67	9.74	0.46

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4254	2453	pre-peak	I	non-deficient

NGC4254: observational record and environmental sequence

Adopted mechanism. high-speed tidal encounter/harassment. Secondary/contextual process: weak ongoing ram pressure.

H I evidence (direct morphology first). M99 has a strongly asymmetric disk and an approximately 250 kpc star-poor H I stream. The main disk remains H I rich rather than classically truncated.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. ALFALFA and numerical work connect the stream to a rapid fly-by; low-frequency radio asymmetry and VESTIGE Halpha knots in stripped gas indicate that ICM interaction may operate simultaneously.

Best-supported event sequence. A fast gravitational encounter generated the one-armed response and long tidal debris; subsequent/ongoing weak ram pressure modified diffuse gas and radio plasma without producing severe global deficiency.

Published timing and its meaning. Tail models disagree in detail: a combined encounter model gives about 280 Myr since closest passage, whereas the VIRGOHI21 high-speed-tail model gives about 750 Myr. The difference is model-dependent, not a measurement error bar.

Competing interpretations and classification tension. Pure RPS cannot readily explain the star-poor 250 kpc feature, while pure tide underpredicts some radio/edge signatures. The weight of evidence favours a compound history; the identity of the perturber remains uncertain.

History confidence. high for a past fast encounter; moderate for current RPS.

gFMR/FMR implication — new atlas inference. Tidal gas rearrangement and weak stripping can change gas mass, SFE, and local dilution in different directions. A single pre-peak code is therefore insufficient for FMR interpretation.

Specific MAUVE-MUSE/VERTICO test. Map abundance and SFE along the one-arm/tail-launch geometry; test whether compressed radio edges correspond to enhanced CO or merely aged cosmic rays.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4254	3	3	3	2	3	1	3

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* CO(2-1); VERTICO ACA survey. **Locator:** Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim:** Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type:* survey measurement. *Timing:* none. *Uncertainty/alternative:* Survey selection is environmentally targeted and not Virgo-population complete.

- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: M99 has a strongly asymmetric disk and an approximately 250 kpc star-poor H I stream. The main disk remains H I rich rather than classically truncated. *Claim type*: observed. *Timing*: Tail models disagree in detail: a combined encounter model gives about 280 Myr since closest passage, whereas the VIRGOHI21 high-speed-tail model gives about 750 Myr. The difference is model-dependent, not a measurement error bar.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Duc et al. (2008), Tidal debris from high-velocity collisions as fake dark galaxies: a numerical model of VIRGOHI 21.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: N-body/sticky-particle simulation; NGC4254 and VIRGOHI21 tail. **Locator**: Secs. 2-5. **Paper-level environmental claim**: A roughly 1100 km/s fly-by can reproduce the long H I tail as tidal debris without a dark galaxy; NGC4192 is a candidate interloper. *Claim type*: model-derived. *Timing*: about 750 Myr after model pericentre. *Uncertainty/alternative*: Perturber identity and timing are non-unique; another model gives about 280 Myr.
- **Haynes et al. (2007), NGC 4254: an act of harassment uncovered by the Arecibo Legacy Fast ALFA survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: ALFALFA H I; NGC4254 and 242-250 kpc tail. **Locator**: pp. L20-L22. **Paper-level environmental claim**: NGC4254 retains near-normal H I while feeding a roughly 5.0e8 Msun star-poor stream, favouring tidal harassment over catastrophic RPS. *Claim type*: observed plus inferred. *Timing*: tail is recent/intermediate; no direct clock. *Uncertainty/alternative*: Single-dish beam and lack of a visible perturber make the encounter non-unique.
- **Kantharia et al. (2007), Low radio frequency signatures of ram pressure stripping in Virgo spiral NGC 4254.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: GMRT radio continuum and polarization; NGC4254 disk and envelope. **Locator**: Secs. 3-5. **Paper-level environmental claim**: A steep-spectrum envelope, sharp southern edge and polarized asymmetry support weak young ram pressure superposed on tidal disturbance. *Claim type*: observed plus inferred. *Timing*: onset no earlier than roughly 200 Myr in model-stage reasoning. *Uncertainty/alternative*: The onset clock is inferred, not directly measured.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: non-deficient; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: Tail models disagree in detail: a combined encounter model gives about 280 Myr since closest passage, whereas the VIRGOHI21 high-speed-tail model gives about 750 Myr. The difference is model-dependent, not a measurement error bar.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Vollmer et al. (2005), NGC 4254: a spiral galaxy entering the Virgo cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: sticky-particle simulation and H I comparison; NGC4254. **Locator**: Secs. 3-6. **Paper-level environmental claim**: A rapid encounter plus ongoing nearly face-on weak RPS reproduces NGC4254's one-armed disk and gas asymmetry. *Claim type*: tailored model-derived. *Timing*: encounter about 280 Myr ago. *Uncertainty/alternative*: NGC4262 is suggested but not uniquely identified as perturber.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class I; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.8 NGC4293

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4293	VCC 460; UGC 7405; NGC 4293	10.53	-0.24	7.45	8.73	-0.48

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4293	717	post-peak	III	past

NGC4293: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. past strong gas stripping. Secondary/contextual process: bar/interaction-driven central gas.

H I evidence (direct morphology first). The galaxy is extremely H I deficient with gas confined centrally; absorption and non-circular motions are prominent.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. A misaligned/disturbed stellar-gas configuration and strong bar/central activity point to gravitational/internal redistribution in addition to environmental removal.

Best-supported event sequence. Outer atomic gas was removed in an earlier passage; surviving gas settled or was driven into the centre by non-axisymmetric dynamics.

Published timing and its meaning. No direct stellar clock in the core set; analytical work calls it a past stripper/possible backplash case.

Competing interpretations and classification tension. The gas deficit supports past removal, but the central kinematics cannot be assigned to RPS. Phase-space backplash is probabilistic.

History confidence. moderate-high.

gFMR/FMR implication — new atlas inference. Loss of metal-poor outskirts and central aperture weighting can elevate measured metallicity without requiring closed-box enrichment; bar-driven SFE is a separate variable.

Specific MAUVE-MUSE/VERTICO test. Measure age/metallicity at the truncation boundary and model bar versus external gas kinematics; avoid AGN/DIG spaxels.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4293	3	3	3	2	2	1	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Brown et al. (2026), MAUVE-MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: MUSE emission-line ratios and velocity dispersion; resolved spaxels in 12 early-release MAUVE galaxies. **Locator**: Secs. 3-6; Table 2. **Paper-level environmental claim**: Non-HII excitation and high-dispersion gas are more prevalent than in field controls; diffuse ionized gas dominates much of the excess and shocks are secondary. *Claim type*: observed survey comparison. *Timing*: current ionized-gas state. *Uncertainty/alternative*: Early-release subset; excitation fractions depend on selections and spatial resolution.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The galaxy is extremely H I deficient with gas confined centrally; absorption and non-circular motions are prominent. *Claim type*: observed. *Timing*: No direct stellar clock in the core set; analytical work calls it a past stripper/possible backplash case.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Huang et al. (2026), MAUVE-MUSE: when metallicity follows or fights star formation - a mass-dependent inversion in Virgo galaxies.** [DOI](#); [ADS](#). *Status/access*: peer reviewed; full text or accepted manuscript. *Tracer/method*: MUSE metallicity and Sigma_SFR; resolved HII-region spaxels in 14 MAUVE galaxies. **Locator**: Secs. 3-6; Figs. 3-6. **Paper-level environmental claim**: The residual metallicity-SFR relation changes sign with calibration and within galaxies; NGC4383 is a robust 0.1-0.2 dex low-metallicity outlier and NGC4192 changes internal sign. *Claim type*: observed analysis. *Timing*: current resolved state. *Uncertainty/alternative*: Strong-line calibration, DIG/HII selection and hierarchical sampling dominate interpretation.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: past; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: No direct stellar clock in the core set; analytical work calls it a past stripper/possible backplash case.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class III; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.9 NGC4294

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4294	VCC 465; UGC 7407; NGC 4294; KPG 330; KPG 330b	9.36	-0.41	9.24	7.61	0.12

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4294	421	pre-peak	I	active

NGC4294: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. early ram pressure stripping. Secondary/contextual process: tidal interaction with NGC 4299.

H I evidence (direct morphology first). A roughly 27 kpc gas-only H I tail and lopsided disk identify recent removal of diffuse gas.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Nearby NGC 4299 and the disturbed stellar/gas geometry leave a tidal contribution plausible; large-scale radio work is consistent with an environmentally perturbed magnetized ISM.

Best-supported event sequence. A mild gravitational perturbation may have loosened outer gas, followed by or concurrent with an ICM wind that formed the one-sided tail.

Published timing and its meaning. No tailored clock; the tail is a present/recent indicator rather than a quantified time to peak.

Competing interpretations and classification tension. Yoon I and Köppen active agree on ongoing/early processing but cannot apportion tide and RPS.

History confidence. moderate.

gFMR/FMR implication — new atlas inference. Gas loss can reduce H I before SFR responds, moving the galaxy off an H I-based gFMR while leaving the SFR-FMR temporarily less affected.

Specific MAUVE-MUSE/VERTICO test. Compare tail-side H α /metallicity to the leading half; model the NGC4294/4299 relative orbit before attributing all asymmetry to pressure.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4294	3	2	2	1	2	0	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2007), Virgo Galaxies with Long One-sided H I Tails.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: VLA H I; resolved one-sided tails. **Locator**: pp. L116-L118. **Paper-level environmental claim**: Long one-sided H I tails at intermediate Virgo radii show that diffuse gas removal begins outside the core; tides can first loosen gas in systems with neighbours. *Claim type*: observed plus inferred. *Timing*: recent arrival/active stripping context. *Uncertainty/alternative*: Tail direction is not a unique 3-D orbit and some tails are compound tidal+RPS features.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: A roughly 27 kpc gas-only H I tail and lopsided disk identify recent removal of diffuse gas. *Claim type*: observed. *Timing*: No tailored clock; the tail is a present/recent indicator rather than a quantified time to peak. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: active; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: No tailored clock; the tail is a present/recent indicator rather

than a quantified time to peak.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.

- **Vollmer et al. (2013), Large-scale radio continuum properties of 19 Virgo cluster galaxies. The influence of tidal interactions, ram pressure stripping, and accreting gas envelopes.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: radio continuum/polarization survey plus models; 19 Virgo galaxies. **Locator**: Secs. 3-5; Tables 2-3. **Paper-level environmental claim**: Polarized radio ridges distinguish ICM compression from tidal shear and identify compound or residual processing, including NGC4424, NGC4568 and NGC4654. *Claim type*: observed plus model-derived. *Timing*: current magnetized-ISM state. *Uncertainty/alternative*: Radio asymmetry is not uniquely environmental without gas and stellar context.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class I; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.10 NGC4298

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4298	VCC 483; UGC 7412; NGC 4298; KPG 332a	10.09	-0.28	8.72	8.95	-0.23

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4298	1122	near-peak	II	active

NGC4298: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. ram pressure plus interaction with NGC4302. Secondary/contextual process: pair tide.

H I evidence (direct morphology first). The H I disk is sharply cut on one side and diffuse on the other, with a disturbed distribution in the physical pair.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. A compressed radio edge supports an ICM wind; pair geometry and shared disturbances require a tidal contribution.

Best-supported event sequence. Mutual gravity perturbed the disks while ram pressure acted on exposed diffuse gas; neither mechanism alone explains every asymmetry.

Published timing and its meaning. No tailored event time. The pair's current proximity is observed; a pericentric clock is not.

Competing interpretations and classification tension. Yoon II and Köppen active agree on current processing but cannot separate pair and cluster forces.

History confidence. high for mixed processing; moderate for chronology.

gFMR/FMR implication — new atlas inference. Gas redistribution/compression may raise local SFE even as H I is removed, creating different trajectories in gas-FMR and SFR-FMR space.

Specific MAUVE-MUSE/VERTICO test. Jointly model both disks with MUSE stellar/gas velocities, radio leading edges and CO compression; avoid treating pair membership as mere noise.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4298	3	3	3	2	3	1	3

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.

- **Brown et al. (2026), MAUVE-MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: MUSE emission-line ratios and velocity dispersion; resolved spaxels in 12 early-release MAUVE galaxies. **Locator**: Secs. 3-6; Table 2. **Paper-level environmental claim**: Non-HII excitation and high-dispersion gas are more prevalent than in field controls; diffuse ionized gas dominates much of the excess and shocks are secondary. *Claim type*: observed survey comparison. *Timing*: current ionized-gas state. *Uncertainty/alternative*: Early-release subset; excitation fractions depend on selections and spatial resolution.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The H I disk is sharply cut on one side and diffuse on the other, with a disturbed distribution in the physical pair. *Claim type*: observed. *Timing*: No tailored event time. The pair's current proximity is observed; a pericentric clock is not.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Huang et al. (2026), MAUVE-MUSE: when metallicity follows or fights star formation - a mass-dependent inversion in Virgo galaxies.** [DOI](#); [ADS](#). *Status/access*: peer reviewed; full text or accepted manuscript. *Tracer/method*: MUSE metallicity and Sigma_SFR; resolved HII-region spaxels in 14 MAUVE galaxies. **Locator**: Secs. 3-6; Figs. 3-6. **Paper-level environmental claim**: The residual metallicity-SFR relation changes sign with calibration and within galaxies; NGC4383 is a robust 0.1-0.2 dex low-metallicity outlier and NGC4192 changes internal sign. *Claim type*: observed analysis. *Timing*: current resolved state. *Uncertainty/alternative*: Strong-line calibration, DIG/HII selection and hierarchical sampling dominate interpretation.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: active; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: No tailored event time. The pair's current proximity is observed; a pericentric clock is not.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Vollmer et al. (2013), Large-scale radio continuum properties of 19 Virgo cluster galaxies. The influence of tidal interactions, ram pressure stripping, and accreting gas envelopes.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: radio continuum/polarization survey plus models; 19 Virgo galaxies. **Locator**: Secs. 3-5; Tables 2-3. **Paper-level environmental claim**: Polarized radio ridges distinguish ICM compression from tidal shear and identify compound or residual processing, including NGC4424, NGC4568 and NGC4654. *Claim type*: observed plus model-derived. *Timing*: current magnetized-ISM state. *Uncertainty/alternative*: Radio asymmetry is not uniquely environmental without gas and stellar context.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class II; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.11 NGC4302

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4302	VCC 497; UGC 7418; NGC 4302; KPG 332b	10.45	-0.30	9.20	8.96	-0.49

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4302	1111	near-peak	II	active

NGC4302: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. active ram pressure stripping. Secondary/contextual process: interaction with NGC4298 and disk-halo flows.

H I evidence (direct morphology first). The edge-on disk has a southern truncation and northern H I extension/tail.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Diffuse ionized-gas halo and extraplanar material are well measured; radio data support environmental perturbation, while the companion can affect disk structure.

Best-supported event sequence. The ICM is displacing outer gas, superposed on an interacting edge-on pair and ordinary fountain/accretion flows.

Published timing and its meaning. No robust stripping clock; morphology indicates a current/recent event.

Competing interpretations and classification tension. Yoon II and Köppen active converge on ongoing processing. Extraplanar gas is not automatically stripped gas because disk-halo circulation is an alternative.

History confidence. moderate-high.

gFMR/FMR implication — new atlas inference. Separating fountain gas from stripped gas is essential: recycled metal-rich extraplanar gas and externally removed low-density gas have opposite metallicity implications.

Specific MAUVE-MUSE/VERTICO test. Use line ratios, rotation lag and metallicity of extraplanar gas to distinguish fountain, accretion and ICM displacement.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4302	3	3	2	2	2	1	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The edge-on disk has a southern truncation and northern H I extension/tail. *Claim type*: observed. *Timing*: No robust stripping clock; morphology indicates a current/recent event.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: active; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: No robust stripping clock; morphology indicates a current/recent event.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class II; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.12 NGC4321

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4321	M 100; VCC 596; UGC 7450; NGC 4321	10.78	0.61	9.49	9.70	0.21

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4321	1579	pre-peak	0	active

NGC4321: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. weak tidal perturbation. Secondary/contextual process: possible weak ram pressure.

H I evidence (direct morphology first). M100 is H I rich with a southwestern extension/warp and a faint bridge toward NGC 4323; there is no severe truncation.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Rich CO and star-formation data mostly diagnose internal bar/spiral physics. Environmental survey studies supply context but no dedicated stripping history.

Best-supported event sequence. A mild companion interaction and/or weak ICM contact perturbed outer atomic gas while leaving the molecular/star-forming disk broadly normal.

Published timing and its meaning. No event clock was found.

Competing interpretations and classification tension. Yoon 0 classifies no definite stripping, whereas Köppen's analytical selection lists it as low-deficiency active; the observations carry more weight than the analytical tag.

History confidence. low-moderate.

gFMR/FMR implication — new atlas inference. Internal bar/spiral SFE variations dominate many resolved relations; environmental interpretations require an outer-disk signature that is not yet isolated.

Specific MAUVE-MUSE/VERTICO test. Use MAUVE to separate bar/arm/interarm metallicity structure from the southwest H I anomaly and possible companion bridge.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4321	3	3	2	2	1	1	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: M100 is H I rich with a southwestern extension/warp and a faint bridge toward NGC 4323; there is no severe truncation. *Claim type*: observed. *Timing*: No event clock was found.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: active; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: No event clock was found.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class 0; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.13 NGC4330

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4330	VCC 630; UGC 7456; NGC 4330	9.61	-0.86	8.67	8.20	-0.49

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4330	1567	near-peak	II	active

NGC4330: observational record and environmental sequence

Adopted mechanism. active pre-peak ram pressure stripping. Secondary/contextual process: minor internal asymmetry.

H I evidence (direct morphology first). A sharply truncated H I disk and one-sided tail, with undisturbed outer stars, directly identify ongoing outside-in stripping.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. GALEX/VESTIGE show a radial quenching gradient; CO is also affected but less strongly. Ionized-gas kinematics trace stripped diffuse material.

Best-supported event sequence. Outer gas began disappearing first; the stripping front progressed inward while the galaxy approached peak pressure and continues to quench the disk outside-in.

Published timing and its meaning. Tailored modelling places the galaxy about 100 Myr before peak. Resolved stellar-population fits find outer regions quenched roughly 400-600 Myr ago, the Halpha truncation radius about 100 Myr ago, and characteristic local SFR decay times near 50 Myr. These clocks measure different radii/processes.

Competing interpretations and classification tension. The model time to pressure peak is not the same as time since quenching at a given radius. Their apparent difference is physically expected in progressive outside-in stripping.

History confidence. high.

gFMR/FMR implication — new atlas inference. NGC4330 provides a time-ordered test: gas fraction falls first, local SFR follows quickly at each radius, and metallicity can rise through halted dilution and radial-selection effects.

Specific MAUVE-MUSE/VERTICO test. Measure abundance/SFE across the dated quenching front and along the tail, with explicit radius-dependent clocks rather than one galaxy-wide stage.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H ₂	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4330	3	3	3	3	3	1	3

Annotated environmental bibliography

- **Abramson et al. (2011), Caught in the act: strong, active ram pressure stripping in Virgo cluster spiral NGC 4330.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: Halpha, UV and optical colours; resolved leading/trailing disk; stellar-population comparison. **Locator**: Secs. 3-5. **Paper-level environmental claim**: NGC4330 shows outside-in stripping, a UV-bright extraplanar tail and recent outer-disk quenching before the modelled pressure maximum. *Claim type*: observed plus inferred. *Timing*: recent, pre-peak; refined by Fossati et al. 2018. *Uncertainty/alternative*: SFH and wind geometry are model dependent.
- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: A sharply truncated H I disk and one-sided tail, with undisturbed outer stars, directly identify ongoing outside-in stripping. *Claim type*: observed. *Timing*: Tailored modelling places the galaxy about 100 Myr before peak. Resolved stellar-population fits find outer regions quenched roughly 400-600 Myr ago, the Halpha truncation radius about 100 Myr ago, and characteristic local SFR decay times near 50 Myr. These clocks measure different radii/processes.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Fossati et al. (2018), A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE). II. Constraining the quenching time in the stripped galaxy NGC 4330.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: VESTIGE Halpha, UV-optical SED and dynamical model; resolved NGC4330 disk and tail. **Locator**: Secs. 3-5. **Paper-level environmental claim**: NGC4330 began outer-disk suppression 400-600 Myr ago, underwent sharp Halpha truncation about 100 Myr ago and is modelled about 100 Myr before peak pressure. *Claim type*: observed plus model-derived. *Timing*: 400-600 Myr onset; about 100 Myr pre-peak. *Uncertainty/alternative*: Different disk zones and SFH parameterizations produce different clocks.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: active; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: Tailored modelling places the galaxy about 100 Myr before peak. Resolved stellar-population fits find outer regions quenched roughly 400-600 Myr ago, the Halpha truncation radius about 100 Myr ago, and characteristic local SFR decay times near 50 Myr. These clocks measure different radii/processes.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Vollmer et al. (2009), A holistic view on ram pressure stripping in the Virgo cluster. The first complete model-based time sequence.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I, radio and multiwavelength models; five Virgo spirals; tailored time sequence. **Locator**: Secs. 2-3; Table 3. **Paper-level environmental claim**: Tailored simulations order NGC4501 and NGC4330 before pressure maximum and NGC4522, NGC4388 and NGC4569 after it. *Claim type*: model-derived. *Timing*: approximately -250, -100, +50, +150 and +300 Myr relative to peak.

Uncertainty/alternative: Eye-fit/model-family times and local ICM normalization are not uniform statistical measurements.

- **Vollmer et al. (2012), Ram pressure stripping of the multiphase ISM and star formation in the Virgo spiral galaxy NGC 4330.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* UV, H α , H I and gas-dynamical model; NGC4330. **Locator:** Secs. 3-5. **Paper-level environmental claim:** A wind nearly in the disk plane reproduces NGC4330's leading-edge truncation and multiwavelength tail. *Claim type:* tailored model-derived. *Timing:* roughly 100 Myr before peak. *Uncertainty/alternative:* Wind angle, gas physics and SF prescription are degenerate.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator:** Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim:** Yoon class II; provides a statistical stripping-stage context, not a unique orbit. *Claim type:* inferred. *Timing:* stage-scale only. *Uncertainty/alternative:* Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.14 NGC4351

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4351	VCC 692; UGC 7476; NGC 4351; NGC 4354	9.35	-0.93	8.50	7.71	-0.39

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4351	2388	pre-peak	I	past

NGC4351: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. active/early ram pressure stripping. Secondary/contextual process: possible gravitational perturbation.

H I evidence (direct morphology first). A sharp H I cutoff and short one-sided tail, together with a compressed edge, indicate current pressure.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Radio compression supports an ICM wind; shell-like/offset stellar structure means gravity may also have disturbed the disk.

Best-supported event sequence. A galaxy on a high-velocity radial approach is losing its outer atomic gas now; a prior gravitational perturbation may explain the stellar asymmetry.

Published timing and its meaning. No tailored numerical time; the morphology constrains an active stage only.

Competing interpretations and classification tension. Yoon I indicates early stripping, while Köppen assigns a past-strip solution. The disagreement shows how analytical deficiency fits can conflict with direct present morphology.

History confidence. moderate-high for active RPS; low for prior tide.

gFMR/FMR implication — new atlas inference. A leading-edge compression episode can transiently raise local SFR/SFE while global H I falls, so local and global FMR residuals need not share a sign.

Specific MAUVE-MUSE/VERTICO test. Test shocked/DIG gas at the compressed edge and derive stellar-versus-gas kinematic asymmetry to isolate a gravitational component.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4351	3	2	2	2	2	0	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* CO(2-1); VERTICO ACA survey. **Locator:** Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim:** Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type:* survey measurement. *Timing:* none. *Uncertainty/alternative:* Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* H I 21-cm; VLA imaging/atlas. **Locator:** Appendix pp. 14-26;

Tables 1-4. **Paper-level environmental claim:** A sharp H I cutoff and short one-sided tail, together with a compressed edge, indicate current pressure. *Claim type:* observed. *Timing:* No tailored numerical time; the morphology constrains an active stage only.. *Uncertainty/alternative:* H I morphology is direct; mechanism and viewing geometry remain interpretive.

- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator:** Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim:** Analytical classification: past; compare with direct morphology before adoption. *Claim type:* model-derived. *Timing:* No tailored numerical time; the morphology constrains an active stage only.. *Uncertainty/alternative:* Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator:** Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim:** Yoon class I; provides a statistical stripping-stage context, not a unique orbit. *Claim type:* inferred. *Timing:* stage-scale only. *Uncertainty/alternative:* Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.15 NGC4380

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4380	VCC 792; UGC 7503; NGC 4380	10.09	-0.79	8.13	8.46	-0.74

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4380	935	post-peak	IV	past

NGC4380: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. old/slow gas depletion. Secondary/contextual process: starvation or weak past stripping.

H I evidence (direct morphology first). H I is mildly asymmetric, low surface brightness and deficient but lacks an active one-sided wind morphology.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. The disk is anemic; CO and radio surveys provide measurements without a dedicated event reconstruction.

Best-supported event sequence. A slow process or relaxed remnant of an older stripping event reduced diffuse gas and star formation without leaving a strong current tail.

Published timing and its meaning. No direct clock.

Competing interpretations and classification tension. Yoon IV supports slow processes; Köppen labels past stripping. Current observations do not discriminate starvation, evaporation, or relaxed RPS.

History confidence. moderate for processed status; low for mechanism.

gFMR/FMR implication — new atlas inference. A slow fall in gas supply can move the system closer to equilibrium than impulsive stripping; radial aperture loss can still mimic metallicity enhancement.

Specific MAUVE-MUSE/VERTICO test. Use stellar ages and outer-disk abundance gradients to distinguish prolonged starvation from an abrupt old truncation.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4380	3	2	1	1	1	0	1

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: H I is mildly asymmetric, low surface brightness and deficient but lacks an active one-sided wind morphology. *Claim type*: observed. *Timing*: No direct clock.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: past; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: No direct clock.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class IV; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.16 NGC4383

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4383	VCC 801; UGC 7507; NGC 4383	9.42	-0.01	9.49	8.24	0.48

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4383	1663	pre-peak	0	non-deficient

NGC4383: observational record and environmental sequence

Adopted mechanism. recent gas acquisition and starburst feedback. Secondary/contextual process: tidal perturbation.

H I evidence (direct morphology first). The H I disk is exceptionally massive and extended, with irregular outer clouds rather than a stripped/truncated morphology.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. MAUVE-MUSE resolves a roughly 6 kpc bipolar ionized outflow. The galaxy is a metallicity outlier below the ensemble rMZR; this is consistent with recent gas-supply disturbance but does not identify the donor or orbit.

Best-supported event sequence. External gas acquisition or a tidal perturbation plausibly fuelled a central starburst, which launched the observed wind; direct Virgo ICM stripping is not the dominant observed process.

Published timing and its meaning. Outflow dynamical times are Myr-scale and source-dependent; no secure cluster-pericentre time exists. Do not equate the wind age with the interaction age.

Competing interpretations and classification tension. Yoon 0 and Köppen non-deficient agree against strong stripping. The accretion trigger is an inference from H I richness, morphology and metallicity, not a directly identified encounter.

History confidence. high for feedback outflow; moderate for recent gas supply; low for orbital history.

gFMR/FMR implication — new atlas inference. This is the clearest case where gas accretion/dilution and non-disk ionized emission can drive a gFMR/FMR residual independently of cluster stage; H α may not be a clean in-situ SFR proxy in the wind.

Specific MAUVE-MUSE/VERTICO test. Use CO/H I metallicity mixing maps, separate disk HII regions from outflow spaxels, and search deep imaging/kinematics for an accretion donor.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4383	3	3	3	2	1	1	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The H I disk is exceptionally massive and extended, with irregular outer clouds rather than a stripped/truncated morphology. *Claim type*: observed. *Timing*: Outflow dynamical times are Myr-scale and source-dependent; no secure cluster-pericentre time exists. Do not equate the wind age with the interaction age.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Huang et al. (2026), MAUVE-MUSE: when metallicity follows or fights star formation - a mass-dependent inversion in Virgo galaxies.** [DOI](#); [ADS](#). *Status/access*: peer reviewed; full text or accepted manuscript. *Tracer/method*: MUSE metallicity and Sigma_SFR; resolved HII-region spaxels in 14 MAUVE galaxies. **Locator**: Secs. 3-6; Figs. 3-6. **Paper-level environmental claim**: The residual metallicity-SFR relation changes sign with calibration and within galaxies; NGC4383 is a robust 0.1-0.2 dex low-metallicity outlier and NGC4192 changes internal sign. *Claim type*: observed analysis. *Timing*: current resolved state. *Uncertainty/alternative*: Strong-line calibration, DIG/HII selection and hierarchical sampling dominate interpretation.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: non-deficient; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: Outflow dynamical times are Myr-scale and source-dependent; no secure cluster-pericentre time exists. Do not equate the wind age with the interaction age.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Watts et al. (2024), MAUVE: a 6 kpc bipolar outflow launched from NGC 4383, one of the most H I-rich galaxies in the Virgo cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: MUSE ionized-gas kinematics; NGC4383 bipolar structure. **Locator**: Secs. 3-6. **Paper-level environmental claim**: A large metal-enriched ionized outflow in NGC4383 is powered by central star formation and is not evidence for an ICM-stripped tail. *Claim type*: observed plus model-derived. *Timing*: current/recent outflow. *Uncertainty/alternative*: Outflow geometry, electron density and ionization corrections affect mass loading.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class 0; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.17 NGC4388

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4388	VCC 836; UGC 7520; NGC 4388	10.05	0.38	8.59	8.75	0.46

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4388	2538	near-peak	II	active

NGC4388: observational record and environmental sequence

Adopted mechanism. strong recent ram pressure stripping. Secondary/contextual process: AGN illumination/outflow.

H I evidence (direct morphology first). The severely truncated disk feeds a roughly 110 x 25 kpc, star-poor H I plume containing about 3.4e8 Msun; this is among the clearest direct gas-removal cases.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. The 35 kpc ionized VEELR is mostly stripped gas photoionized by the Seyfert nucleus, with shocks in outer/high-velocity regions. X-ray absorption places the H I in front of M86 and favours the main Virgo/M87 atmosphere; CO about 70 kpc down-tail shows inefficient in-situ molecule formation.

Best-supported event sequence. A high-pressure encounter removed most outer gas; stripped multiphase material persists far downstream while the active nucleus illuminates, shocks or drives physically distinct ionized components.

Published timing and its meaning. The 2009 model is about 150 Myr after peak; updated tailored work gives about 240 Myr after peak, while outer-disk SFHs give 190+/-30 Myr since quenching. These are different clocks, and the tailored peak has no

formal statistical error.

Competing interpretations and classification tension. MAUVE/Yoon near-peak grouping is morphological, while tailored work is post-peak. Later X-ray geometry weakens the earlier tentative M86-halo origin, though local structure can contribute. No single model fits every disk/tail observable; AGN emission must not be counted as stripped-gas star formation.

History confidence. very high for stripping; high-moderate for exact ICM/orbit.

gFMR/FMR implication — new atlas inference. Gas loss is extreme, but line ratios are AGN contaminated; SFR-FMR work needs strict HII selection. Tail depletion times of about 1.6×10^{10} to $>1.4 \times 10^{12}$ yr show that gas presence does not imply efficient star formation.

Specific MAUVE-MUSE/VERTICO test. Separate AGN cone, shocks, extraplanar tail and disk HII emission spectroscopically; use metallicity/velocity continuity to identify stripped parcels.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4388	3	3	3	2	3	3	3

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The severely truncated disk feeds a roughly 110 x 25 kpc, star-poor H I plume containing about 3.4×10^8 Msun; this is among the clearest direct gas-removal cases. *Claim type*: observed. *Timing*: The 2009 model is about 150 Myr after peak; updated tailored work gives about 240 Myr after peak, while outer-disk SFHs give 190+/-30 Myr since quenching. These are different clocks, and the tailored peak has no formal statistical error.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Crowl et al. (2008), The stellar populations of stripped spiral galaxies in the Virgo cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: optical spectroscopy plus UV; truncated outer disks of ten Virgo spirals. **Locator**: Secs. 3-5; Tables 3-5. **Paper-level environmental claim**: Resolved outer-disk stellar populations yield galaxy-specific cessation ages from less than 200 to about 500 Myr, demonstrating rapid outside-in quenching. *Claim type*: model-derived from observations. *Timing*: component-specific values in Tables 3-5. *Uncertainty/alternative*: SFH priors, dust and aperture placement contribute uncertainty.
- **Gu et al. (2013), Multi-wavelength studies of spectacular ram pressure stripping of a galaxy: discovery of an X-ray absorption feature.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: X-ray absorption/emission plus H I; NGC4388 plume projected on M86. **Locator**: Sects. 3-4. **Paper-level environmental claim**: X-ray absorption places the H I plume in front of M86; hot stripped gas and pressure estimates favour the main Virgo/M87 ICM over an M86-halo origin. *Claim type*: observed plus inferred. *Timing*: present tail; no independent stripping clock. *Uncertainty/alternative*: A secondary contribution from local M86 gas cannot be fully excluded.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: active; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: The 2009 model is about 150 Myr after peak; updated tailored work gives about 240 Myr after peak, while outer-disk SFHs give 190+/-30 Myr since quenching. These are different clocks, and the tailored peak has no formal statistical error.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Oosterloo et al. (2005), A large H I cloud near the centre of the Virgo cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: WSRT H I; 110 x 25 kpc NGC4388 plume. **Locator**: pp. L20-L22. **Paper-level environmental claim**: A 3.4×10^8 Msun star-free H I plume continuous with NGC4388 directly establishes removal of disk gas. *Claim type*: observed. *Timing*: recent; later models date peak. *Uncertainty/alternative*: The paper's tentative M86-halo origin is weakened by later X-ray geometry.
- **Pappalardo et al. (2010), Pinning down the ram-pressure-induced halt of star formation in the Virgo cluster spiral galaxy NGC 4388. A joint inversion of spectroscopic and photometric data.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: UV-to-NIR outer-disk SFH; NGC4388. **Locator**: Sect. 5; Fig. 10; Sect. 6.1. **Paper-level environmental claim**: Outer-disk colours and spectra give a characteristic quenching half-time of 190+/-30 Myr. *Claim type*:

model-derived from observations. *Timing*: 190+/-30 Myr quenching lookback. *Uncertainty/alternative*: This dates star-formation decline, not exactly peak pressure.

- **Verdugo et al. (2015), Ram pressure stripping in the Virgo Cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: IRAM CO and Halpha; four positions about 70 kpc down NGC4388 tail. **Locator**: Table 2; Sects. 4-5. **Paper-level environmental claim**: Molecular gas formed in situ far down the NGC4388 tail, but depletion times of 1.6e10 to >1.4e12 yr show extremely inefficient star formation. *Claim type*: observed plus inferred. *Timing*: present tail; no new peak clock. *Uncertainty/alternative*: Sparse pointings and CO conversion limit total H2 inventory.
- **Vollmer et al. (2009), A holistic view on ram pressure stripping in the Virgo cluster. The first complete model-based time sequence.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I, radio and multiwavelength models; five Virgo spirals; tailored time sequence. **Locator**: Sects. 2-3; Table 3. **Paper-level environmental claim**: Tailored simulations order NGC4501 and NGC4330 before pressure maximum and NGC4522, NGC4388 and NGC4569 after it. *Claim type*: model-derived. *Timing*: approximately -250, -100, +50, +150 and +300 Myr relative to peak. *Uncertainty/alternative*: Eye-fit/model-family times and local ICM normalization are not uniform statistical measurements.
- **Vollmer et al. (2018), Two uneven sisters. I. NGC 4388 - a strongly constrained ram-pressure stripping event.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I, CO, radio, UV and dynamical model; NGC4388. **Locator**: Table 2; Sects. 3-5. **Paper-level environmental claim**: A strongly constrained model favours a high-pressure event about 240 Myr ago, but no single realization fits all disk and far-tail observables. *Claim type*: tailored multiwavelength model-derived. *Timing*: about 240 Myr after peak. *Uncertainty/alternative*: No formal peak-time error; model tensions are explicitly retained.
- **Weżgowiec et al. (2011), Hot gas in Mach cones around Virgo cluster spiral galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: XMM-Newton X-ray imaging/spectroscopy; NGC4388, NGC4501 and NGC4569. **Locator**: Sects. 3-5. **Paper-level environmental claim**: Hot extraplanar gas and pressure structures provide independent evidence for ISM-ICM interaction, but AGN/outflow components differ among galaxies. *Claim type*: observed plus inferred. *Timing*: current hot-gas state. *Uncertainty/alternative*: Projection, low counts and nuclear contamination limit Mach-cone interpretation.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Sects. 4-5. **Paper-level environmental claim**: Yoon class II; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.
- **Yoshida et al. (2002), Discovery of a very extended emission-line region around the Seyfert 2 galaxy NGC 4388.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: deep Halpha imaging and spectroscopy; NGC4388 VEELR. **Locator**: Sects. 3-5. **Paper-level environmental claim**: A very extended emission-line region around NGC4388 contains high-velocity gas; the early paper considered both tidal/merger and stripping origins. *Claim type*: observed plus speculative interpretation. *Timing*: current tail. *Uncertainty/alternative*: Later spectroscopy rejects the preferred tidal interpretation for most VEELR gas.
- **Yoshida et al. (2004), Subaru deep spectroscopy of the very extended emission-line region of NGC 4388: ram pressure stripped gas ionized by nuclear radiation.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: deep optical spectroscopy; NGC4388 VEELR. **Locator**: Sects. 3-5. **Paper-level environmental claim**: Velocity, excitation and geometry show that most VEELR gas was ram-pressure stripped and then AGN-photoionized, with shocks in outer/high-velocity regions. *Claim type*: observed plus inferred. *Timing*: recent stripping; no independent peak clock. *Uncertainty/alternative*: AGN radiation complicates any SFR or metallicity use of the emission lines.

5.18 NGC4394

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4394	VCC 857; UGC 7523; NGC 4394; KPG 334b	10.32	-0.81	8.67	7.84	-0.91

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4394	772	post-peak	IV	not included

NGC4394: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. slow/old gas depletion. Secondary/contextual process: possible weak M85 tide.

H I evidence (direct morphology first). The H I disk is deficient and mildly truncated but strikingly symmetric.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. The galaxy is anemic. Although projected near M85, VIVA reports no decisive tidal signature linking the two.

Best-supported event sequence. Gas supply/removal acted gently or long enough ago for the disk to relax; no active stripping evidence remains.

Published timing and its meaning. No published event clock and no Köppen entry.

Competing interpretations and classification tension. Yoon IV supplies only a morphology class. Association with M85 is contextual, not causal evidence.

History confidence. low-moderate.

gFMR/FMR implication — new atlas inference. This is a useful test of slow gas exhaustion: lower gas and SFR may remain coupled, unlike impulsive stripping, but aperture/radial effects must be controlled.

Specific MAUVE-MUSE/VERTICO test. Derive spatially resolved SFHs and gas-star misalignment; search for subtle M85-facing age or metallicity asymmetry.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4394	3	2	1	1	1	0	1

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The H I disk is deficient and mildly truncated but strikingly symmetric. *Claim type*: observed. *Timing*: No published event clock and no Köppen entry.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class IV; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.19 NGC4396

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4396	VCC 865; UGC 7526; NGC 4396	9.34	-0.69	8.96	7.92	-0.14

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4396	-115	pre-peak	I	active

NGC4396: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. early ram pressure stripping. Secondary/contextual process: possible high-speed tide.

H I evidence (direct morphology first). A long northwestern H I tail and gas-free outer stellar disk favour removal of diffuse gas.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. The radio continuum does not show every expected wind diagnostic, so the tail geometry is not uniquely hydrodynamic; MAUVE ionized-gas data provide a current disk-state measurement.

Best-supported event sequence. Recent outer-disk gas loss is likely; a fast encounter may have contributed, but no perturber is securely tied to the event.

Published timing and its meaning. No tailored timing constraint.

Competing interpretations and classification tension. Yoon I and Köppen active support ongoing processing, while the incomplete radio signature lowers mechanism confidence.

History confidence. moderate.

gFMR/FMR implication — new atlas inference. Outer gas removal should first change gas fraction and resolved radial sampling; an integrated metallicity increase could reflect loss of low-metallicity outskirts rather than rapid enrichment.

Specific MAUVE-MUSE/VERTICO test. Measure tail-side gas excitation and metallicity, and compare gas/stellar residual fields for a tidal impulse.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H ₂	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4396	3	2	3	1	2	0	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Brown et al. (2026), MAUVE-MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: MUSE emission-line ratios and velocity dispersion; resolved spaxels in 12 early-release MAUVE galaxies. **Locator**: Secs. 3-6; Table 2. **Paper-level environmental claim**: Non-HII excitation and high-dispersion gas are more prevalent than in field controls; diffuse ionized gas dominates much of the excess and shocks are secondary. *Claim type*: observed survey comparison. *Timing*: current ionized-gas state. *Uncertainty/alternative*: Early-release subset; excitation fractions depend on selections and spatial resolution.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: A long northwestern H I tail and gas-free outer stellar disk favour removal of diffuse gas. *Claim type*: observed. *Timing*: No tailored timing constraint.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Edler et al. (2024), ViCTORIA project: The LOFAR view of environmental effects in Virgo cluster star-forming galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: LOFAR radio continuum; resolved Virgo-galaxy survey. **Locator**: Secs. 3-6; Tables 1-3. **Paper-level environmental claim**: Low-frequency asymmetries identify candidate ongoing RPS and retain signatures after H I morphology has partly relaxed. *Claim type*: survey observation plus inferred. *Timing*: current/recent radio state. *Uncertainty/alternative*: Radio morphology can also reflect tides, star formation and viewing angle.
- **Huang et al. (2026), MAUVE-MUSE: when metallicity follows or fights star formation - a mass-dependent inversion in Virgo galaxies.** [DOI](#); [ADS](#). *Status/access*: peer reviewed; full text or accepted manuscript. *Tracer/method*: MUSE metallicity and Sigma_SFR; resolved HII-region spaxels in 14 MAUVE galaxies. **Locator**: Secs. 3-6; Figs. 3-6. **Paper-level environmental claim**: The residual metallicity-SFR relation changes sign with calibration and within galaxies; NGC4383 is a robust 0.1-0.2 dex low-metallicity outlier and NGC4192 changes internal sign. *Claim type*: observed analysis. *Timing*: current resolved state. *Uncertainty/alternative*: Strong-line calibration, DIG/HII selection and hierarchical sampling dominate interpretation.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: active; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: No tailored timing constraint.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class I; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.20 NGC4405

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4405	VCC 874; UGC 7529; IC 788; NGC 4405	9.73	-0.90	7.68	8.13	-0.61

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4405	1751	post-peak	III	past

NGC4405: observational record and environmental sequence

Adopted mechanism. past ram pressure stripping. Secondary/contextual process: none clearly established.

H I evidence (direct morphology first). The H I disk is severely and symmetrically truncated while the stellar disk is regular, with little evidence for current pressure.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Outer-disk stellar populations are quenched; broad CO/survey data do not identify a competing gravitational event.

Best-supported event sequence. An earlier ICM passage removed outer gas; the remaining inner disk has since relaxed.

Published timing and its meaning. Outer-disk quenching is about 450+/-150 Myr.

Competing interpretations and classification tension. Yoon III and Köppen past agree. The clock dates cessation of local star formation, not necessarily exact pressure maximum.

History confidence. high.

gFMR/FMR implication — new atlas inference. A post-stripping system should show low gas fraction and radially censored star formation; metallicity offsets can arise from outside-in removal and halted dilution.

Specific MAUVE-MUSE/VERTICO test. Measure metallicity/ages immediately inside and outside the old truncation radius and search for fallback gas.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4405	3	2	1	3	1	0	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The H I disk is severely and symmetrically truncated while the stellar disk is regular, with little evidence for current pressure. *Claim type*: observed. *Timing*: Outer-disk quenching is about 450+/-150 Myr.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Crowl et al. (2008), The stellar populations of stripped spiral galaxies in the Virgo cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: optical spectroscopy plus UV; truncated outer disks of ten Virgo spirals. **Locator**: Secs. 3-5; Tables 3-5. **Paper-level environmental claim**: Resolved outer-disk stellar populations yield galaxy-specific cessation ages from less than 200 to about 500 Myr, demonstrating rapid outside-in quenching. *Claim type*: model-derived from observations. *Timing*: component-specific values in Tables 3-5. *Uncertainty/alternative*: SFH priors, dust and aperture placement contribute uncertainty.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: past; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: Outer-disk quenching is about 450+/-150 Myr.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.

- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class III; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.21 NGC4402

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4402	VCC 873; UGC 7528; NGC 4402	10.11	-0.21	8.59	8.97	-0.17

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4402	190	near-peak	II	active

NGC4402: observational record and environmental sequence

Adopted mechanism. active ram pressure stripping. Secondary/contextual process: dense-cloud decoupling.

H I evidence (direct morphology first). A compressed southeastern edge, sharp H I truncation and northwestern extraplanar tail show an ICM wind acting now.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Dust filaments reveal ablation and dense-cloud decoupling; ALMA detects compressed and extraplanar molecular gas. Radio polarization/deficit patterns independently locate the windward edge.

Best-supported event sequence. Diffuse outer gas was stripped first; denser clouds partially decoupled and are now ablated, while even molecular gas is compressed/displaced.

Published timing and its meaning. Outer-disk stellar work gives less than about 200 Myr since local quenching; no single tailored time to peak is as secure as for NGC4330/4501/4522.

Competing interpretations and classification tension. The current wind is certain, but individual extraplanar dense clouds may be transiently bound or fall back; molecular removal is not uniform.

History confidence. high.

gFMR/FMR implication — new atlas inference. This is a direct demonstration that environment can alter both gas mass and local SFE. Dense surviving clouds and stripped diffuse gas should occupy different resolved gFMR pathways.

Specific MAUVE-MUSE/VERTICO test. Pair cloud-scale CO, extinction and metallicity with leading/trailing MUSE maps; identify bound versus escaping components kinematically.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4402	3	3	3	3	3	1	3

Annotated environmental bibliography

- **Abramson et al. (2014), HUBBLE SPACE TELESCOPEIMAGING OF DECOUPLED DUST CLOUDS IN THE RAM PRESSURE STRIPPED VIRGO SPIRALS NGC 4402 AND NGC 4522.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: HST dust silhouettes and young stars; resolved upturn/decoupled-cloud regions. **Locator**: Secs. 3-5. **Paper-level environmental claim**: Dense clouds and dust fronts in NGC4402 and NGC4522 resist removal while lower-density gas is swept downstream, directly resolving selective stripping. *Claim type*: observed. *Timing*: active/recent stripping; no uniform clock. *Uncertainty/alternative*: Small fields sample selected leading-edge regions.
- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: A compressed southeastern edge, sharp H I truncation and northwestern extraplanar tail show an ICM wind acting now. *Claim type*: observed. *Timing*: Outer-disk stellar work gives less than about 200

Myr since local quenching; no single tailored time to peak is as secure as for NGC4330/4501/4522.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.

- **W. J. et al. (2020), ALMA evidence for ram pressure compression and stripping of molecular gas in the Virgo Cluster galaxy NGC 4402.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: resolved CO and multiwavelength imaging; NGC4402 leading edge. **Locator**: Secs. 3-5. **Paper-level environmental claim**: Molecular gas on the windward side of NGC4402 is compressed and reorganized while lower-density material is removed, resolving how RPS couples to dense clouds. *Claim type*: observed plus inferred. *Timing*: ongoing stripping. *Uncertainty/alternative*: Local cloud-scale conditions need not represent the global disk.
- **Crowl et al. (2005), Dense cloud ablation and ram pressure stripping of the Virgo spiral NGC 4402.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: HST dust and stellar morphology; NGC4402 leading edge and extraplanar dust. **Locator**: Secs. 3-5. **Paper-level environmental claim**: Dust filaments and decoupled dense clouds show active stripping of a multiphase ISM in NGC4402. *Claim type*: observed. *Timing*: ongoing/recent. *Uncertainty/alternative*: Morphology alone does not provide peak-pressure time.
- **Crowl et al. (2008), The stellar populations of stripped spiral galaxies in the Virgo cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: optical spectroscopy plus UV; truncated outer disks of ten Virgo spirals. **Locator**: Secs. 3-5; Tables 3-5. **Paper-level environmental claim**: Resolved outer-disk stellar populations yield galaxy-specific cessation ages from less than 200 to about 500 Myr, demonstrating rapid outside-in quenching. *Claim type*: model-derived from observations. *Timing*: component-specific values in Tables 3-5. *Uncertainty/alternative*: SFH priors, dust and aperture placement contribute uncertainty.
- **Edler et al. (2024), ViCTORIA project: The LOFAR view of environmental effects in Virgo cluster star-forming galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: LOFAR radio continuum; resolved Virgo-galaxy survey. **Locator**: Secs. 3-6; Tables 1-3. **Paper-level environmental claim**: Low-frequency asymmetries identify candidate ongoing RPS and retain signatures after H I morphology has partly relaxed. *Claim type*: survey observation plus inferred. *Timing*: current/recent radio state. *Uncertainty/alternative*: Radio morphology can also reflect tides, star formation and viewing angle.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: active; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: Outer-disk stellar work gives less than about 200 Myr since local quenching; no single tailored time to peak is as secure as for NGC4330/4501/4522.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Lee et al. (2017), The effect of ram pressure on the molecular gas of galaxies: three case studies in the Virgo cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: resolved CO; NGC4330, NGC4402 and NGC4522. **Locator**: Secs. 3-5. **Paper-level environmental claim**: Molecular-gas morphology and depletion patterns show that dense gas is altered but survives longer than diffuse H I during RPS. *Claim type*: observed survey comparison. *Timing*: active/recent stripping. *Uncertainty/alternative*: CO conversion and resolution limit cloud-scale SFE claims.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class II; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.22 NGC4419

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4419	VCC 958; UGC 7551; NGC 4419	10.23	0.05	7.79	8.97	0.01

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4419	-228	post-peak	III	past

NGC4419: observational record and environmental sequence

Adopted mechanism. past strong ram pressure stripping. Secondary/contextual process: central bar/nuclear processing.

H I evidence (direct morphology first). The H I disk is extremely small relative to the stars; extraplanar/absorption features suggest residual displacement or fallback.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. The stellar disk is comparatively undisturbed. CO is centrally concentrated and historical observations show asymmetry; MAUVE ionized-gas data sample a strongly processed disk.

Best-supported event sequence. A strong event removed outer gas; surviving/fallback material and central dynamics now dominate the inner disk.

Published timing and its meaning. Outer-disk quenching is about 500+/-150 Myr (individual indicators vary). No current pressure maximum is inferred.

Competing interpretations and classification tension. The regular stellar disk favours hydrodynamic removal, while central non-axisymmetry may be internal. Negative line-of-sight velocity makes phase-space placement projection sensitive.

History confidence. high.

gFMR/FMR implication — new atlas inference. Strong radial truncation changes which metallicities enter integrated measurements; central gas retention can preserve SFR at very low H I mass.

Specific MAUVE-MUSE/VERTICO test. Search for kinematically distinct fallback gas and compare central molecular SFE to the quenched outer-disk age gradient.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4419	3	3	3	3	2	1	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Brown et al. (2026), MAUVE-MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: MUSE emission-line ratios and velocity dispersion; resolved spaxels in 12 early-release MAUVE galaxies. **Locator**: Secs. 3-6; Table 2. **Paper-level environmental claim**: Non-HII excitation and high-dispersion gas are more prevalent than in field controls; diffuse ionized gas dominates much of the excess and shocks are secondary. *Claim type*: observed survey comparison. *Timing*: current ionized-gas state. *Uncertainty/alternative*: Early-release subset; excitation fractions depend on selections and spatial resolution.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The H I disk is extremely small relative to the stars; extraplanar/absorption features suggest residual displacement or fallback. *Claim type*: observed. *Timing*: Outer-disk quenching is about 500+/-150 Myr (individual indicators vary). No current pressure maximum is inferred.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Crowl et al. (2008), The stellar populations of stripped spiral galaxies in the Virgo cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: optical spectroscopy plus UV; truncated outer disks of ten Virgo spirals. **Locator**: Secs. 3-5; Tables 3-5. **Paper-level environmental claim**: Resolved outer-disk stellar populations yield galaxy-specific cessation ages from less than 200 to about 500 Myr, demonstrating rapid outside-in quenching. *Claim type*: model-derived from observations. *Timing*: component-specific values in Tables 3-5. *Uncertainty/alternative*: SFH priors, dust and aperture placement contribute uncertainty.
- **Huang et al. (2026), MAUVE-MUSE: when metallicity follows or fights star formation - a mass-dependent inversion in Virgo galaxies.** [DOI](#); [ADS](#). *Status/access*: peer reviewed; full text or accepted manuscript. *Tracer/method*: MUSE metallicity and Sigma_SFR; resolved HII-region spaxels in 14 MAUVE galaxies. **Locator**: Secs. 3-6; Figs. 3-6. **Paper-level environmental claim**: The residual metallicity-SFR relation changes sign with calibration and within galaxies; NGC4383 is a robust 0.1-0.2 dex low-metallicity outlier and NGC4192 changes internal sign. *Claim type*: observed analysis. *Timing*: current resolved state. *Uncertainty/alternative*: Strong-line calibration, DIG/HII selection and hierarchical sampling dominate interpretation.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: past; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: Outer-disk quenching is about 500+/-150 Myr (individual indicators vary). No current pressure maximum is inferred.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class III; provides a

statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.23 NGC4424

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4424	VCC 979; UGC 7561; NGC 4424	9.90	-0.51	8.31	8.18	-0.34

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4424	447	near-peak	II	active

NGC4424: observational record and environmental sequence

Adopted mechanism. intermediate-ratio merger remnant plus ongoing ram pressure. Secondary/contextual process: central starburst outflow.

H I evidence (direct morphology first). A roughly 110 kpc one-sided H I tail and strongly disturbed central gas coexist with a shell/banana-shaped stellar body.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Low V/sigma, possible gas counter-rotation and non-circular CO establish a recent roughly 1:4-1:10 merger. VESTIGE finds outside-in Halpha truncation and an RPS tail, while the opposite 10 kpc NW ionized feature is a shock-excited starburst outflow supported by radio polarization.

Best-supported event sequence. A gravitational encounter transformed the stellar disk and concentrated gas; ongoing near-face-on ram pressure removed low-density material, while the central burst powered a distinct outflow.

Published timing and its meaning. West/east outer-disk quenching times are 281+/-27 and 251+/-14 Myr; a simple central SFH event is 179+/-68 Myr. The merger is inferred within roughly 500 Myr, not dynamically dated.

Competing interpretations and classification tension. Yoon II and Köppen active classify the gas morphology but omit the prior merger. The southern stripping tail and NW outflow are physically different; treating all extraplanar gas as one mechanism is wrong.

History confidence. high for compound history; moderate for detailed ordering.

gFMR/FMR implication — new atlas inference. Merger-driven inflow/mixing and selective outer removal can be simultaneous; shock/outflow spaxels and non-HII excitation must be excluded before interpreting metallicity residuals.

Specific MAUVE-MUSE/VERTICO test. Reconstruct stellar populations and metallicity along shells, central inflow and both tails; use gas-star misalignment to time gravity separately from stripping.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4424	3	3	3	3	3	1	3

Annotated environmental bibliography

- **Boselli et al. (2018), A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE). IV. A tail of ionised gas in the merger remnant NGC4424.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: VESTIGE Halpha plus MUSE/UV; NGC4424 disk, southern tail and NW outflow. Locator: Secs. 3-6. Paper-level environmental claim: NGC4424 combines a long stripping tail and outside-in quenching with a physically distinct shock-excited central outflow in a merger remnant. Claim type: observed plus model-derived. Timing: outer quenching 251+/-14 and 281+/-27 Myr; merger less than roughly 500 Myr. Uncertainty/alternative: Simple central SFHs should not be over-interpreted.
- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: CO(2-1); VERTICO ACA survey. Locator: Sec. 2; Tables 1-2; Sec. 6. Paper-level environmental claim: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. Claim type: survey measurement. Timing: none. Uncertainty/alternative: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: H I 21-cm; VLA imaging/atlas. Locator: Appendix pp. 14-26; Tables 1-4. Paper-level environmental claim: A roughly 110 kpc one-sided H I tail and strongly disturbed central gas coexist with a shell/banana-shaped stellar body. Claim type: observed. Timing: West/east outer-disk quenching times are 281+/-27 and 251+/-14 Myr; a simple central SFH event is 179+/-68 Myr. The merger is inferred within roughly 500 Myr, not dynamically

dated.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.

- **Cortés et al. (2006), The nature of the peculiar Virgo cluster galaxies NGC 4064 and NGC 4424.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: stellar, H α and CO kinematics; resolved NGC4064 and NGC4424 disks. **Locator**: Secs. 3-6. **Paper-level environmental claim**: Both peculiar disks show gravitational disturbance plus gas truncation; NGC4424 is best explained by an intermediate-ratio merger, while NGC4064 is consistent with tidal/bar inflow after stripping. *Claim type*: observed plus inferred. *Timing*: recent merger/encounter; no formal orbital time. *Uncertainty/alternative*: Ordering gravity and ICM stripping remains non-unique.
- **Crowl et al. (2008), The stellar populations of stripped spiral galaxies in the Virgo cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: optical spectroscopy plus UV; truncated outer disks of ten Virgo spirals. **Locator**: Secs. 3-5; Tables 3-5. **Paper-level environmental claim**: Resolved outer-disk stellar populations yield galaxy-specific cessation ages from less than 200 to about 500 Myr, demonstrating rapid outside-in quenching. *Claim type*: model-derived from observations. *Timing*: component-specific values in Tables 3-5. *Uncertainty/alternative*: SFH priors, dust and aperture placement contribute uncertainty.
- **Kenney et al. (1996), Evidence for a merger in the peculiar Virgo cluster Sa galaxy NGC 4424.** [DOI](#); [ADS](#). *Status/access*: peer reviewed; full text or accepted manuscript. *Tracer/method*: optical morphology and kinematics; NGC4424. **Locator**: Secs. 3-5. **Paper-level environmental claim**: Strong stellar asymmetry and disturbed kinematics identify NGC4424 as a gravitationally transformed system rather than a pure stripped spiral. *Claim type*: observed plus inferred. *Timing*: recent; no tailored clock. *Uncertainty/alternative*: Later work better separates merger, RPS and outflow components.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: active; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: West/east outer-disk quenching times are 281 $\pm$ 27 and 251 $\pm$ 14 Myr; a simple central SFH event is 179 $\pm$ 68 Myr. The merger is inferred within roughly 500 Myr, not dynamically dated.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Vollmer et al. (2013), Large-scale radio continuum properties of 19 Virgo cluster galaxies. The influence of tidal interactions, ram pressure stripping, and accreting gas envelopes.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: radio continuum/polarization survey plus models; 19 Virgo galaxies. **Locator**: Secs. 3-5; Tables 2-3. **Paper-level environmental claim**: Polarized radio ridges distinguish ICM compression from tidal shear and identify compound or residual processing, including NGC4424, NGC4568 and NGC4654. *Claim type*: observed plus model-derived. *Timing*: current magnetized-ISM state. *Uncertainty/alternative*: Radio asymmetry is not uniquely environmental without gas and stellar context.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class II; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.24 NGC4450

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4450	VCC 1110; UGC 7594; NGC 4450	10.68	-0.57	8.48	8.54	-0.91

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4450	2048	post-peak	IV	past

NGC4450: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. old/slow environmental depletion. Secondary/contextual process: bar/AGN/internal secular evolution.

H I evidence (direct morphology first). H I is moderately truncated and low surface density but symmetric, with no current tail or compressed edge.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Anemic star formation coexists with a bar and low-luminosity AGN; these internal processes complicate ionized-gas interpretation.

Best-supported event sequence. Diffuse gas was slowly removed/starved or stripped long ago; the present gas disk is relaxed.

Published timing and its meaning. No direct quenching or pressure clock.

Competing interpretations and classification tension. Yoon IV and Köppen past agree on an evolved state but not on mechanism. AGN emission is not evidence of environment.

History confidence. moderate-low.

gFMR/FMR implication — new atlas inference. Slow depletion and AGN/DIG contamination make H α -based SFR especially vulnerable; gas-based relations may better record the environmental deficit.

Specific MAUVE-MUSE/VERTICO test. Use HII-only masks, stellar ages and outer CO/H I limits to distinguish starvation from old impulsive stripping.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4450	3	3	1	2	2	1	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: H I is moderately truncated and low surface density but symmetric, with no current tail or compressed edge. *Claim type*: observed. *Timing*: No direct quenching or pressure clock.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: past; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: No direct quenching or pressure clock.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class IV; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.25 NGC4457

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4457	VCC 1145; UGC 7609; NGC 4457	10.47	-0.44	8.31	8.88	-0.64

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4457	738	post-peak	III	not included

NGC4457: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. past/relaxed gas stripping. Secondary/contextual process: warp or minor accretion.

H I evidence (direct morphology first). The H I disk is small and asymmetric/warped but lacks a strong current wind signature.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Radio and MAUVE ionized-gas surveys reveal a processed disk but no dedicated orbital model; internal structure remains plausible.

Best-supported event sequence. An earlier mild-to-moderate removal episode and/or minor accretion disturbed gas; the present event status is relaxed.

Published timing and its meaning. No published clock; Köppen does not include the galaxy.

Competing interpretations and classification tension. Yoon III implies severe/past morphology, but VIVA describes a less decisive active signature. Mechanism confidence is limited.

History confidence. moderate-low.

gFMR/FMR implication — new atlas inference. Gas warps/accretion can change metallicity and kinematics independently of stripping, so a post label alone cannot set the expected FMR residual.

Specific MAUVE-MUSE/VERTICO test. Test gas-star misalignment and azimuthal metallicity asymmetry; determine whether the warp carries distinct abundance.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4457	3	3	3	2	2	0	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Brown et al. (2026), MAUVE-MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: MUSE emission-line ratios and velocity dispersion; resolved spaxels in 12 early-release MAUVE galaxies. **Locator**: Secs. 3-6; Table 2. **Paper-level environmental claim**: Non-HII excitation and high-dispersion gas are more prevalent than in field controls; diffuse ionized gas dominates much of the excess and shocks are secondary. *Claim type*: observed survey comparison. *Timing*: current ionized-gas state. *Uncertainty/alternative*: Early-release subset; excitation fractions depend on selections and spatial resolution.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The H I disk is small and asymmetric/warped but lacks a strong current wind signature. *Claim type*: observed. *Timing*: No published clock; Köppen does not include the galaxy.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Huang et al. (2026), MAUVE-MUSE: when metallicity follows or fights star formation - a mass-dependent inversion in Virgo galaxies.** [DOI](#); [ADS](#). *Status/access*: peer reviewed; full text or accepted manuscript. *Tracer/method*: MUSE metallicity and Sigma_SFR; resolved HII-region spaxels in 14 MAUVE galaxies. **Locator**: Secs. 3-6; Figs. 3-6. **Paper-level environmental claim**: The residual metallicity-SFR relation changes sign with calibration and within galaxies; NGC4383 is a robust 0.1-0.2 dex low-metallicity outlier and NGC4192 changes internal sign. *Claim type*: observed analysis. *Timing*: current resolved state. *Uncertainty/alternative*: Strong-line calibration, DIG/HII selection and hierarchical sampling dominate interpretation.
- **Vollmer et al. (2013), Large-scale radio continuum properties of 19 Virgo cluster galaxies. The influence of tidal interactions, ram pressure stripping, and accreting gas envelopes.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: radio continuum/polarization survey plus models; 19 Virgo galaxies. **Locator**: Secs. 3-5; Tables 2-3. **Paper-level environmental claim**: Polarized radio ridges distinguish ICM compression from tidal shear and identify compound or residual processing, including NGC4424, NGC4568 and NGC4654. *Claim type*: observed plus model-derived. *Timing*: current magnetized-ISM state. *Uncertainty/alternative*: Radio asymmetry is not uniquely environmental without gas and stellar context.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class III; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.26 NGC4501

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4501	M 88; VCC 1401; UGC 7675; NGC 4501	10.98	0.41	9.25	9.56	-0.13

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4501	2120	near-peak	II	active

NGC4501: observational record and environmental sequence

Adopted mechanism. pre-peak ram pressure stripping. Secondary/contextual process: internal spiral/bar response.

H I evidence (direct morphology first). A compressed southwest/southeast H I edge and extended opposite side provide a textbook early-stage wind geometry.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Polarized radio ridge and MAUVE ionized-gas asymmetry support current compression; CO studies show multiphase response rather than simple atomic-only loss.

Best-supported event sequence. The galaxy is approaching maximum pressure; the leading side is compressed while diffuse gas is displaced to the trailing side, before severe symmetric truncation.

Published timing and its meaning. Tailored models place it roughly 200-300 Myr before peak (Vollmer 2009 adopts -250 Myr). Model variants span about 190-320 Myr and depend on pressure profile/orientation.

Competing interpretations and classification tension. Projected phase space cannot by itself fix the clock; the tailored multi-tracer fit is stronger but still model-dependent. Compression need not imply globally enhanced star formation.

History confidence. high.

gFMR/FMR implication — new atlas inference. This is the clean pre-peak comparator: pressure can redistribute gas and DIG before major gas depletion, potentially altering local SFE/SFR while total gas remains substantial.

Specific MAUVE-MUSE/VERTICO test. Test leading-versus-trailing metallicity, DIG fraction and CO excitation against the model wind vector; compare with NGC4522 near peak.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4501	3	3	3	2	3	2	3

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Brown et al. (2026), MAUVE-MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: MUSE emission-line ratios and velocity dispersion; resolved spaxels in 12 early-release MAUVE galaxies. **Locator**: Secs. 3-6; Table 2. **Paper-level environmental claim**: Non-HII excitation and high-dispersion gas are more prevalent than in field controls; diffuse ionized gas dominates much of the excess and shocks are secondary. *Claim type*: observed survey comparison. *Timing*: current ionized-gas state. *Uncertainty/alternative*: Early-release subset; excitation fractions depend on selections and spatial resolution.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: A compressed southwest/southeast H I edge and extended opposite side provide a textbook early-stage wind geometry. *Claim type*: observed. *Timing*: Tailored models place it roughly 200-300 Myr before peak (Vollmer 2009 adopts -250 Myr). Model variants span about 190-320 Myr and depend on pressure profile/orientation.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: active; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: Tailored models place it roughly 200-300 Myr before peak (Vollmer 2009 adopts -250 Myr). Model variants span about 190-320 Myr and depend on pressure profile/orientation.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Nehlig et al. (2016), Effects of environmental gas compression on the multiphase ISM and star formation. The Virgo spiral galaxies NGC 4501 and NGC 4567/68.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: multiwavelength ISM comparison; NGC4501 and NGC4567/4568. **Locator**: Secs. 3-5. **Paper-level environmental claim**: Gas

phases and star formation respond differently to compression in an RPS disk versus an interacting pair, providing a controlled mechanism comparison. *Claim type*: observed plus model-derived. *Timing*: current/pre-peak contexts. *Uncertainty/alternative*: Cross-system comparison does not supply a new orbital clock.

- **Vollmer et al. (2008), Pre-peak ram pressure stripping in the Virgo cluster spiral galaxy NGC 4501.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: multiwavelength dynamical model; NGC4501. **Locator**: Secs. 3-5. **Paper-level environmental claim**: The asymmetric compressed edge and diffuse opposite-side gas in NGC4501 are reproduced shortly before peak RPS. *Claim type*: tailored model-derived. *Timing*: roughly 190-320 Myr before peak; representative about 250 Myr. *Uncertainty/alternative*: Model variants span wind angle and time.
- **Vollmer et al. (2009), A holistic view on ram pressure stripping in the Virgo cluster. The first complete model-based time sequence.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I, radio and multiwavelength models; five Virgo spirals; tailored time sequence. **Locator**: Secs. 2-3; Table 3. **Paper-level environmental claim**: Tailored simulations order NGC4501 and NGC4330 before pressure maximum and NGC4522, NGC4388 and NGC4569 after it. *Claim type*: model-derived. *Timing*: approximately -250, -100, +50, +150 and +300 Myr relative to peak. *Uncertainty/alternative*: Eye-fit/model-family times and local ICM normalization are not uniform statistical measurements.
- **Weżgowiec et al. (2011), Hot gas in Mach cones around Virgo cluster spiral galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: XMM-Newton X-ray imaging/spectroscopy; NGC4388, NGC4501 and NGC4569. **Locator**: Secs. 3-5. **Paper-level environmental claim**: Hot extraplanar gas and pressure structures provide independent evidence for ISM-ICM interaction, but AGN/outflow components differ among galaxies. *Claim type*: observed plus inferred. *Timing*: current hot-gas state. *Uncertainty/alternative*: Projection, low counts and nuclear contamination limit Mach-cone interpretation.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class II; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.27 NGC4522

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4522	VCC 1516; UGC 7711; NGC 4522	9.64	-0.80	8.56	8.17	-0.45

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4522	2332	near-peak	II	active

NGC4522: observational record and environmental sequence

Adopted mechanism. near/post-peak ram pressure stripping. Secondary/contextual process: dense-cloud decoupling and ICM-ISM mixing.

H I evidence (direct morphology first). The H I disk is truncated to roughly 0.35 optical radii with abundant extraplanar gas while the stellar disk is undisturbed.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. HST dust, CO, radio and MUSE all show direct multiphase stripping. New high-resolution radio spectra indicate shock re-acceleration and mixing; extraplanar thermal emission is not solely star formation.

Best-supported event sequence. A strong ICM encounter compressed the leading edge, removed diffuse and some molecular gas, and left dense clouds temporarily decoupled in the extraplanar flow.

Published timing and its meaning. Tailored modelling gives about 50 Myr after pressure maximum; outer-disk stellar quenching is about 100+/-50 Myr. The pressure excess requires local ICM substructure or relative ICM motion.

Competing interpretations and classification tension. The adopted near-peak label covers an object that best-fit models place just after peak. Static smooth-ICM models underpredict the required pressure, so orbit-only interpretations are inadequate.

History confidence. very high.

gFMR/FMR implication — new atlas inference. Because shocks/mixing add ionized emission and CO-bearing clouds survive, both H α SFR and gas mass require phase-aware measurement. The galaxy is a benchmark for non-equilibrium gFMR tracks.

Specific MAUVE-MUSE/VERTICO test. Separate HII, shock/mixing and DIG components; measure cloud-by-cloud metallicity, CO, and escape/fallback kinematics across the wind interface.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4522	3	3	3	3	3	2	3

Annotated environmental bibliography

- **Abramson et al. (2014), HUBBLE SPACE TELESCOPE IMAGING OF DECOUPLED DUST CLOUDS IN THE RAM PRESSURE STRIPPED VIRGO SPIRALS NGC 4402 AND NGC 4522.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: HST dust silhouettes and young stars; resolved upturn/decoupled-cloud regions. **Locator**: Secs. 3-5. **Paper-level environmental claim**: Dense clouds and dust fronts in NGC4402 and NGC4522 resist removal while lower-density gas is swept downstream, directly resolving selective stripping. *Claim type*: observed. *Timing*: active/recent stripping; no uniform clock. *Uncertainty/alternative*: Small fields sample selected leading-edge regions.
- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Brown et al. (2026), MAUVE-MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: MUSE emission-line ratios and velocity dispersion; resolved spaxels in 12 early-release MAUVE galaxies. **Locator**: Secs. 3-6; Table 2. **Paper-level environmental claim**: Non-HII excitation and high-dispersion gas are more prevalent than in field controls; diffuse ionized gas dominates much of the excess and shocks are secondary. *Claim type*: observed survey comparison. *Timing*: current ionized-gas state. *Uncertainty/alternative*: Early-release subset; excitation fractions depend on selections and spatial resolution.
- **Choi et al. (2026), The Impact of Ram Pressure on the Radio Spectral Index and Magnetic Field of NGC 4522: A High-resolution VLA Continuum Study.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: resolved radio continuum and polarization; NGC4522 disk and extraplanar regions; MHD comparison. **Locator**: Secs. 3-6; Table 3. **Paper-level environmental claim**: Magnetic-field compression/draping and extraplanar radio structure constrain the present wind geometry in NGC4522. *Claim type*: observed plus model-derived. *Timing*: recent post-peak event. *Uncertainty/alternative*: Cosmic-ray transport and field evolution broaden the timing inference.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The H I disk is truncated to roughly 0.35 optical radii with abundant extraplanar gas while the stellar disk is undisturbed. *Claim type*: observed. *Timing*: Tailored modelling gives about 50 Myr after pressure maximum; outer-disk stellar quenching is about 100+/-50 Myr. The pressure excess requires local ICM substructure or relative ICM motion.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Crowl et al. (2006), The Stellar Population of Stripped Cluster Spiral NGC 4522: A Local Analog to K+A Galaxies?.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: optical/UV stellar populations; NGC4522 outer disk. **Locator**: pp. L76-L78. **Paper-level environmental claim**: Outer-disk star formation in NGC4522 ceased about 100+/-50 Myr ago, consistent with a recent strong stripping event. *Claim type*: model-derived from observations. *Timing*: outer quenching 100+/-50 Myr. *Uncertainty/alternative*: Quenching time is not exactly time from peak pressure.
- **Kenney et al. (2004), VLA HiObservations of Gas Stripping in the Virgo Cluster Spiral NGC 4522.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: VLA H I and radio continuum; NGC4522 disk and extraplanar gas. **Locator**: Secs. 3-5. **Paper-level environmental claim**: Extraplanar H I and a sharply truncated disk show active strong RPS despite the galaxy's projected distance from M87, implying a locally enhanced/structured ICM. *Claim type*: observed plus inferred. *Timing*: near or just after peak. *Uncertainty/alternative*: Projected radius does not measure 3-D ICM density.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: active; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: Tailored modelling gives about 50 Myr after pressure maximum; outer-disk stellar quenching is about 100+/-50 Myr. The pressure excess requires local ICM substructure or relative ICM motion.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Lee et al. (2017), The effect of ram pressure on the molecular gas of galaxies: three case studies in the Virgo cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: resolved CO; NGC4330, NGC4402 and NGC4522. **Locator**: Secs. 3-5. **Paper-level environmental claim**: Molecular-gas morphology and depletion patterns show that dense gas is altered but survives longer than diffuse H I during RPS. *Claim type*: observed survey comparison. *Timing*: active/recent stripping. *Uncertainty/alternative*: CO conversion and resolution limit cloud-scale SFE claims.

- **Vollmer et al. (2006), A dynamical model for the heavily ram pressure stripped Virgo spiral galaxy NGC 4522.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: gas-dynamical and polarized-radio model; NGC4522. **Locator**: Secs. 3-5. **Paper-level environmental claim**: A recent pressure maximum and subsequent re-accretion reproduce the disk/tail morphology and polarized ridge of NGC4522. *Claim type*: tailored model-derived. *Timing*: about 50 Myr after peak. *Uncertainty/alternative*: Local ICM enhancement and wind geometry are required.
- **Vollmer et al. (2009), A holistic view on ram pressure stripping in the Virgo cluster. The first complete model-based time sequence.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I, radio and multiwavelength models; five Virgo spirals; tailored time sequence. **Locator**: Secs. 2-3; Table 3. **Paper-level environmental claim**: Tailored simulations order NGC4501 and NGC4330 before pressure maximum and NGC4522, NGC4388 and NGC4569 after it. *Claim type*: model-derived. *Timing*: approximately -250, -100, +50, +150 and +300 Myr relative to peak. *Uncertainty/alternative*: Eye-fit/model-family times and local ICM normalization are not uniform statistical measurements.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class II; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.28 NGC4535

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4535	VCC 1555; UGC 7727; NGC 4535	10.53	0.35	9.54	9.28	0.11

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4535	1973	pre-peak	I	active

NGC4535: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. weak ongoing ram pressure. Secondary/contextual process: internal spiral structure.

H I evidence (direct morphology first). The H I disk is extended and close to normal but has a cutoff on the M87-facing side and an extension on the opposite side.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Radio asymmetry offers weak support for pressure; abundant PHANGS/VERTICO measurements otherwise emphasize internal disk dynamics.

Best-supported event sequence. A low-level ICM wind is beginning to reshape outer atomic gas without wholesale depletion.

Published timing and its meaning. No tailored clock.

Competing interpretations and classification tension. Yoon I and Köppen active are consistent, but both evidence and effect size are modest.

History confidence. moderate-low.

gFMR/FMR implication — new atlas inference. This system tests the earliest regime where gas geometry changes before large SFR or metallicity offsets emerge.

Specific MAUVE-MUSE/VERTICO test. Target leading/trailing HII-region abundances and DIG, and compare CO/H I edge radii at matched sensitivity.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4535	3	3	2	2	2	1	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The H I disk is extended and close to normal but has a cutoff on the M87-facing side and an extension on the opposite side. *Claim type*: observed. *Timing*: No tailored clock.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: active; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: No tailored clock.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class I; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.29 NGC4548

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4548	M 91; VCC 1615; UGC 7753; NGC 4548	10.67	-0.26	8.83	8.86	-0.59

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4548	498	post-peak	IV	past

NGC4548: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. slow or old stripping/starvation. Secondary/contextual process: bar dynamics.

H I evidence (direct morphology first). The H I disk is mildly truncated and low surface density; it is an anemic galaxy without a convincing present wind tail.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. A strong bar dominates gas kinematics. LOFAR supplies environmental radio context but does not yield a unique stripping history.

Best-supported event sequence. Gas supply was reduced gradually or in an older event; the surviving disk evolved secularly under the bar.

Published timing and its meaning. No robust time. Earlier kinematic work explicitly questioned whether stripping is still active.

Competing interpretations and classification tension. Yoon IV and Köppen past support an evolved state, while direct evidence cannot choose starvation versus relaxed RPS.

History confidence. moderate-low.

gFMR/FMR implication — new atlas inference. Bar-driven SFE and slow gas depletion are degenerate in SFR-based FMR projections; spatial gas mass and stellar ages are needed.

Specific MAUVE-MUSE/VERTICO test. Model bar streaming before searching for weak wind residuals; compare outer-disk ages to gas truncation.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4548	3	3	1	2	2	0	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The H I disk is mildly truncated and low surface density; it is an anemic galaxy without a convincing present wind tail. *Claim type*: observed. *Timing*: No robust time. Earlier kinematic work explicitly questioned whether stripping is still active.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Edler et al. (2024), ViCTORIA project: The LOFAR view of environmental effects in Virgo cluster star-forming galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: LOFAR radio continuum; resolved Virgo-galaxy survey. **Locator**: Secs. 3-6; Tables 1-3. **Paper-level environmental claim**: Low-frequency asymmetries identify candidate ongoing RPS and retain signatures after H I morphology has partly relaxed. *Claim type*: survey observation plus inferred. *Timing*: current/recent radio state. *Uncertainty/alternative*: Radio morphology can also reflect tides, star formation and viewing angle.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: past; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: No robust time. Earlier kinematic work explicitly questioned whether stripping is still active.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class IV; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.30 NGC4567_8

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4567	VCC 1673; UGC 7777; NGC 4567; KPG 347a; VV 219a	10.23	0.01	9.00	8.70	-0.03
NGC4568	VCC 1676; UGC 7776; NGC 4568; KPG 347b; VV 219b	10.45	0.27	9.21	9.28	0.08

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4567	2213	pre-peak	0	not included
NGC4568	2260	pre-peak	0	not included

This is one MAUVE target system but two physical galaxies. Both are Yoon class 0/pre-peak in the adopted bookkeeping. Every evidence and property record remains component-specific; only the shared collision geometry is synthesized jointly.

NGC4567: observational record and environmental sequence

Adopted mechanism. direct galaxy-galaxy collision. Secondary/contextual process: no strong ICM stripping established.

H I evidence (direct morphology first). H I connects the pair; assigning integrated H I flux to either component is uncertain in the overlap.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. CO and H I show enhanced molecular fraction in the overlap. ALMA resolves a continuous roughly 50 km/s connection and a 1.8 x 0.35 kpc molecular collision-front filament, but later line ratios indicate low excitation rather than a mature dense starburst ridge.

Best-supported event sequence. NGC4567 and NGC4568 are undergoing first close contact; localized gas compression and phase conversion precede any merger remnant or global starburst.

Published timing and its meaning. The encounter is early/pre-coalescence, but no defensible absolute pericentre time or Virgo orbital clock is published; cloud free-fall times are not encounter ages.

Competing interpretations and classification tension. The system is Yoon 0 and absent from Köppen. Integrated SFR/SFE remains normal within scatter, so a collision front is not equivalent to global enhancement; cluster RPS is not securely established.

History confidence. high for pair interaction; no secure pair or cluster clock.

gFMR/FMR implication — new atlas inference. Localized H I-to-H₂ conversion and mixing can affect resolved gFMR projections, but there is no direct abundance-dilution measurement and no demonstrated galaxy-wide SFE enhancement.

Specific MAUVE-MUSE/VERTICO test. Decompose MUSE velocities and metallicities by component and overlap; keep NGC4567 and NGC4568 gas budgets separate wherever the data allow.

NGC4568: observational record and environmental sequence

Adopted mechanism. direct galaxy-galaxy collision. Secondary/contextual process: tidal compression/shear; no strong ICM stripping established.

H I evidence (direct morphology first). H I bridges the pair, but component fluxes are blended; NGC4568 has its own catalogued gas and stellar properties.

CO/H₂, ionized gas/UV, stars/dust, radio, X-ray and dynamics. The molecular collision front is gravitationally bound but low excitation (maximum CO(3-2)/CO(1-0) about 0.17+/-0.04). Radio polarization is consistent with tidal compression/shear rather than strong ongoing RPS.

Best-supported event sequence. NGC4568 participates in the same first-contact event, with localized phase conversion/compression rather than evidence for global stripping or a mature overlap starburst.

Published timing and its meaning. The encounter is early/pre-coalescence; no secure absolute encounter time or Virgo pericentre time is published.

Competing interpretations and classification tension. Yoon 0 and no Köppen entry support leaving the ICM history unclassified. Resolved SFE is at most slightly enhanced in the overlap and the pair-wide value is normal within uncertainty.

History confidence. high for pair interaction; no secure pair or cluster clock.

gFMR/FMR implication — new atlas inference. Local compression/mixing can move spaxels across resolved FMR projections without implying an ICM sequence; direct oxygen-abundance evidence is still needed.

Specific MAUVE-MUSE/VERTICO test. Fit two stellar/gas velocity systems and exclude overlap spaxels when establishing each component's baseline resolved MZR and SFE.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H ₂	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4567	3	3	2	2	2	0	3
NGC4568	3	3	2	2	3	0	3

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: H I connects the pair; assigning integrated H I flux to either component is uncertain in the overlap. *Claim type*: observed. *Timing*: The encounter is early/pre-coalescence, but no defensible absolute pericentre time or Virgo orbital clock is published; cloud free-fall times are not encounter ages.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Kaneko et al. (2017), Properties of molecular gas in galaxies in the early and mid stages of interaction. II. Molecular gas fraction.** [DOI](#). *Status/access*: peer reviewed; full text or accepted manuscript. *Tracer/method*: CO plus H I; NGC4567/4568 disks and overlap. **Locator**: Secs. 3-5. **Paper-level environmental claim**: The overlap has locally elevated molecular fraction, consistent with collision pressure converting H I to H₂ before a global starburst. *Claim type*: observed plus inferred. *Timing*: early interaction; no absolute clock. *Uncertainty/alternative*: Component H I separation is uncertain in the overlap.

- **Kaneko et al. (2018), Discovery of a molecular collision front in interacting galaxies NGC 4567/4568 with ALMA.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: ALMA CO; 1.8 x 0.35 kpc overlap filament. Locator: Secs. 2-4. **Paper-level environmental claim:** A continuous roughly 50 km/s gas connection and turbulent molecular filament directly identify a collision front in NGC4567/4568. Claim type: observed. Timing: early/first-contact; no pericentre date. Uncertainty/alternative: Cloud free-fall times are not encounter ages.
- **Kaneko et al. (2022), Properties of molecular gas in galaxies in early and mid stages of Interaction. III. Resolved Kennicutt-Schmidt law.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: resolved CO and SFR; pair disks and overlap. Locator: Pair discussion; summary. **Paper-level environmental claim:** Galaxy-integrated SFR/SFE is normal within comparison scatter and the overlap is only slightly enhanced, despite the collision front. Claim type: observed analysis. Timing: current early interaction. Uncertainty/alternative: Resolved SFE uncertainties are of order 55-60 percent.
- **Kaneko et al. (2023), Investigating physical states of molecular gas in the overlapping region of interacting galaxies NGC 4567/4568 using ALMA.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: ALMA multi-transition CO; NGC4567/4568 overlap. Locator: Secs. 3-5. **Paper-level environmental claim:** Collision-front clouds are bound but low excitation, with maximum CO(3-2)/CO(1-0) about 0.17+/-0.04, arguing against a mature dense starburst ridge. Claim type: observed. Timing: current early interaction. Uncertainty/alternative: Line-ratio conversion and beam matching affect excitation estimates.
- **Nehlig et al. (2016), Effects of environmental gas compression on the multiphase ISM and star formation. The Virgo spiral galaxies NGC 4501 and NGC 4567/68.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: multiwavelength ISM comparison; NGC4501 and NGC4567/4568. Locator: Secs. 3-5. **Paper-level environmental claim:** Gas phases and star formation respond differently to compression in an RPS disk versus an interacting pair, providing a controlled mechanism comparison. Claim type: observed plus model-derived. Timing: current/pre-peak contexts. Uncertainty/alternative: Cross-system comparison does not supply a new orbital clock.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. Locator: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim:** Yoon class 0; provides a statistical stripping-stage context, not a unique orbit. Claim type: inferred. Timing: stage-scale only. Uncertainty/alternative: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.
- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: CO(2-1); VERTICO ACA survey. Locator: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim:** Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. Claim type: survey measurement. Timing: none. Uncertainty/alternative: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: H I 21-cm; VLA imaging/atlas. Locator: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim:** H I bridges the pair, but component fluxes are blended; NGC4568 has its own catalogued gas and stellar properties. Claim type: observed. Timing: The encounter is early/pre-coalescence; no secure absolute encounter time or Virgo pericentre time is published.. Uncertainty/alternative: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Kaneko et al. (2017), Properties of molecular gas in galaxies in the early and mid stages of interaction. II. Molecular gas fraction.** DOI. Status/access: peer reviewed; full text or accepted manuscript. Tracer/method: CO plus H I; NGC4567/4568 disks and overlap. Locator: Secs. 3-5. **Paper-level environmental claim:** The overlap has locally elevated molecular fraction, consistent with collision pressure converting H I to H₂ before a global starburst. Claim type: observed plus inferred. Timing: early interaction; no absolute clock. Uncertainty/alternative: Component H I separation is uncertain in the overlap.
- **Kaneko et al. (2018), Discovery of a molecular collision front in interacting galaxies NGC 4567/4568 with ALMA.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: ALMA CO; 1.8 x 0.35 kpc overlap filament. Locator: Secs. 2-4. **Paper-level environmental claim:** A continuous roughly 50 km/s gas connection and turbulent molecular filament directly identify a collision front in NGC4567/4568. Claim type: observed. Timing: early/first-contact; no pericentre date. Uncertainty/alternative: Cloud free-fall times are not encounter ages.
- **Kaneko et al. (2022), Properties of molecular gas in galaxies in early and mid stages of Interaction. III. Resolved Kennicutt-Schmidt law.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: resolved CO and SFR; pair disks and overlap. Locator: Pair discussion; summary. **Paper-level environmental claim:** Galaxy-integrated SFR/SFE is normal within comparison scatter and the overlap is only slightly enhanced, despite the collision front. Claim type: observed analysis. Timing: current early interaction. Uncertainty/alternative: Resolved SFE uncertainties are of order 55-60 percent.
- **Kaneko et al. (2023), Investigating physical states of molecular gas in the overlapping region of interacting galaxies NGC 4567/4568 using ALMA.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: ALMA multi-transition CO; NGC4567/4568 overlap. Locator: Secs. 3-5. **Paper-level environmental claim:** Collision-front clouds are bound but low excitation, with maximum CO(3-2)/CO(1-0) about 0.17+/-0.04, arguing against a mature dense starburst ridge. Claim type: observed. Timing: current early interaction. Uncertainty/alternative: Line-ratio conversion and beam matching affect

excitation estimates.

- **Nehlig et al. (2016), Effects of environmental gas compression on the multiphase ISM and star formation. The Virgo spiral galaxies NGC 4501 and NGC 4567/68.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: multiwavelength ISM comparison; NGC4501 and NGC4567/4568. Locator: Secs. 3-5. Paper-level environmental claim: Gas phases and star formation respond differently to compression in an RPS disk versus an interacting pair, providing a controlled mechanism comparison. Claim type: observed plus model-derived. Timing: current/pre-peak contexts. Uncertainty/alternative: Cross-system comparison does not supply a new orbital clock.
- **Vollmer et al. (2013), Large-scale radio continuum properties of 19 Virgo cluster galaxies. The influence of tidal interactions, ram pressure stripping, and accreting gas envelopes.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: radio continuum/polarization survey plus models; 19 Virgo galaxies. Locator: Secs. 3-5; Tables 2-3. Paper-level environmental claim: Polarized radio ridges distinguish ICM compression from tidal shear and identify compound or residual processing, including NGC4424, NGC4568 and NGC4654. Claim type: observed plus model-derived. Timing: current magnetized-ISM state. Uncertainty/alternative: Radio asymmetry is not uniquely environmental without gas and stellar context.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** DOI: ADS; arXiv. Status/access: peer reviewed; full text. Tracer/method: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. Locator: Sec. 2.1; Table 1; Secs. 4-5. Paper-level environmental claim: Yoon class 0; provides a statistical stripping-stage context, not a unique orbit. Claim type: inferred. Timing: stage-scale only. Uncertainty/alternative: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.31 NGC4569

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4569	M 90; VCC 1690; UGC 7786; NGC 4569	10.84	0.14	8.82	9.45	-0.30

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4569	-220	post-peak	III	past

NGC4569: observational record and environmental sequence

Adopted mechanism. past strong ram pressure stripping. Secondary/contextual process: nuclear starburst/outflow.

H I evidence (direct morphology first). The H I disk is only about one third of the stellar extent and is highly deficient; extraplanar material and a newly traced 230 kpc Halpha tail record removed gas.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. A magnetized/nuclear outflow is distinct from the stripping tail. Spectroscopy along the 230 kpc tail shows a smooth velocity gradient toward the Virgo systemic velocity, directly tracing deceleration of stripped ISM.

Best-supported event sequence. A major pressure event removed the outer disk about 0.3 Gyr ago; a central wind developed from surviving gas while stripped clouds mixed with and decelerated in the ICM.

Published timing and its meaning. Tailored gas models and outer stellar populations both give about 300 Myr since peak/quenching (stellar uncertainty about 50 Myr). The far Halpha tail implies cloud survival at least about 0.5 Gyr under model assumptions.

Competing interpretations and classification tension. The nuclear outflow contains far less mass than the stripped component and must not be used to explain the global deficiency. The 230 kpc tail strengthens RPS but cloud survival/mixing physics remain uncertain.

History confidence. very high.

gFMR/FMR implication — new atlas inference. This galaxy combines genuine gas removal, long-lived ionized tail emission and a central wind; Halpha outside the disk is not in-situ SFR, while gas fraction strongly departs from equilibrium expectations.

Specific MAUVE-MUSE/VERTICO test. Use tail metallicity/line ratios and MUSE disk SFHs to connect the 300 Myr event to present central SFE; exclude wind/tail emission from disk FMR fits.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4569	3	3	3	3	3	3	3

Annotated environmental bibliography

- **Boselli et al. (2016), Spectacular tails of ionized gas in the Virgo cluster galaxy NGC 4569.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: deep H α imaging; disk and approximately 85 kpc tail. **Locator**: Secs. 3-6. **Paper-level environmental claim**: NGC4569 has an exceptionally long ionized tail whose morphology and energetics favour ram-pressure removal over a purely nuclear wind. *Claim type*: observed plus inferred. *Timing*: roughly 300 Myr after peak in the associated stripping model. *Uncertainty/alternative*: Ionization and survival of diffuse clouds require assumptions.
- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The H I disk is only about one third of the stellar extent and is highly deficient; extraplanar material and a newly traced 230 kpc H α tail record removed gas. *Claim type*: observed. *Timing*: Tailored gas models and outer stellar populations both give about 300 Myr since peak/quenching (stellar uncertainty about 50 Myr). The far H α tail implies cloud survival at least about 0.5 Gyr under model assumptions.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Chyży et al. (2006), Large-scale magnetized outflows from the Virgo Cluster spiral NGC 4569. A galactic wind in a ram pressure wind.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: radio continuum and polarization; NGC4569 disk and extraplanar lobes. **Locator**: Secs. 3-5. **Paper-level environmental claim**: Large magnetized lobes in NGC4569 trace a nuclear/starburst outflow distinct from the earlier outside-in stripping of the disk. *Claim type*: observed plus inferred. *Timing*: current/recent outflow; stripping clock separate. *Uncertainty/alternative*: Radio aging and wind speed assumptions affect timescales.
- **Crowl et al. (2008), The stellar populations of stripped spiral galaxies in the Virgo cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: optical spectroscopy plus UV; truncated outer disks of ten Virgo spirals. **Locator**: Secs. 3-5; Tables 3-5. **Paper-level environmental claim**: Resolved outer-disk stellar populations yield galaxy-specific cessation ages from less than 200 to about 500 Myr, demonstrating rapid outside-in quenching. *Claim type*: model-derived from observations. *Timing*: component-specific values in Tables 3-5. *Uncertainty/alternative*: SFH priors, dust and aperture placement contribute uncertainty.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: past; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: Tailored gas models and outer stellar populations both give about 300 Myr since peak/quenching (stellar uncertainty about 50 Myr). The far H α tail implies cloud survival at least about 0.5 Gyr under model assumptions.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Sun et al. (2026), A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE) XIX. The discovery of a spectacular 230 kpc H α tail following NGC 4569 in the Virgo cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: VESTIGE H α imaging and spectroscopy; NGC4569 disk-to-230 kpc tail. **Locator**: Secs. 3-7. **Paper-level environmental claim**: A roughly 230 kpc ionized tail with smooth kinematic deceleration extends the known RPS debris of NGC4569; the nuclear wind is distinct. *Claim type*: observed plus model-derived. *Timing*: stripping about 300 Myr ago; cloud survival roughly 0.5 Gyr. *Uncertainty/alternative*: Tail ionization and drag/deceleration depend on ambient-medium assumptions.
- **Vollmer et al. (2004), NGC 4569: Recent evidence for a past ram pressure stripping event.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: gas-dynamical simulation; NGC4569. **Locator**: Secs. 3-5. **Paper-level environmental claim**: A strong near-core RPS event reproduces the truncated gas disk of NGC4569. *Claim type*: tailored model-derived. *Timing*: about 300 Myr after peak. *Uncertainty/alternative*: Adopted ICM profile and orbit control the inferred peak pressure.
- **Vollmer et al. (2009), A holistic view on ram pressure stripping in the Virgo cluster. The first complete model-based time sequence.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I, radio and multiwavelength models; five Virgo spirals; tailored time sequence. **Locator**: Secs. 2-3; Table 3. **Paper-level environmental claim**: Tailored simulations order NGC4501 and NGC4330 before pressure maximum and NGC4522, NGC4388 and NGC4569 after it. *Claim type*: model-derived. *Timing*: approximately -250, -100, +50, +150 and +300 Myr relative to peak. *Uncertainty/alternative*: Eye-fit/model-family times and local ICM normalization are not uniform statistical measurements.
- **Weżgowiec et al. (2011), Hot gas in Mach cones around Virgo cluster spiral galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: XMM-Newton X-ray imaging/spectroscopy; NGC4388, NGC4501 and NGC4569. **Locator**: Secs. 3-5. **Paper-level environmental claim**: Hot extraplanar gas and pressure structures provide independent evidence for

ISM-ICM interaction, but AGN/outflow components differ among galaxies. *Claim type*: observed plus inferred. *Timing*: current hot-gas state. *Uncertainty/alternative*: Projection, low counts and nuclear contamination limit Mach-cone interpretation.

- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class III; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.32 NGC4579

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4579	M 58; VCC 1727; UGC 7796; NGC 4579	10.90	0.06	8.78	9.18	-0.42

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4579	1627	post-peak	IV	past

NGC4579: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. old/slow gas depletion. Secondary/contextual process: bar and AGN.

H I evidence (direct morphology first). The H I disk is moderately and symmetrically truncated, without a clear current tail.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. A bar and LINER/Seyfert nucleus dominate central molecular and ionized-gas behavior.

Best-supported event sequence. Environmental gas loss/starvation acted in the past or slowly; the present disk is relaxed and central secular/AGN processes now dominate.

Published timing and its meaning. No environmental clock.

Competing interpretations and classification tension. Yoon IV and Köppen past agree only at the broad level. AGN/torque studies are not environmental evidence.

History confidence. moderate-low.

gFMR/FMR implication — new atlas inference. AGN/DIG and bar inflow contaminate SFR/metallicity diagnostics; gas-based deficits should be interpreted separately from central emission.

Specific MAUVE-MUSE/VERTICO test. Apply conservative HII masks and compare outer stellar ages to H I/CO truncation, after subtracting bar streaming.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4579	3	3	1	2	2	1	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The H I disk is moderately and symmetrically truncated, without a clear current tail. *Claim type*: observed. *Timing*: No environmental clock.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model.

Locator: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim:** Analytical classification: past; compare with direct morphology before adoption. *Claim type:* model-derived. *Timing:* No environmental clock.. *Uncertainty/alternative:* Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.

- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator:** Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim:** Yoon class IV; provides a statistical stripping-stage context, not a unique orbit. *Claim type:* inferred. *Timing:* stage-scale only. *Uncertainty/alternative:* Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.33 NGC4580

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4580	VCC 1730; UGC 7794; NGC 4580	9.97	-0.87	7.47	8.42	-0.74

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4580	1227	post-peak	III	past

NGC4580: observational record and environmental sequence

Adopted mechanism. past ram pressure stripping/backsplash candidate. Secondary/contextual process: none clearly established.

H I evidence (direct morphology first). A severely truncated H I disk sits inside a regular stellar disk with no present tail.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Outer-disk stellar populations are quenched; the retained central disk continues limited star formation.

Best-supported event sequence. A past ICM event stripped the outer disk, after which the galaxy moved to its current projected location and the gas morphology relaxed.

Published timing and its meaning. Outer-disk quenching is about 475+/-100 Myr. Backsplash status is a phase-space/model inference, not directly observed.

Competing interpretations and classification tension. Yoon III and Köppen past converge. The current large projected radius does not uniquely prove a core passage.

History confidence. high for past stripping; moderate for backsplash.

gFMR/FMR implication — new atlas inference. The galaxy is a strong post-event test of how metallicity and SFE evolve for roughly half a Gyr after abrupt outside-in gas loss.

Specific MAUVE-MUSE/VERTICO test. Derive a resolved age-metallicity profile across the fossil truncation and search for reaccreted/fallback gas.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4580	3	2	1	3	1	0	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* CO(2-1); VERTICO ACA survey. **Locator:** Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim:** Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type:* survey measurement. *Timing:* none. *Uncertainty/alternative:* Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* H I 21-cm; VLA imaging/atlas. **Locator:** Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim:** A severely truncated H I disk sits inside a regular stellar disk with no present tail. *Claim type:* observed. *Timing:* Outer-disk quenching is about 475+/-100 Myr. Backsplash status is a phase-space/model inference, not directly observed.. *Uncertainty/alternative:* H I morphology is direct; mechanism and viewing geometry remain interpretive.

- **Crowl et al. (2008), The stellar populations of stripped spiral galaxies in the Virgo cluster.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: optical spectroscopy plus UV; truncated outer disks of ten Virgo spirals. **Locator**: Secs. 3-5; Tables 3-5. **Paper-level environmental claim**: Resolved outer-disk stellar populations yield galaxy-specific cessation ages from less than 200 to about 500 Myr, demonstrating rapid outside-in quenching. *Claim type*: model-derived from observations. *Timing*: component-specific values in Tables 3-5. *Uncertainty/alternative*: SFH priors, dust and aperture placement contribute uncertainty.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: past; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: Outer-disk quenching is about 475 $\pm$ 100 Myr. Backsplash status is a phase-space/model inference, not directly observed.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class III; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.34 NGC4606

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4606	VCC 1859; UGC 7839; NGC 4606	9.85	-1.09	7.42	8.01	-0.88

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4606	1653	post-peak	III	past

NGC4606: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. past severe gas stripping. Secondary/contextual process: gravitational disturbance.

H I evidence (direct morphology first). H I occupies only a small fraction of the stellar disk and is strongly deficient.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. The stellar morphology is disturbed, so tides/harassment likely contributed; no dedicated tailored model fixes the balance.

Best-supported event sequence. A gravitational perturbation altered the stellar disk and environmental pressure removed most diffuse gas, in uncertain order.

Published timing and its meaning. No direct clock; Köppen treats it as a past stripper.

Competing interpretations and classification tension. Yoon III and Köppen past support an evolved gas state, but a pure-RPS narrative is inconsistent with the disturbed stars.

History confidence. moderate.

gFMR/FMR implication — new atlas inference. Tidal inflow and gas stripping can offset one another in SFR and metallicity, making an integrated post-peak residual non-unique.

Specific MAUVE-MUSE/VERTICO test. Use stellar/gas kinematic misalignment and SFHs to order the tidal and stripping events; constrain remaining central CO SFE.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4606	3	2	1	2	1	0	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: H I occupies only a small fraction of the stellar disk and is strongly deficient. *Claim type*: observed. *Timing*: No direct clock; Köppen treats it as a past stripper.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Edler et al. (2024), ViCTORIA project: The LOFAR view of environmental effects in Virgo cluster star-forming galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: LOFAR radio continuum; resolved Virgo-galaxy survey. **Locator**: Secs. 3-6; Tables 1-3. **Paper-level environmental claim**: Low-frequency asymmetries identify candidate ongoing RPS and retain signatures after H I morphology has partly relaxed. *Claim type*: survey observation plus inferred. *Timing*: current/recent radio state. *Uncertainty/alternative*: Radio morphology can also reflect tides, star formation and viewing angle.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: past; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: No direct clock; Köppen treats it as a past stripper.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class III; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.35 NGC4607

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4607	VCC 1868; UGC 7843; NGC 4607	9.62	-0.60	8.37	8.44	-0.24

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4607	2284	post-peak	III	past

NGC4607: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. past or ongoing ram pressure stripping. Secondary/contextual process: projected association with NGC4606.

H I evidence (direct morphology first). The edge-on H I disk is truncated and deficient.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. The approximately 600 km/s velocity difference and lack of a clear tidal bridge weaken the idea that NGC4606 is a bound perturber. LOFAR identifies NGC4607 as a new radio-RPS candidate.

Best-supported event sequence. ICM interaction likely truncated the disk; apparent pairing with NGC4606 may be mostly projection rather than the cause.

Published timing and its meaning. No tailored time.

Competing interpretations and classification tension. Yoon III/Köppen past imply a relaxed event, whereas current radio asymmetry may indicate residual/renewed pressure. This is a real status tension, not an error.

History confidence. moderate.

gFMR/FMR implication — new atlas inference. Radio evidence can persist or reveal current pressure after H I truncation; gas mass and SFR clocks may therefore be asynchronous.

Specific MAUVE-MUSE/VERTICO test. Use MUSE DIG/shock asymmetry and radio spectral aging to decide whether pressure is still active; test the pair hypothesis with stellar velocities.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4607	3	2	2	1	3	1	2

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The edge-on H I disk is truncated and deficient. *Claim type*: observed. *Timing*: No tailored time.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Edler et al. (2024), ViCTORIA project: The LOFAR view of environmental effects in Virgo cluster star-forming galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: LOFAR radio continuum; resolved Virgo-galaxy survey. **Locator**: Secs. 3-6; Tables 1-3. **Paper-level environmental claim**: Low-frequency asymmetries identify candidate ongoing RPS and retain signatures after H I morphology has partly relaxed. *Claim type*: survey observation plus inferred. *Timing*: current/recent radio state. *Uncertainty/alternative*: Radio morphology can also reflect tides, star formation and viewing angle.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: past; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: No tailored time.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class III; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.36 NGC4654

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4654	VCC 1987; UGC 7902; IC 3708; NGC 4654	10.24	0.29	9.50	9.19	0.24

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4654	1035	near-peak	II	non-deficient

NGC4654: observational record and environmental sequence

Adopted mechanism. tidal interaction plus weak ram pressure. Secondary/contextual process: stellar feedback.

H I evidence (direct morphology first). A long southeastern H I tail and northwestern compressed edge cannot be reproduced simultaneously by a single simple perturbation.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Models and polarized radio data favour a close tidal encounter with NGC4639 about 500 Myr ago plus weak ongoing pressure; resolved CO/SF work adds feedback and gas-compression constraints.

Best-supported event sequence. Tides pulled gas outward and generated asymmetric stellar/gas structure; weak ram pressure then acted efficiently on the loosened gas and continues to shape the outer arm.

Published timing and its meaning. The preferred model places the NGC4639 encounter about 500 Myr ago; ongoing pressure is weak. This is a tidal clock, not time from Virgo pressure peak.

Competing interpretations and classification tension. Köppen lists the galaxy as non-deficient despite Yoon II morphology; this is expected because strong asymmetry can occur without large integrated deficiency in a mixed tide+RPS event.

History confidence. high.

gFMR/FMR implication — new atlas inference. Tides first change gas geometry, then weak pressure removes exposed gas; local SFE/metallicity effects can precede a large global H I deficit.

Specific MAUVE-MUSE/VERTICO test. Map abundance and SFE separately in the tidal inner arm, pressure-dominated outer arm and tail; compare to the published wind/encounter geometry.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H ₂	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4654	3	3	2	2	3	1	3

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: A long southeastern H I tail and northwestern compressed edge cannot be reproduced simultaneously by a single simple perturbation. *Claim type*: observed. *Timing*: The preferred model places the NGC4639 encounter about 500 Myr ago; ongoing pressure is weak. This is a tidal clock, not time from Virgo pressure peak.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Chung et al. (2014), H I and H₂ properties and star formation efficiency of NGC 4654 in the cluster environment.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I and CO; resolved NGC4654 disk and tail. **Locator**: Secs. 3-5. **Paper-level environmental claim**: Atomic and molecular asymmetries in NGC4654 support a combined tidal encounter and ongoing ram pressure rather than either mechanism alone. *Claim type*: observed plus inferred. *Timing*: encounter of order 500 Myr; weak pressure ongoing. *Uncertainty/alternative*: Gas conversion factor and interaction geometry are model dependent.
- **Edler et al. (2024), ViCTORIA project: The LOFAR view of environmental effects in Virgo cluster star-forming galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: LOFAR radio continuum; resolved Virgo-galaxy survey. **Locator**: Secs. 3-6; Tables 1-3. **Paper-level environmental claim**: Low-frequency asymmetries identify candidate ongoing RPS and retain signatures after H I morphology has partly relaxed. *Claim type*: survey observation plus inferred. *Timing*: current/recent radio state. *Uncertainty/alternative*: Radio morphology can also reflect tides, star formation and viewing angle.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: non-deficient; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: The preferred model places the NGC4639 encounter about 500 Myr ago; ongoing pressure is weak. This is a tidal clock, not time from Virgo pressure peak.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Lizée et al. (2021), Gas compression and stellar feedback in the tidally interacting and ram-pressure stripped Virgo spiral galaxy NGC 4654.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: multiwavelength gas modelling; NGC4654. **Locator**: Secs. 3-6. **Paper-level environmental claim**: A combined past tidal encounter and present weak ram pressure reproduces the asymmetric multiphase gas better than either process alone. *Claim type*: tailored model-derived. *Timing*: encounter roughly 500 Myr; pressure ongoing. *Uncertainty/alternative*: Model geometry and ICM normalization remain degenerate.
- **Soida et al. (2006), NGC 4654: polarized radio continuum emission as a diagnostic tool for a galaxy-cluster interaction. Models versus observations.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: radio polarization plus MHD model; NGC4654. **Locator**: Secs. 3-5. **Paper-level environmental claim**: Magnetic-field asymmetry supports combined tidal shear/compression and weak ram pressure in NGC4654. *Claim type*: observed plus model-derived. *Timing*: post-encounter/current pressure. *Uncertainty/alternative*: Field evolution depends on cosmic-ray transport and adopted velocity field.
- **Vollmer et al. (2003), NGC 4654: Gravitational interaction or ram pressure stripping?.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: sticky-particle dynamical simulation; NGC4654. **Locator**: Secs. 3-6. **Paper-level**

environmental claim: A tidal encounter with NGC4639 plus ongoing weak RPS reproduces NGC4654's one-sided H I tail and compressed edge. *Claim type:* tailored model-derived. *Timing:* encounter about 500 Myr ago; RPS ongoing. *Uncertainty/alternative:* Perturber orbit and pressure profile are not unique.

- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator:** Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim:** Yoon class II; provides a statistical stripping-stage context, not a unique orbit. *Claim type:* inferred. *Timing:* stage-scale only. *Uncertainty/alternative:* Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.37 NGC4689

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4689	VCC 2058; UGC 7965; NGC 4689	10.14	-0.31	8.70	8.89	-0.29

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4689	1522	post-peak	IV	past

NGC4689: observational record and environmental sequence

Dedicated-literature status: No dedicated modern environmental-processing paper was located in the saturated curated search. The history below is constrained by VIVA/phase-space/analytical and multi-survey evidence; confidence is correspondingly limited.

Adopted mechanism. old/slow gas depletion. Secondary/contextual process: internal disk evolution.

H I evidence (direct morphology first). The H I disk is mildly and symmetrically truncated with no clear current wind feature.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Rich CO/cloud measurements characterize the inner disk but do not supply a dedicated environmental event clock.

Best-supported event sequence. A slow process or old weak stripping reduced the outer gas; the present disk is relaxed.

Published timing and its meaning. No direct clock.

Competing interpretations and classification tension. Yoon IV and Köppen past agree broadly but cannot distinguish starvation from old RPS.

History confidence. moderate-low.

gFMR/FMR implication — new atlas inference. This is a slow-processing comparison where gas and SFR may evolve together more nearly than in active strippers; resolved cloud physics remains an internal confounder.

Specific MAUVE-MUSE/VERTICO test. Compare cloud-scale SFE and stellar ages across the H I truncation, using PHANGS/VERTICO resolution matching.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4689	3	3	2	2	1	0	1

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* CO(2-1); VERTICO ACA survey. **Locator:** Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim:** Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type:* survey measurement. *Timing:* none. *Uncertainty/alternative:* Survey selection is environmentally targeted and not Virgo-population complete.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* H I 21-cm; VLA imaging/atlas. **Locator:** Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim:** The H I disk is mildly and symmetrically truncated with no clear current wind feature. *Claim type:* observed. *Timing:* No direct clock.. *Uncertainty/alternative:* H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* H I deficiency and projected cluster context; analytical peak/time-integrated pressure model.

Locator: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim:** Analytical classification: past; compare with direct morphology before adoption. *Claim type:* model-derived. *Timing:* No direct clock.. *Uncertainty/alternative:* Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.

- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator:** Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim:** Yoon class IV; provides a statistical stripping-stage context, not a unique orbit. *Claim type:* inferred. *Timing:* stage-scale only. *Uncertainty/alternative:* Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.38 NGC4694

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4694	VCC 2066; UGC 7969; NGC 4694	9.92	-0.87	8.43	8.09	-0.71

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4694	1211	near-peak	II	past

NGC4694: observational record and environmental sequence

Adopted mechanism. tidal accretion/merger debris. Secondary/contextual process: possible secondary ICM reshaping.

H I evidence (direct morphology first). A coherent 38-50 kpc, roughly 1.1e9 Msun H I bridge/debris complex connects NGC4694 with the tidal-dwarf candidate VCC2062; its structure, high column density and stellar bridge favour gravity over a clean wind tail.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. CO, young stars and 12+log(O/H) about 8.6-8.7 in VCC2062 prove recycled, pre-enriched tidal material. Molecular depletion times are roughly 1-2e10 yr and can reach 6e10 yr locally.

Best-supported event sequence. A slow unequal-mass gas-rich merger produced the bridge and tidal dwarf; ram pressure may reshape the debris but is not the primary explanation.

Published timing and its meaning. The completed single-nucleus merger is inferred to be several hundred Myr old. CIGALE burst ages are 376+/-99 Myr globally and 313+/-86 Myr in the CO region; neither is a pressure-peak clock.

Competing interpretations and classification tension. Yoon II is contaminated by counting a tidal bridge as a one-sided stripping feature, and Yoon notes the within-disk morphology resembles class III. Köppen's analytical past-strip label is weaker than the dedicated tidal evidence.

History confidence. very high for tidal history; low for ICM history.

gFMR/FMR implication — new atlas inference. VCC2062 is pre-enriched recycled gas, not pristine inflow or ordinary dwarf chemistry. Merger mixing/inflow may affect NGC4694, but its resolved abundance history remains unmeasured.

Specific MAUVE-MUSE/VERTICO test. Use MUSE metallicity/kinematic continuity from disk to bridge and combine with CO to distinguish recycled accretion from any stripped component.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4694	3	3	3	3	1	0	3

Annotated environmental bibliography

- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* CO(2-1); VERTICO ACA survey. **Locator:** Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim:** Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type:* survey measurement. *Timing:* none. *Uncertainty/alternative:* Survey selection is environmentally targeted and not Virgo-population complete.
- **Brown et al. (2026), MAUVE-MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks.** [DOI](#); [ADS](#); [arXiv](#). *Status/access:* peer reviewed; full text. *Tracer/method:* MUSE emission-line ratios and velocity dispersion; resolved spaxels in 12 early-release MAUVE galaxies. **Locator:** Secs. 3-6; Table 2. **Paper-level environmental claim:** Non-HII excitation and high-dispersion gas are more prevalent than in field controls; diffuse ionized gas dominates much of the excess and shocks are secondary. *Claim type:* observed survey comparison. *Timing:* current ionized-gas state. *Uncertainty/alternative:* Early-release subset; excitation fractions depend on selections and spatial resolution.

- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: A coherent 38-50 kpc, roughly 1.1e9 Msun H I bridge/debris complex connects NGC4694 with the tidal-dwarf candidate VCC2062; its structure, high column density and stellar bridge favour gravity over a clean wind tail. *Claim type*: observed. *Timing*: The completed single-nucleus merger is inferred to be several hundred Myr old. CIGALE burst ages are 376+/-99 Myr globally and 313+/-86 Myr in the CO region; neither is a pressure-peak clock.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Duc et al. (2007), VCC 2062: an old tidal dwarf galaxy in the Virgo cluster?** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: VLA H I, CO and optical spectroscopy; NGC4694-VCC2062 bridge. **Locator**: Secs. 3-6. **Paper-level environmental claim**: A coherent gas+stellar bridge and pre-enriched, CO-bearing VCC2062 establish recycled tidal debris from an NGC4694 merger. *Claim type*: observed plus inferred. *Timing*: several hundred Myr scenario. *Uncertainty/alternative*: A weak ICM contribution may reshape debris but is not required to form it.
- **Huang et al. (2026), MAUVE-MUSE: when metallicity follows or fights star formation - a mass-dependent inversion in Virgo galaxies.** [DOI](#); [ADS](#). *Status/access*: peer reviewed; full text or accepted manuscript. *Tracer/method*: MUSE metallicity and Sigma_SFR; resolved HII-region spaxels in 14 MAUVE galaxies. **Locator**: Secs. 3-6; Figs. 3-6. **Paper-level environmental claim**: The residual metallicity-SFR relation changes sign with calibration and within galaxies; NGC4383 is a robust 0.1-0.2 dex low-metallicity outlier and NGC4192 changes internal sign. *Claim type*: observed analysis. *Timing*: current resolved state. *Uncertainty/alternative*: Strong-line calibration, DIG/HII selection and hierarchical sampling dominate interpretation.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: past; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: The completed single-nucleus merger is inferred to be several hundred Myr old. CIGALE burst ages are 376+/-99 Myr globally and 313+/-86 Myr in the CO region; neither is a pressure-peak clock.. *Uncertainty/alternative*: Peak pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.
- **Lisenfeld et al. (2016), Molecular gas and star formation in the tidal dwarf galaxy VCC 2062.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO, UV-optical SED and deep imaging; VCC2062 and NGC4694 bridge. **Locator**: Table 4; Sect. 6.1. **Paper-level environmental claim**: The stellar bridge confirms tidal origin; burst ages are 376+/-99 Myr globally and 313+/-86 Myr over the CO region, with very long molecular depletion times. *Claim type*: observed plus model-derived. *Timing*: several-hundred-Myr tidal event. *Uncertainty/alternative*: SED burst age is not a unique merger pericentre.
- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class II; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

5.39 NGC4698

Identity, catalogue properties and adopted status

Component	Preferred aliases	log Mstar	log SFR	log MHI	log MH2	DMS
NGC4698	VCC 2070; UGC 7970; NGC 4698	10.47	-0.85	9.24	7.81	-1.05

Component	v (km/s)	MAUVE stage	Yoon	Köppen
NGC4698	1032	post-peak	I	past

NGC4698: observational record and environmental sequence

Adopted mechanism. ancient external accretion plus uncertain weak cluster stripping. Secondary/contextual process: possible M49-substructure context.

H I evidence (direct morphology first). The H I disk extends to roughly twice the stellar disk and is mostly regular; two short SE tails and NW compression are only tentative weak/incipient RPS indicators, not a classic post-stripping remnant.

CO/H2, ionized gas/UV, stars/dust, radio, X-ray and dynamics. Orthogonal bulge/disk and a polar stellar+gas nuclear disk directly prove ancient external acquisition. Its stellar population requires star formation to have ended more than 5 Gyr ago, unrelated to any demonstrated present Virgo event.

Best-supported event sequence. An ancient accretion event built the decoupled central structure; a separate incipient, weak or past ICM event remains possible but is not reconstructed by a dedicated modern processing study.

Published timing and its meaning. No tailored RPS clock. The polar event is older than 5 Gyr; Köppen's past-strip label is analytical and carries at least factor-of-two pressure uncertainty.

Competing interpretations and classification tension. This is the second reclassification audit: Yoon I/VIVA indicate weak or early processing with projection ambiguity, while Köppen labels it past. MAUVE retains post-peak as a sample code, but confidence in that ICM chronology is low; Corsini supports ancient accretion, not post-peak RPS or proven preprocessing.

History confidence. high for ancient accretion; low for adopted post-peak ICM status.

gFMR/FMR implication — new atlas inference. The ancient polar component may cause central chemical/kinematic decoupling but is not evidence for current dilution. Very low SFR and uncertain gas processing make either FMR projection non-unique.

Specific MAUVE-MUSE/VERTICO test. Date separate stellar components, map metallicity/kinematics of the extended H I-linked disk, and test M49-substructure membership with distances rather than projected phase space alone.

Evidence coverage by tracer

Grades: 0=no material source; 1=contextual; 2=survey/resolved constraint; 3=direct/tailored strong constraint.

Component	H I	CO/H2	Ionized/UV	Stars/dust	Radio	X-ray	Dynamics
NGC4698	3	2	3	3	1	1	3

Annotated environmental bibliography

- **Bertola et al. (1999), The bulge-disk orthogonal decoupling in galaxies: NGC 4698.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: stellar and ionized-gas kinematics; nuclear and main-disk axes. **Locator**: pp. L128-L130. **Paper-level environmental claim**: NGC4698 contains orthogonally rotating/structured central components, demonstrating external acquisition rather than a simple axisymmetric disk history. *Claim type*: observed plus inferred. *Timing*: ancient; no Virgo clock. *Uncertainty/alternative*: Structural decoupling does not by itself identify cluster processing.
- **Brown et al. (2021), VERTICO: the Virgo Environment Traced in CO survey.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: CO(2-1); VERTICO ACA survey. **Locator**: Sec. 2; Tables 1-2; Sec. 6. **Paper-level environmental claim**: Provides resolved molecular-gas measurement and radial profiles; the survey paper alone does not identify this component's mechanism or orbit. *Claim type*: survey measurement. *Timing*: none. *Uncertainty/alternative*: Survey selection is environmentally targeted and not Virgo-population complete.
- **Brown et al. (2026), MAUVE-MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: MUSE emission-line ratios and velocity dispersion; resolved spaxels in 12 early-release MAUVE galaxies. **Locator**: Secs. 3-6; Table 2. **Paper-level environmental claim**: Non-HII excitation and high-dispersion gas are more prevalent than in field controls; diffuse ionized gas dominates much of the excess and shocks are secondary. *Claim type*: observed survey comparison. *Timing*: current ionized-gas state. *Uncertainty/alternative*: Early-release subset; excitation fractions depend on selections and spatial resolution.
- **Chung et al. (2009), VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I 21-cm; VLA imaging/atlas. **Locator**: Appendix pp. 14-26; Tables 1-4. **Paper-level environmental claim**: The H I disk extends to roughly twice the stellar disk and is mostly regular; two short SE tails and NW compression are only tentative weak/incipient RPS indicators, not a classic post-stripping remnant. *Claim type*: observed. *Timing*: No tailored RPS clock. The polar event is older than 5 Gyr; Köppen's past-strip label is analytical and carries at least factor-of-two pressure uncertainty.. *Uncertainty/alternative*: H I morphology is direct; mechanism and viewing geometry remain interpretive.
- **Corsini et al. (2012), Polar bulges and polar nuclear discs: the case of NGC 4698.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: HST/UKIRT photometry and stellar populations; polar bulge and nuclear disk. **Locator**: Secs. 2-4. **Paper-level environmental claim**: The polar stellar+gas nuclear disk in NGC4698 is an ancient externally acquired component whose star formation ended more than 5 Gyr ago. *Claim type*: observed plus inferred. *Timing*: >5 Gyr for polar component. *Uncertainty/alternative*: This is an assembly clock, not a current Virgo-processing clock.
- **Huang et al. (2026), MAUVE-MUSE: when metallicity follows or fights star formation - a mass-dependent inversion in Virgo galaxies.** [DOI](#); [ADS](#). *Status/access*: peer reviewed; full text or accepted manuscript. *Tracer/method*: MUSE metallicity and Sigma_SFR; resolved HII-region spaxels in 14 MAUVE galaxies. **Locator**: Secs. 3-6; Figs. 3-6. **Paper-level environmental claim**: The residual metallicity-SFR relation changes sign with calibration and within galaxies; NGC4383 is a robust 0.1-0.2 dex low-metallicity outlier and NGC4192 changes internal sign. *Claim type*: observed analysis. *Timing*: current resolved state. *Uncertainty/alternative*: Strong-line calibration, DIG/HII selection and hierarchical sampling dominate interpretation.
- **Köppen et al. (2018), Ram pressure stripping made easy: an analytical approach.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I deficiency and projected cluster context; analytical peak/time-integrated pressure model. **Locator**: Secs. 7.2-7.3; Tables E1-E5. **Paper-level environmental claim**: Analytical classification: past; compare with direct morphology before adoption. *Claim type*: model-derived. *Timing*: No tailored RPS clock. The polar event is older than 5 Gyr; Köppen's past-strip label is analytical and carries at least factor-of-two pressure uncertainty.. *Uncertainty/alternative*: Peak

pressures are uncertain at roughly factor-of-two level; model assumptions can misclassify tidal systems.

- **Yoon et al. (2017), A history of H I stripping in Virgo: a phase-space view of VIVA galaxies.** [DOI](#); [ADS](#); [arXiv](#). *Status/access*: peer reviewed; full text. *Tracer/method*: H I morphology plus projected phase space; VIVA class and projected phase-space comparison. **Locator**: Sec. 2.1; Table 1; Secs. 4-5. **Paper-level environmental claim**: Yoon class I; provides a statistical stripping-stage context, not a unique orbit. *Claim type*: inferred. *Timing*: stage-scale only. *Uncertainty/alternative*: Projected radius and line-of-sight velocity are degenerate with 3-D orbital state.

6. Cross-sample synthesis

6.1 Stage--mechanism conflicts

The three-stage sequence performs best for clean RPS cases with matched morphology and clocks. It performs poorly when stellar gravity dominates (NGC4216, NGC4567/4568, NGC4694), when a merger precedes stripping (NGC4424), when feedback generates extraplanar emission (NGC4064, NGC4383, NGC4569), or when projected phase space conflicts with the gas disk (NGC4192, NGC4698). These are physical exceptions, not bookkeeping failures to erase.

6.2 Mixed pathways and subgroup context

Virgo's dynamical youth permits group infall and locally enhanced ICM pressure. Yet preprocessing should be asserted only when companion/group evidence exists. NGC4254 and NGC4654 have strong compound interaction+RPS cases; NGC4216 has direct satellite disruption but an uncertain relationship to H I deficiency; NGC4698 has possible M49 context but no secure current-processing reconstruction.

6.3 Timing synthesis

Resolved outer-disk stellar populations commonly date rapid quenching to about 0.1--0.5 Gyr, while tailored models place selected galaxies on either side of peak pressure. Merger/accretion clocks can be hundreds of Myr or >5 Gyr. Tail survival and current radio morphology add still different response times. The atlas therefore stores event-specific clocks instead of one universal time since infall.

6.4 Implications for resolved metallicity--SFR work

A robust MAUVE chemical analysis should stratify by mechanism and excitation, not only stage. H II-region metallicities require DIG/shock/AGN masks; leading/trailing and inside/outside-truncation comparisons should be reported; pair-overlap spaxels require two-component kinematics; gas mass should be separated into H I and H₂; and within-galaxy trends should not be treated as independent population points. NGC4383 is a high-priority low-metallicity outflow/inflow case, while NGC4192's internal sign reversal makes it a calibration and geometry stress test.

6.5 Descriptive mechanism counts

Atlas mechanism group	Components
RPS-dominated	21
mixed RPS + gravity	4
gravity/accretion	4
slow/old depletion	6
feedback/outflow	1
uncertain/weak	4

These counts are heuristic summaries of the 40 selected components, not inferred Virgo population fractions.

7. Prioritized follow-up programme

1. **Excitation-safe metallicity atlas:** build consistent H II/DIG/shock/AGN masks, repeat multiple strong-line calibrations, and publish leading/trailing plus inside/outside-truncation residuals.
2. **Joint gas-phase accounting:** combine VIVA H I and matched-resolution VERTICO CO to test whether metallicity residuals follow total gas, molecular fraction, SFE or removal geometry.
3. **Event-specific SFHs:** fit outer-disk and leading/trailing stellar populations for survey-only galaxies, explicitly separating quenching time from peak-pressure time.
4. **Compound-history modelling:** prioritize NGC4216, NGC4254, NGC4424, NGC4567/4568, NGC4654, NGC4694 and NGC4698 with stellar+gas dynamics; do not fit an ICM-only model to a gravitationally disturbed disk.
5. **Radio/X-ray persistence tests:** use spectral aging, polarization and hot-gas morphology to determine whether apparently post-peak disks retain active pressure signatures.
6. **Distance/substructure audit:** combine surface-brightness-fluctuation/Tully--Fisher distances and group velocities before assigning M49/M86 preprocessing or backsplash histories.
7. **Pair-specific NGC4567/4568 products:** publish component-deblended mass/SFR/metallicity and a separately masked overlap-front analysis.

8. Appendices

Appendix A. Target/component/alias crosswalk

This is the acceptance crosswalk: each of the 39 targets appears once; each of the 40 physical components appears exactly once in the component field.

Target identifier	Physical component (s)	Preferred aliases	Stage	Yoon	Köppen
IC3392	IC3392	IC3392: VCC 1126; UGC 7602; IC 3392	post-peak	IC3392=III	IC3392=past
NGC4064	NGC4064	NGC4064: UGC 7054; NGC 4064	post-peak	NGC4064=III	NGC4064=past
NGC4189	NGC4189	NGC4189: VCC 89; UGC 7235; IC 3050; NGC 4189	pre-peak	NGC4189=0	NGC4189=not included
NGC4192	NGC4192	NGC4192: M 98; VCC 92; UGC 7231; NGC 4192	pre-peak	NGC4192=IV	NGC4192=active
NGC4216	NGC4216	NGC4216: VCC 167; UGC 7284; NGC 4216	post-peak	NGC4216=IV	NGC4216=past
NGC4222	NGC4222	NGC4222: VCC 187; UGC 7291; IC 3087; NGC 4222	pre-peak	NGC4222=0	NGC4222=active
NGC4254	NGC4254	NGC4254: M 99; VCC 307; UGC 7345; NGC 4254	pre-peak	NGC4254=I	NGC4254=non-deficient
NGC4293	NGC4293	NGC4293: VCC 460; UGC 7405; NGC 4293	post-peak	NGC4293=III	NGC4293=past
NGC4294	NGC4294	NGC4294: VCC 465; UGC 7407; NGC 4294; KPG 330; KPG 330b	pre-peak	NGC4294=I	NGC4294=active
NGC4298	NGC4298	NGC4298: VCC 483; UGC 7412; NGC 4298; KPG 332a	near-peak	NGC4298=II	NGC4298=active
NGC4302	NGC4302	NGC4302: VCC 497; UGC 7418; NGC 4302; KPG 332b	near-peak	NGC4302=II	NGC4302=active
NGC4321	NGC4321	NGC4321: M 100; VCC 596; UGC 7450; NGC 4321	pre-peak	NGC4321=0	NGC4321=active
NGC4330	NGC4330	NGC4330: VCC 630; UGC 7456; NGC 4330	near-peak	NGC4330=II	NGC4330=active
NGC4351	NGC4351	NGC4351: VCC 692; UGC 7476; NGC 4351; NGC 4354	pre-peak	NGC4351=I	NGC4351=past
NGC4380	NGC4380	NGC4380: VCC 792; UGC 7503; NGC 4380	post-peak	NGC4380=IV	NGC4380=past
NGC4383	NGC4383	NGC4383: VCC 801; UGC 7507; NGC 4383	pre-peak	NGC4383=0	NGC4383=non-deficient
NGC4388	NGC4388	NGC4388: VCC 836; UGC 7520; NGC 4388	near-peak	NGC4388=II	NGC4388=active
NGC4394	NGC4394	NGC4394: VCC 857; UGC 7523; NGC 4394; KPG 334b	post-peak	NGC4394=IV	NGC4394=not included
NGC4396	NGC4396	NGC4396: VCC 865; UGC 7526; NGC 4396	pre-peak	NGC4396=I	NGC4396=active
NGC4405	NGC4405	NGC4405: VCC 874; UGC 7529; IC 788; NGC 4405	post-peak	NGC4405=III	NGC4405=past
NGC4402	NGC4402	NGC4402: VCC 873; UGC 7528; NGC 4402	near-peak	NGC4402=II	NGC4402=active
NGC4419	NGC4419	NGC4419: VCC 958; UGC 7551; NGC 4419	post-peak	NGC4419=III	NGC4419=past
NGC4424	NGC4424	NGC4424: VCC 979; UGC 7561; NGC 4424	near-peak	NGC4424=II	NGC4424=active
NGC4450	NGC4450	NGC4450: VCC 1110; UGC 7594; NGC 4450	post-peak	NGC4450=IV	NGC4450=past
NGC4457	NGC4457	NGC4457: VCC 1145; UGC 7609; NGC 4457	post-peak	NGC4457=III	NGC4457=not included
NGC4501	NGC4501	NGC4501: M 88; VCC 1401; UGC 7675; NGC 4501	near-peak	NGC4501=II	NGC4501=active
NGC4522	NGC4522	NGC4522: VCC 1516; UGC 7711; NGC 4522	near-peak	NGC4522=II	NGC4522=active
NGC4535	NGC4535	NGC4535: VCC 1555; UGC 7727; NGC 4535	pre-peak	NGC4535=I	NGC4535=active
NGC4548	NGC4548	NGC4548: M 91; VCC 1615; UGC 7753; NGC 4548	post-peak	NGC4548=IV	NGC4548=past
NGC4567_8	NGC4567; NGC4568	NGC4567: VCC 1673; UGC 7777; NGC 4567; KPG 347a; VV 219a / NGC4568: VCC 1676; UGC 7776; NGC 4568; KPG 347b; VV 219b	pre-peak	NGC4567=0; NGC4568=0	NGC4567=not included; NGC4568=not included
NGC4569	NGC4569	NGC4569: M 90; VCC 1690; UGC 7786; NGC 4569	post-peak	NGC4569=III	NGC4569=past
NGC4579	NGC4579	NGC4579: M 58; VCC 1727; UGC 7796; NGC 4579	post-peak	NGC4579=IV	NGC4579=past
NGC4580	NGC4580	NGC4580: VCC 1730; UGC 7794; NGC 4580	post-peak	NGC4580=III	NGC4580=past
NGC4606	NGC4606	NGC4606: VCC 1859; UGC 7839; NGC 4606	post-peak	NGC4606=III	NGC4606=past
NGC4607	NGC4607	NGC4607: VCC 1868; UGC 7843; NGC 4607	post-peak	NGC4607=III	NGC4607=past
NGC4654	NGC4654	NGC4654: VCC 1987; UGC 7902; IC 3708; NGC 4654	near-peak	NGC4654=II	NGC4654=non-deficient
NGC4689	NGC4689	NGC4689: VCC 2058; UGC 7965; NGC 4689	post-peak	NGC4689=IV	NGC4689=past
NGC4694	NGC4694	NGC4694: VCC 2066; UGC 7969; NGC 4694	near-peak	NGC4694=II	NGC4694=past
NGC4698	NGC4698	NGC4698: VCC 2070; UGC 7970; NGC 4698	post-peak	NGC4698=I	NGC4698=past

The expanded 1,328-row identifier table is [alias_crosswalk.csv](#).

Appendix B. Paper-level evidence ledger

The complete 259-record ledger is [paper_level_evidence_ledger.csv](#). It contains DOI, bibcode, arXiv identifier, status/access, component aliases, tracer, scale, method, locator, claim type, mechanism, timing, uncertainty and gFMR relevance. The narrative chapters render every component-linked ledger record in compact annotated form.

Appendix C. Complete all-mentions inventory snapshot

The complete 19,465-link execution-date inventory is [simbad_all_mentions_20260812.csv](#). It contains every object--bibliography link returned by SIMBAD for the 40 component OIDs, including catalogue-level and incidental appearances. It is a database snapshot, not a guarantee that every scholarly mention is indexed by SIMBAD.

Triage category	Links
substantive multi-galaxy analysis	12,340
incidental catalogue mention	3,703
survey measurement	2,978
dedicated study	444

Search-link counts by component

Component	SIMBAD-linked records	Unique bibcodes
IC3392	186	186
NGC4064	190	190
NGC4189	295	295
NGC4192	505	505
NGC4216	428	428
NGC4222	211	211
NGC4254	1,019	1,019
NGC4293	314	314
NGC4294	321	321
NGC4298	364	364
NGC4302	349	349
NGC4321	1,977	1,977
NGC4330	249	249
NGC4351	241	241
NGC4380	277	277
NGC4383	279	279
NGC4388	1,442	1,442
NGC4394	380	380
NGC4396	209	209
NGC4402	424	424
NGC4405	169	169
NGC4419	423	423
NGC4424	413	413
NGC4450	589	589
NGC4457	350	350
NGC4501	849	849
NGC4522	400	400
NGC4535	690	690
NGC4548	643	643
NGC4567	340	340
NGC4568	397	397
NGC4569	957	957
NGC4579	1,145	1,145
NGC4580	235	235
NGC4606	175	175
NGC4607	186	186
NGC4654	604	604
NGC4689	422	422
NGC4694	306	306
NGC4698	512	512

Appendix D. Curated source registry

The machine-readable registry is [curated_source_metadata.csv](#). Journal/preprint versions are deduplicated as a single work while retaining DOI, bibcode and arXiv relationships.

Year	Citation/title	Journal	DOI/ADS/arXiv	Status/access	Scope	Locator
1972	Gunn et al. (1972) — On the Infall of Matter Into Clusters of Galaxies and Some Effects on Their Evolution	The Astrophysical Journal 176 1	DOI	peer reviewed; full text or accepted manuscript	framework	Sec. II
1996	Kenney et al. (1996) — Evidence for a merger in the peculiar Virgo cluster Sa galaxy NGC 4424	The Astronomical Journal 111 152	DOI ; ADS	peer reviewed; full text or accepted manuscript	NGC4424	Secs. 3-5
1999	Bertola et al. (1999) — The bulge-disk orthogonal decoupling in galaxies: NGC 4698	The Astrophysical Journal 519 L127-L130	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4698	pp. L128-L130
2002	Yoshida et al. (2002) — Discovery of a very extended emission-line region around the Seyfert 2 galaxy NGC 4388	The Astrophysical Journal 567 118-129	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4388	Secs. 3-5
2003	Vollmer et al. (2003) — NGC 4654: Gravitational interaction or ram pressure stripping?	Astronomy & Astrophysics 398 525-539	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4654	Secs. 3-6
2004	Kenney et al. (2004) — VLA HiObservations of Gas Stripping in the Virgo Cluster Spiral NGC 4522	The Astronomical Journal 127 3361-3374	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4522	Secs. 3-5
2004	Koopmann et al. (2004) — H{alpha} morphologies and environmental effects in Virgo cluster spiral galaxies	The Astrophysical Journal 613 866-885	DOI ; ADS ; arXiv	peer reviewed; full text	32 components	Secs. 3-5; Tables 3-5
2004	Vollmer et al. (2004) — NGC 4569: Recent evidence for a past ram pressure stripping event	Astronomy & Astrophysics 419 35-46	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4569	Secs. 3-5
2004	Yoshida et al. (2004) — Subaru deep spectroscopy of the very extended emission-line region of NGC 4388: ram pressure stripped gas ionized by nuclear radiation	The Astronomical Journal 127 90-104	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4388	Secs. 3-5
2005	Crowl et al. (2005) — Dense cloud ablation and ram pressure stripping of the Virgo spiral NGC 4402	The Astronomical Journal 130 65-72	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4402	Secs. 3-5
2005	Oosterloo et al. (2005) — A large H I cloud near the centre of the Virgo cluster	Astronomy & Astrophysics 437 L19-L22	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4388	pp. L20-L22
2005	Vollmer et al. (2005) — NGC 4254: a spiral galaxy entering the Virgo cluster	Astronomy & Astrophysics 439 921-933	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4254	Secs. 3-6
2006	Chyży et al. (2006) — Large-scale magnetized outflows from the Virgo Cluster spiral NGC 4569. A galactic wind in a ram pressure wind	Astronomy & Astrophysics 447 465-472	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4569	Secs. 3-5
2006	Cortés et al. (2006) — The nature of the peculiar Virgo cluster galaxies NGC 4064 and NGC 4424	The Astronomical Journal 131 747-770	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4064; NGC4424	Secs. 3-6
2006	Crowl et al. (2006) — The Stellar Population of Stripped Cluster Spiral NGC 4522: A Local Analog to K+A Galaxies?	The Astrophysical Journal 649 L75-L78	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4522	pp. L76-L78
2006	Soida et al. (2006) — NGC 4654: polarized radio continuum emission as a diagnostic tool for a galaxy-cluster interaction. Models versus observations	Astronomy & Astrophysics 458 727-739	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4654	Secs. 3-5
2006	Vollmer et al. (2006) — A dynamical model for the heavily ram pressure stripped Virgo spiral galaxy NGC 4522	Astronomy & Astrophysics 453 883-893	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4522	Secs. 3-5
2007	Chung et al. (2007) — Virgo Galaxies with Long One-sided H I Tails	The Astrophysical Journal 659 L115-L119	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4294; NGC4302; NGC4330; NGC4396; NGC4424; NGC4654	pp. L116-L118
2007	Duc et al. (2007) — VCC 2062: an old tidal dwarf galaxy in the Virgo cluster?	Astronomy & Astrophysics 475 187-197	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4694	Secs. 3-6
2007	Haynes et al. (2007) — NGC 4254: an act of harassment uncovered by the Arecibo Legacy Fast ALFA survey	The Astrophysical Journal 665 L19-L22	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4254	pp. L20-L22

Year	Citation/title	Journal	DOI/ADS/arXiv	Status/access	Scope	Locator
2007	Kantharia et al. (2007) — Low radio frequency signatures of ram pressure stripping in Virgo spiral NGC 4254	Monthly Notices of the Royal Astronomical Society 383 173-182	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4254	Secs. 3-5
2008	Crowl et al. (2008) — The stellar populations of stripped spiral galaxies in the Virgo cluster	The Astronomical Journal 136 1623-1644	DOI ; ADS ; arXiv	peer reviewed; full text	IC3392;NGC4064;NGC4388;NGC4402;NGC4405;NGC4419;NGC4424;NGC4522;NGC4569;NGC4580	Secs. 3-5; Tables 3-5
2008	Duc et al. (2008) — Tidal debris from high-velocity collisions as fake dark galaxies: a numerical model of VIRGOHI 21	The Astrophysical Journal 673 787-797	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4254	Secs. 2-5
2008	Vollmer et al. (2008) — Pre-peak ram pressure stripping in the Virgo cluster spiral galaxy NGC 4501	Astronomy & Astrophysics 483 89-106	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4501	Secs. 3-5
2009	Chung et al. (2009) — VLA imaging of Virgo spirals in atomic gas (VIVA). I. The atlas and the H I properties	The Astronomical Journal 138 1741-1816	DOI ; ADS ; arXiv	peer reviewed; full text	all	Appendix pp. 14-26; Tables 1-4
2009	Vollmer et al. (2009) — A holistic view on ram pressure stripping in the Virgo cluster. The first complete model-based time sequence	Astronomy & Astrophysics 502 427-435	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4330;NGC4388;NGC4501;NGC4522;NGC4569	Secs. 2-3; Table 3
2010	Mannucci et al. (2010) — A fundamental relation between mass, star formation rate and metallicity in local and high-redshift galaxies	Monthly Notices of the Royal Astronomical Society 408 2115-2127	DOI ; arXiv	peer reviewed; full text	framework	Secs. 2-5
2010	Pappalardo et al. (2010) — Pinning down the ram-pressure-induced halt of star formation in the Virgo cluster spiral galaxy NGC 4388. A joint inversion of spectroscopic and photometric data	Astronomy and Astrophysics 514 A33	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4388	Sect. 5; Fig. 10; Sect. 6.1
2011	Abramson et al. (2011) — Caught in the act: strong, active ram pressure stripping in Virgo cluster spiral NGC 4330	The Astronomical Journal 141 164	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4330	Secs. 3-5
2011	Weżgowiec et al. (2011) — Hot gas in Mach cones around Virgo cluster spiral galaxies	Astronomy & Astrophysics 531 A44	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4388;NGC4501;NGC4569	Secs. 3-5
2012	Corsini et al. (2012) — Polar bulges and polar nuclear discs: the case of NGC 4698	Monthly Notices of the Royal Astronomical Society: Letters 423 L79-L83	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4698	Secs. 2-4
2012	Vollmer et al. (2012) — Ram pressure stripping of the multiphase ISM and star formation in the Virgo spiral galaxy NGC 4330	Astronomy & Astrophysics 537 A143	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4330	Secs. 3-5
2013	Bothwell et al. (2013) — A fundamental relation between the metallicity, gas content and stellar mass of local galaxies	Monthly Notices of the Royal Astronomical Society 433 1425-1435	DOI ; arXiv	peer reviewed; full text	framework	Secs. 3-5
2013	Gu et al. (2013) — Multi-wavelength studies of spectacular ram pressure stripping of a galaxy: discovery of an X-ray absorption feature	The Astrophysical Journal 777 L36	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4388	Sects. 3-4

Year	Citation/title	Journal	DOI/ADS/arXiv	Status/access	Scope	Locator
2013	Paudel et al. (2013) — The next generation Virgo cluster survey. IV. NGC 4216: a bombarded spiral in the Virgo cluster	The Astrophysical Journal 767 133	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4216	Secs. 2-4
2013	Vollmer et al. (2013) — Large-scale radio continuum properties of 19 Virgo cluster galaxies. The influence of tidal interactions, ram pressure stripping, and accreting gas envelopes	Astronomy & Astrophysics 553 A116	DOI ; ADS ; arXiv	peer reviewed; full text	19 Virgo galaxies	Secs. 3-5; Tables 2-3
2014	Abramson et al. (2014) — HUBBLE SPACE TELESCOPEIMAGING OF DECOUPLED DUST CLOUDS IN THE RAM PRESSURE STRIPPED VIRGO SPIRALS NGC 4402 AND NGC 4522	The Astronomical Journal 147 63	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4402 ;NGC4522	Secs. 3-5
2014	Chung et al. (2014) — H I and H2 properties and star formation efficiency of NGC 4654 in the cluster environment	Publications of the Astronomical Society of Japan 66 11	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4654	Secs. 3-5
2015	Verdugo et al. (2015) — Ram pressure stripping in the Virgo Cluster	Astronomy & Astrophysics 582 A6	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4388	Table 2; Sects. 4-5
2016	Boselli et al. (2016) — Spectacular tails of ionized gas in the Virgo cluster galaxy NGC 4569	Astronomy & Astrophysics 587 A68	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4569	Secs. 3-6
2016	Lisenfeld et al. (2016) — Molecular gas and star formation in the tidal dwarf galaxy VCC 2062	Astronomy & Astrophysics 590 A92	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4694	Table 4; Sect. 6.1
2016	Nehlig et al. (2016) — Effects of environmental gas compression on the multiphase ISM and star formation. The Virgo spiral galaxies NGC 4501 and NGC 4567/68	Astronomy & Astrophysics 587 A108	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4501 ;NGC4567;NGC4568	Secs. 3-5
2017	Kaneko et al. (2017) — Properties of molecular gas in galaxies in the early and mid stages of interaction. II. Molecular gas fraction	Publications of the Astronomical Society of Japan 69	DOI	peer reviewed; full text or accepted manuscript	NGC4567 ;NGC4568	Secs. 3-5
2017	Lee et al. (2017) — The effect of ram pressure on the molecular gas of galaxies: three case studies in the Virgo cluster	Monthly Notices of the Royal Astronomical Society 466 1382-1398	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4330 ;NGC4402;NGC4522	Secs. 3-5
2017	Yoon et al. (2017) — A history of H I stripping in Virgo: a phase-space view of VIVA galaxies	The Astrophysical Journal 838 81	DOI ; ADS ; arXiv	peer reviewed; full text	all	Sec. 2.1; Table 1; Sects. 4-5
2018	Boselli et al. (2018) — A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE). IV. A tail of ionised gas in the merger remnant NGC4424	Astronomy & Astrophysics 620 A164	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4424	Secs. 3-6
2018	Fossati et al. (2018) — A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE). II. Constraining the quenching time in the stripped galaxy NGC 4330	Astronomy & Astrophysics 614 A57	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4330	Secs. 3-5
2018	Kaneko et al. (2018) — Discovery of a molecular collision front in interacting galaxies NGC 4567/4568 with ALMA	The Astrophysical Journal Letters 860 L14	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4567 ;NGC4568	Secs. 2-4
2018	Köppen et al. (2018) — Ram pressure stripping made easy: an analytical approach	Monthly Notices of the Royal Astronomical Society 479 4367-4390	DOI ; ADS ; arXiv	peer reviewed; full text	35 components	Secs. 7.2-7.3; Tables E1-E5
2018	Vollmer et al. (2018) — Two uneven sisters. I. NGC 4388 - a strongly constrained ram-pressure stripping event	Astronomy & Astrophysics 620 A108	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4388	Table 2; Sects. 3-5
2020	Taylor et al. (2020) — Faint and fading tails: the fate of stripped H I gas in Virgo cluster galaxies	The Astronomical Journal 159 218	DOI ; ADS ; arXiv	peer reviewed; full text	VIVA tails	Secs. 3-5

Year	Citation/title	Journal	DOI/ADS/arXiv	Status/access	Scope	Locator
2020	W. J. et al. (2020) — ALMA evidence for ram pressure compression and stripping of molecular gas in the Virgo Cluster galaxy NGC 4402	The Astrophysical Journal 901 95	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4402	Secs. 3-5
2021	Brown et al. (2021) — VERTICO: the Virgo Environment Traced in CO survey	The Astrophysical Journal Supplement Series 257 21	DOI ; ADS ; arXiv	peer reviewed; full text	all	Sec. 2; Tables 1-2; Sec. 6
2021	Cortese et al. (2021) — The Dawes Review 9: The role of cold gas stripping on the star formation quenching of satellite galaxies	Publications of the Astronomical Society of Australia 38	DOI ; arXiv	peer reviewed review; full text	framework	Secs. 2-4
2021	Lizée et al. (2021) — Gas compression and stellar feedback in the tidally interacting and ram-pressure stripped Virgo spiral galaxy NGC 4654	Astronomy & Astrophysics 645 A111	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4654	Secs. 3-6
2021	Vollmer et al. (2021) — A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE). VIII. Modeling ram pressure stripping of diffuse gas in the Virgo cluster spiral galaxy NGC 4330	Astronomy & Astrophysics 645 A121	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4330	Secs. 3-5
2022	Boselli et al. (2022) — Ram pressure stripping in high-density environments	The Astronomy and Astrophysics Review 30	DOI ; arXiv	peer reviewed review; full text	framework	Secs. 3-6
2022	Kaneko et al. (2022) — Properties of molecular gas in galaxies in early and mid stages of Interaction. III. Resolved Kennicutt-Schmidt law	Publications of the Astronomical Society of Japan 74 343-363	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4567; NGC4568	Pair discussion; summary
2022	Sardaneta et al. (2022) — A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE). XI. Two-dimensional H α kinematics of the edge-on ram pressure stripped galaxy NGC 4330	Astronomy & Astrophysics 659 A45	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4330	Secs. 3-5
2022	Villanueva et al. (2022) — VERTICO. IV. Environmental Effects on the Gas Distribution and Star Formation Efficiency of Virgo Cluster Spirals	The Astrophysical Journal 940 176	DOI ; ADS ; arXiv	peer reviewed; full text	VERTICO sample	Secs. 3-5
2022	Zabel et al. (2022) — VERTICO II: How H I-identified Environmental Mechanisms Affect the Molecular Gas in Cluster Galaxies	The Astrophysical Journal 933 10	DOI ; ADS ; arXiv	peer reviewed; full text	VERTICO sample	Secs. 3-5; Tables 1-2
2023	Brown et al. (2023) — VERTICO. VII. Environmental Quenching Caused by the Suppression of Molecular Gas Content and Star Formation Efficiency in Virgo Cluster Galaxies	The Astrophysical Journal 956 37	DOI ; ADS ; arXiv	peer reviewed; full text	VERTICO sample	Secs. 3-6
2023	Jiménez-Donaire et al. (2023) — VERTICO III. The Kennicutt-Schmidt relation in Virgo cluster galaxies	Astronomy & Astrophysics 671 A3	DOI ; ADS ; arXiv	peer reviewed; full text	VERTICO sample	Secs. 3-5
2023	Kaneko et al. (2023) — Investigating physical states of molecular gas in the overlapping region of interacting galaxies NGC 4567/4568 using ALMA	Publications of the Astronomical Society of Japan 75 646-659	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4567; NGC4568	Secs. 3-5
2023	Roberts et al. (2023) — VERTICO VI. Cold-gas asymmetries in Virgo cluster galaxies	Astronomy & Astrophysics 675 A78	DOI ; ADS ; arXiv	peer reviewed; full text	VERTICO sample	Secs. 3-6
2023	Watts et al. (2023) — VERTICO V: The environmentally driven evolution of the inner cold gas discs of Virgo cluster galaxies	Publications of the Astronomical Society of Australia 40	DOI ; arXiv	peer reviewed; full text	18 VERTICO disks	Secs. 2, 4-6
2024	Edler et al. (2024) — VICTORIA project: The LOFAR view of environmental effects in Virgo cluster star-forming galaxies	Astronomy & Astrophysics 683 A149	DOI ; ADS ; arXiv	peer reviewed; full text	12 MAUVE components	Secs. 3-6; Tables 1-3
2024	Hou et al. (2024) — X-Ray Constraints on the Hot Gaseous Corona of Edge-on Late-type Galaxies in Virgo	The Astrophysical Journal 961 249	DOI ; ADS ; arXiv	peer reviewed; full text	edge-on Virgo disks	Secs. 3-5

Year	Citation/title	Journal	DOI/ADS/arXiv	Status/access	Scope	Locator
2024	Morgan et al. (2024) — A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE) XVI. The ubiquity of truncated star-forming discs across the Virgo cluster environment	Astronomy & Astrophysics 691 A20	DOI ; ADS ; arXiv	peer reviewed; full text	Virgo late types	Secs. 3-6
2024	Watts et al. (2024) — MAUVE: a 6 kpc bipolar outflow launched from NGC 4383, one of the most H I-rich galaxies in the Virgo cluster	Monthly Notices of the Royal Astronomical Society 530 1968-1983	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4383	Secs. 3-6
2025	Catinella et al. (2025) — Multiphase Astrophysics to Unveil the Virgo Environment (MAUVE)	The Messenger 195 15-18	DOI	peer reviewed; full text or accepted manuscript	all	pp. 15-18
2025	Davis et al. (2025) — VERTICO - IX. Signatures of environmental processing of the gas in Virgo cluster spiral galaxies through mapping of CO isotopologues	Monthly Notices of the Royal Astronomical Society 541 4031-4048	DOI ; ADS ; arXiv	peer reviewed; full text	VERTICO sample	Secs. 3-6
2026	Attwater et al. (2026) — MAUVE-MUSE: A Star-formation-driven Outflow Caught in the Act of Quenching the Stripped Virgo Galaxy NGC 4064	The Astrophysical Journal Letters 996 L40	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4064	Secs. 3-5; Table 1
2026	Brown et al. (2026) — MAUVE-MUSE: Ionization and Kinematic Signatures of Environmental Effects on Virgo Cluster Disks	The Astrophysical Journal 1004 73	DOI ; ADS ; arXiv	peer reviewed; full text	12 MAUVE galaxies	Secs. 3-6; Table 2
2026	Choi et al. (2026) — The Impact of Ram Pressure on the Radio Spectral Index and Magnetic Field of NGC 4522: A High-resolution VLA Continuum Study	The Astrophysical Journal 999 116	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4522	Secs. 3-6; Table 3
2026	Huang et al. (2026) — MAUVE-MUSE: when metallicity follows or fights star formation - a mass-dependent inversion in Virgo galaxies	Monthly Notices of the Royal Astronomical Society 549 1019	DOI ; ADS	peer reviewed; full text or accepted manuscript	14 MAUVE galaxies	Secs. 3-6; Figs. 3-6
2026	Kai Wang et al. (2026) — Inflow-driven galaxy evolution - I. Revealing the physics of the fundamental metallicity relation	arXiv e-prints	ADS ; arXiv	preprint v1; submitted 2026-08-05; not peer reviewed as of 2026-08-12; full text	framework	Secs. 2.1.2, 3.3, 4.4, 5.2; Fig. 12
2026	Sun et al. (2026) — A Virgo Environmental Survey Tracing Ionised Gas Emission (VESTIGE) XIX. The discovery of a spectacular 230 kpc H α tail following NGC 4569 in the Virgo cluster	Astronomy & Astrophysics 705 A139	DOI ; ADS ; arXiv	peer reviewed; full text	NGC4569	Secs. 3-7

Appendix E. Access gaps and negative results

- The SIMBAD all-mentions inventory is complete only for links returned on 2026-08-12; database indexing and object association are outside this atlas's control.
- A NED object identity query succeeded, but its reference-table endpoint stalled during this execution. NED was not silently treated as returning zero references.
- The ESO Messenger DOI did not resolve through Crossref (HTTP 404) but was checked against the publisher/PDF record.
- Several versions of record were gated; green-open-access manuscripts/preprints supplied full text. Where only metadata or abstract-level access was available, the record is labelled and not used for detailed claims.
- No dedicated modern environmental-processing paper was found for several survey-only galaxies; those chapters explicitly say so.
- Wang et al. 2026, arXiv:2608.04784, is a v1 preprint and cannot outweigh peer-reviewed empirical work.

Appendix F. Reproducibility and figure provenance

Derived phase-space values are [phase_space_derived_values.csv](#). Figure constants, palettes and caveats are [figure_provenance.json](#). Timeline inputs are [timeline_model_peak.csv](#) and [timeline_outer_quench.csv](#). Mechanism-stage counts are [mechanism_stage_counts.csv](#).

Appendix G. Citation and claim-integrity statement

No opaque browsing tokens or invented source identifiers are used. Detailed claims are tied to DOI/ADS/arXiv records and source locators. Model-derived histories remain labelled as such; phase-space categories are never promoted to unique orbits. The two reclassification audits and all access limitations remain visible rather than reconciled away.